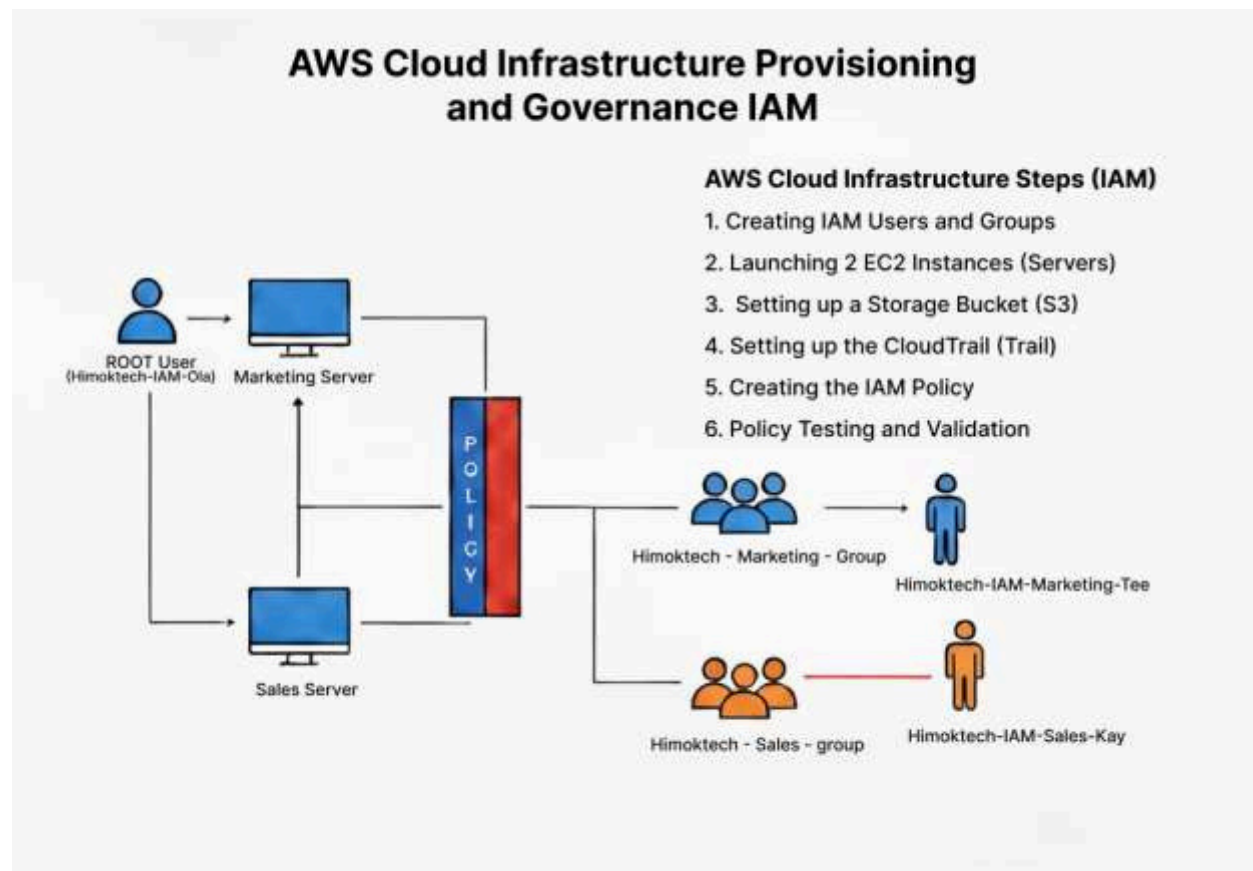


AWS Cloud Infrastructure Provisioning and Governance IAM

This document outlines a phased approach to deploying core infrastructure services, focusing on Identity and Access Management (IAM) and resource provisioning in the AWS Cloud.



Project Goal: To establish a secure, multi-user cloud environment with defined access controls for Sales and Marketing teams, including necessary compute instances and logging capabilities.

Project Location Selection

The AWS Region (location) will be selected based on its proximity to the client to ensure optimal network performance (low latency) when hosting web applications or cloud-based services. For this specific project's configuration, the Austria region is chosen as the deployment location.

Cloud Security and Resource Deployment Steps (IAM Focus)

The overall objective is to implement IAM (Identity and Access Management), the system that manages who can access which resources in your cloud environment.

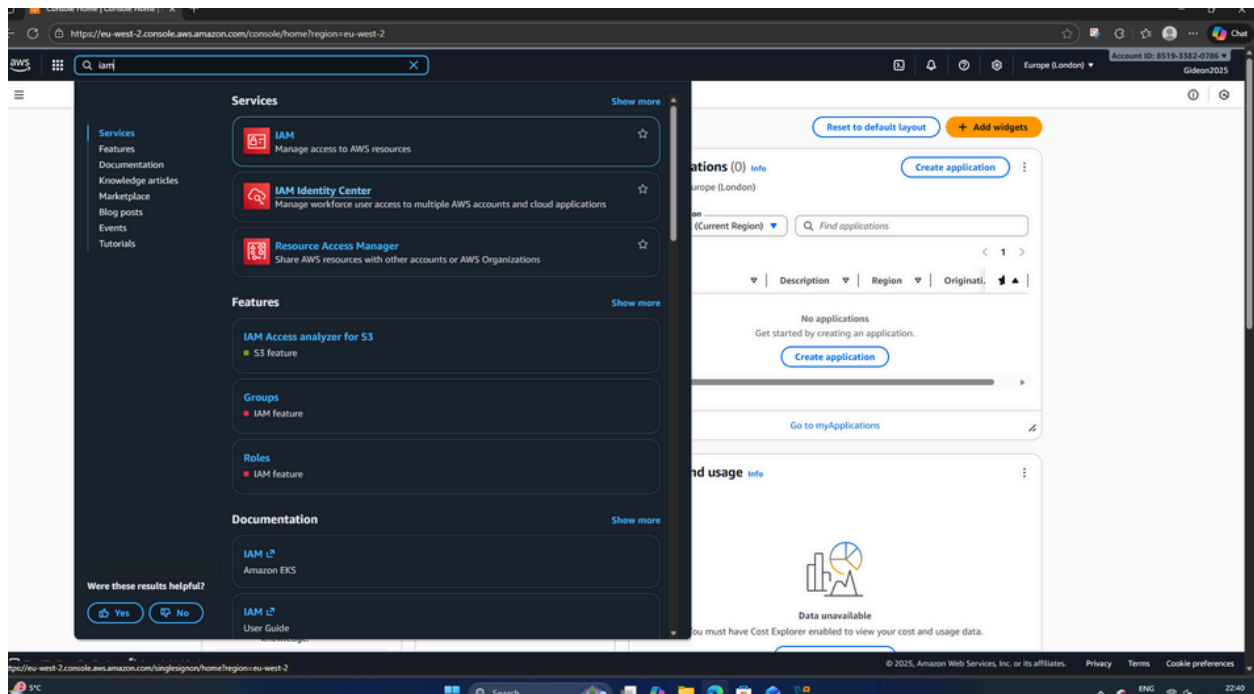
Phase A: Creating IAM Users and Groups

This phase establishes the foundational identities and permission structure.

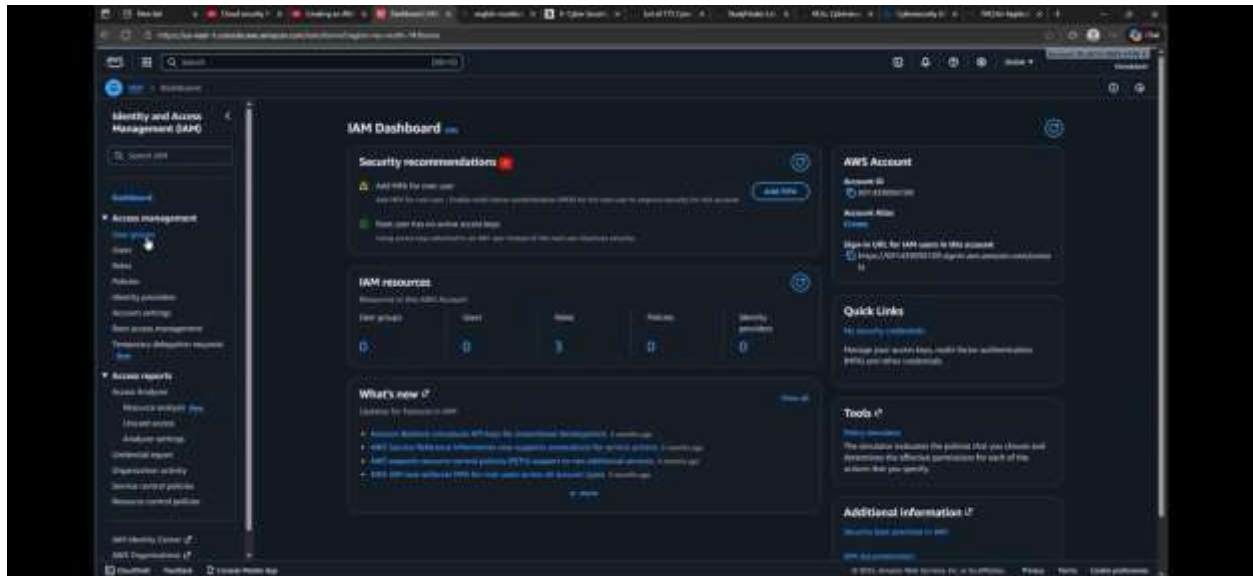
- IAM User: A specific person or application that interacts with AWS.
- IAM Group: A collection of IAM users. Giving permissions to a group automatically gives those permissions to all users in it.
- Policies: Documents that define the permissions (rules) for users, groups, or roles (e.g., "Allow access to this storage bucket").

Step 1: Create IAM Groups

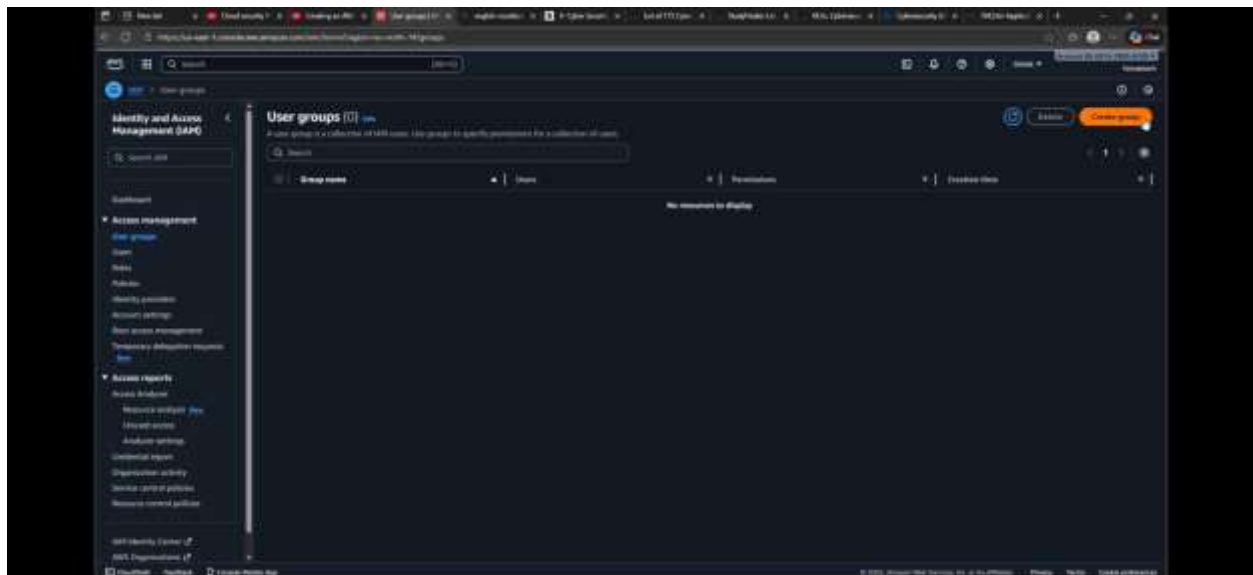
1. Navigate to the IAM service via the search bar.



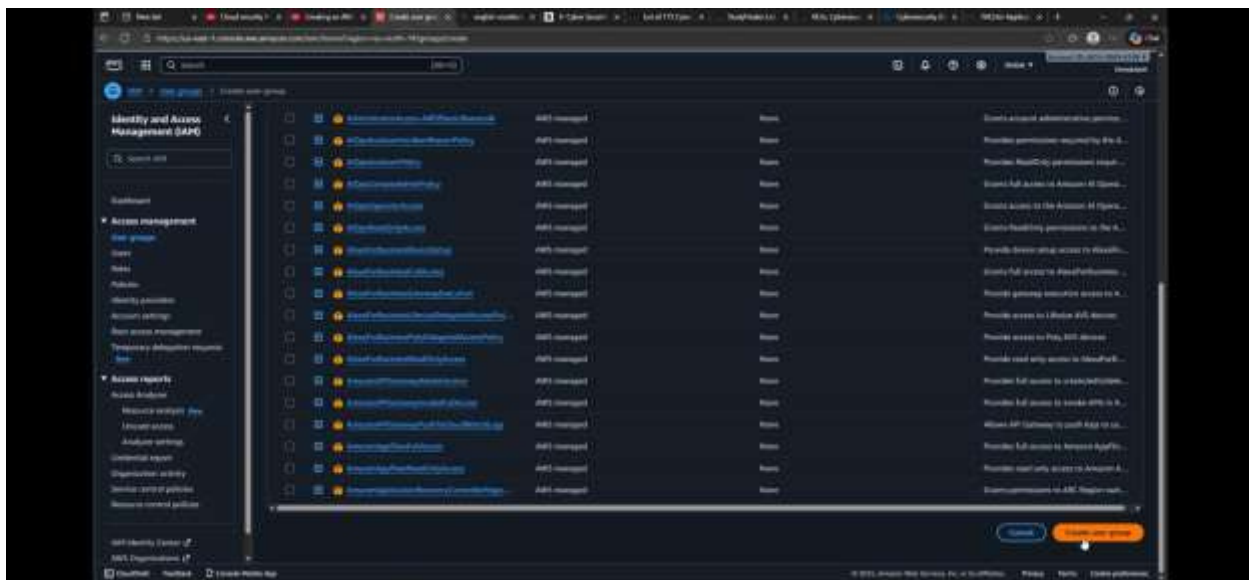
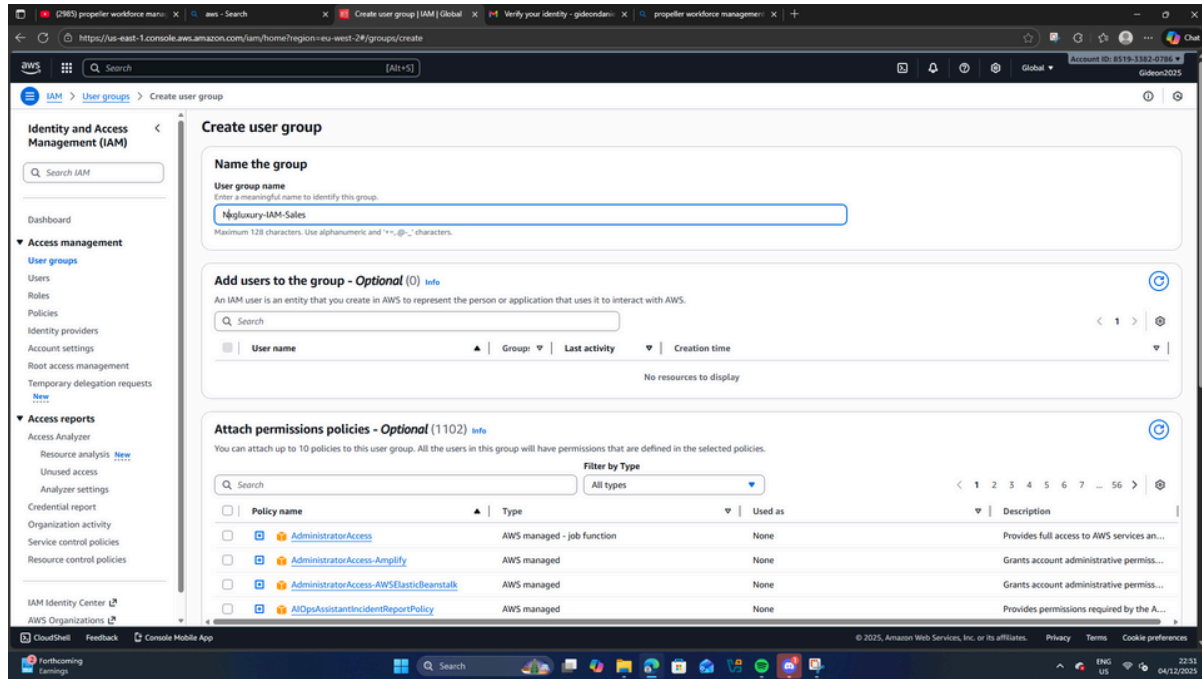
2. On the dashboard interface, select **"user group"** on the left-hand side.

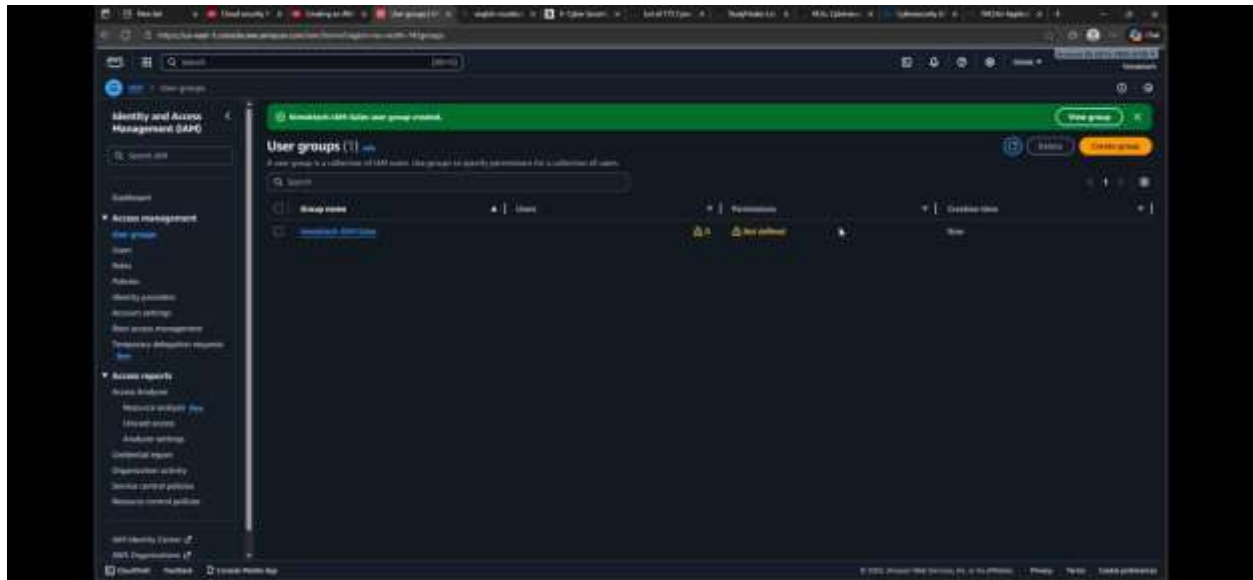


3. Click **"create group"** to establish the first user group.



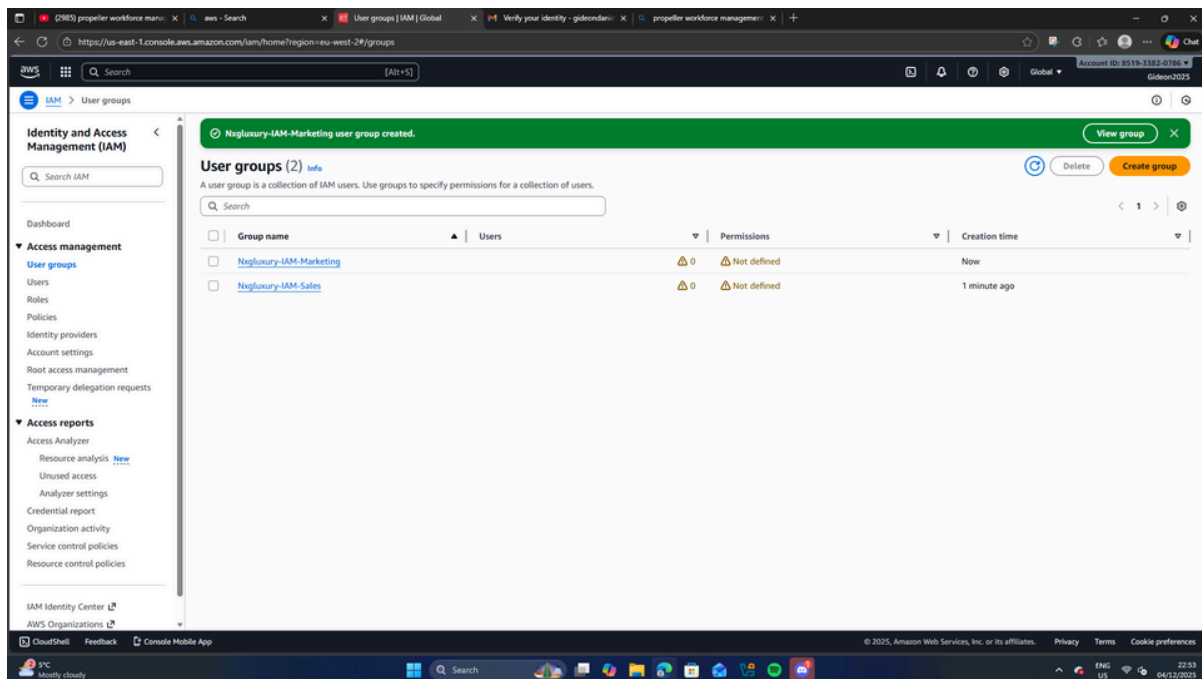
•User Group Name: Nxgluxury-IAM-Sales





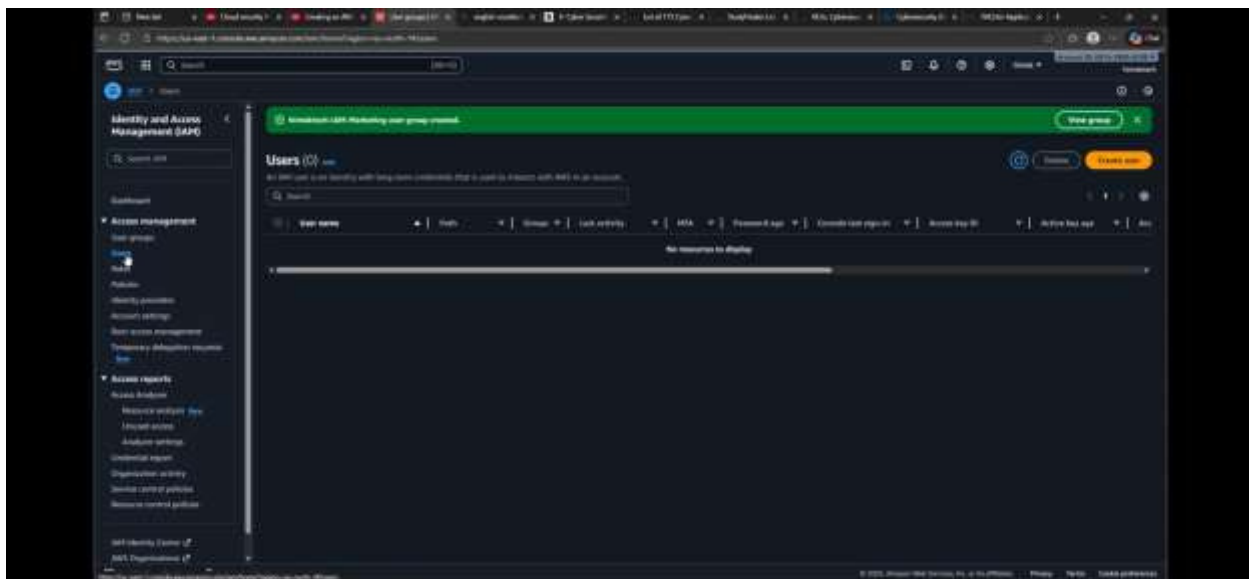
4. Repeat the process to create the second group.

- User Group Name: Nxgluxury-IAM-Marketing



Step 2: Create and Configure IAM Users

1. From the IAM dashboard, click "**users**" on the left and then "**create user**".



2. Create and Configure User 1 (Admin Privilege):

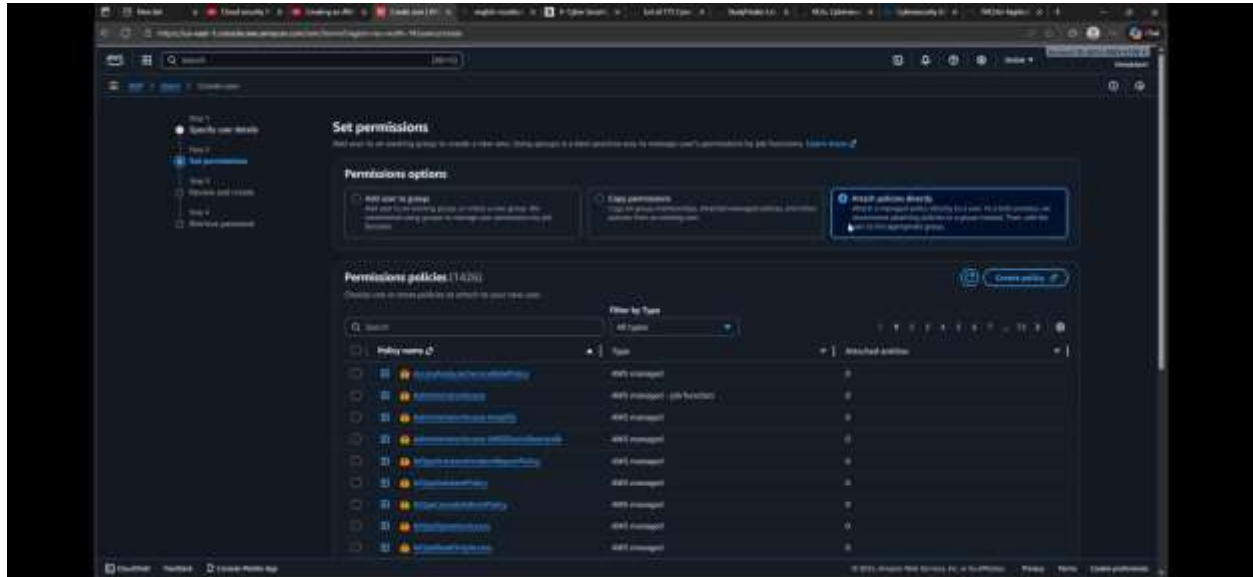
- Username: Nxgluxury-IAM-Gideon • Select the option to provide user access to the AWS Management Console.
- Choose Custom Password and input the required password.

The screenshot shows the AWS IAM console 'Create user' page. The left sidebar indicates the current step is 'Step 1: Specify user details'. The main content area is titled 'Specify user details' and contains the following sections:

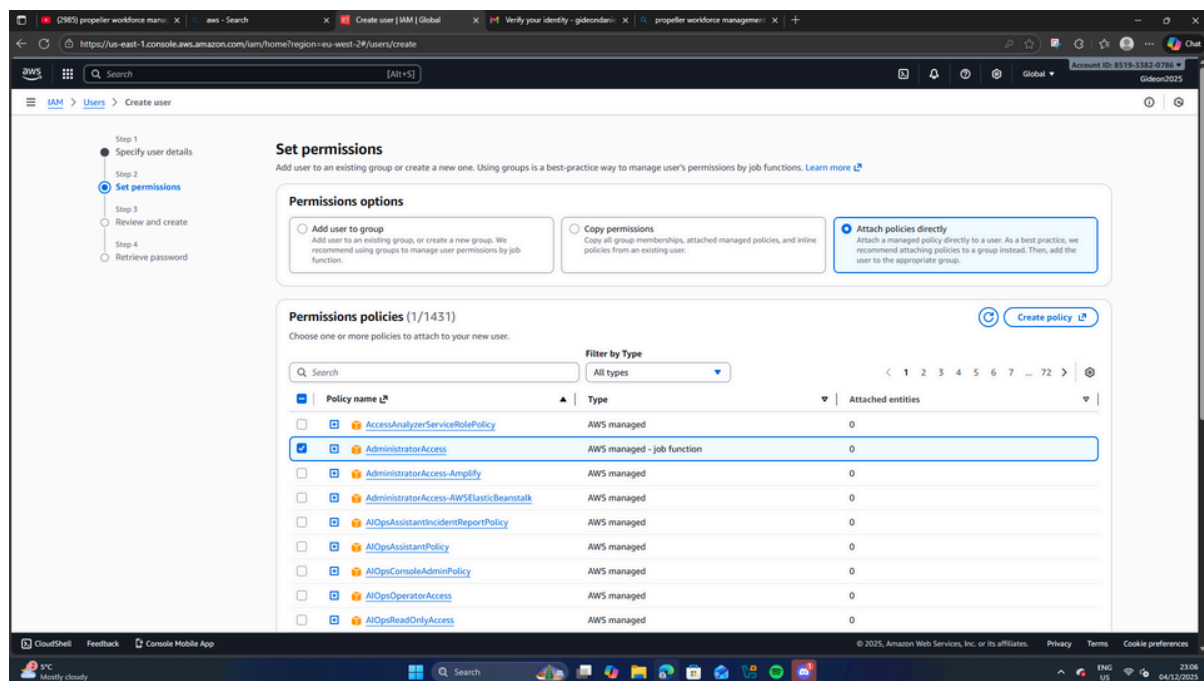
- User details:** A text input field for 'User name' containing 'Nxgluxury-IAM-Gideon'. Below the field, a note states: 'The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and +, -, @, _ (hyphen)'.
- Provide user access to the AWS Management Console - optional:** A checkbox that is checked. Below it, a note states: 'In addition to console access, users with SignInLocalDeveloperAccess permissions can use the same console credentials for programmatic access without the need for access keys.'
- Console password:** Two radio button options: 'Autogenerated password' (unselected) and 'Custom password' (selected). Below the 'Custom password' option, there is a text input field for the password, which is currently empty. A note below the field states: 'Enter a custom password for the user.'
- Must be at least 8 characters long**
- Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols (! @ # \$ % ^ & * () _ + - { } | ' " , . ; : ')**
- Show password:** An unchecked checkbox.
- Users must create a new password at next sign-in - Recommended:** A checked checkbox. Below it, a note states: 'Users automatically get the IAMUserChangePassword policy to allow them to change their own password.'
- Learn more:** A link to learn more about programmatic access.

At the bottom right of the form, there are 'Cancel' and 'Next' buttons. The 'Next' button is highlighted in orange.

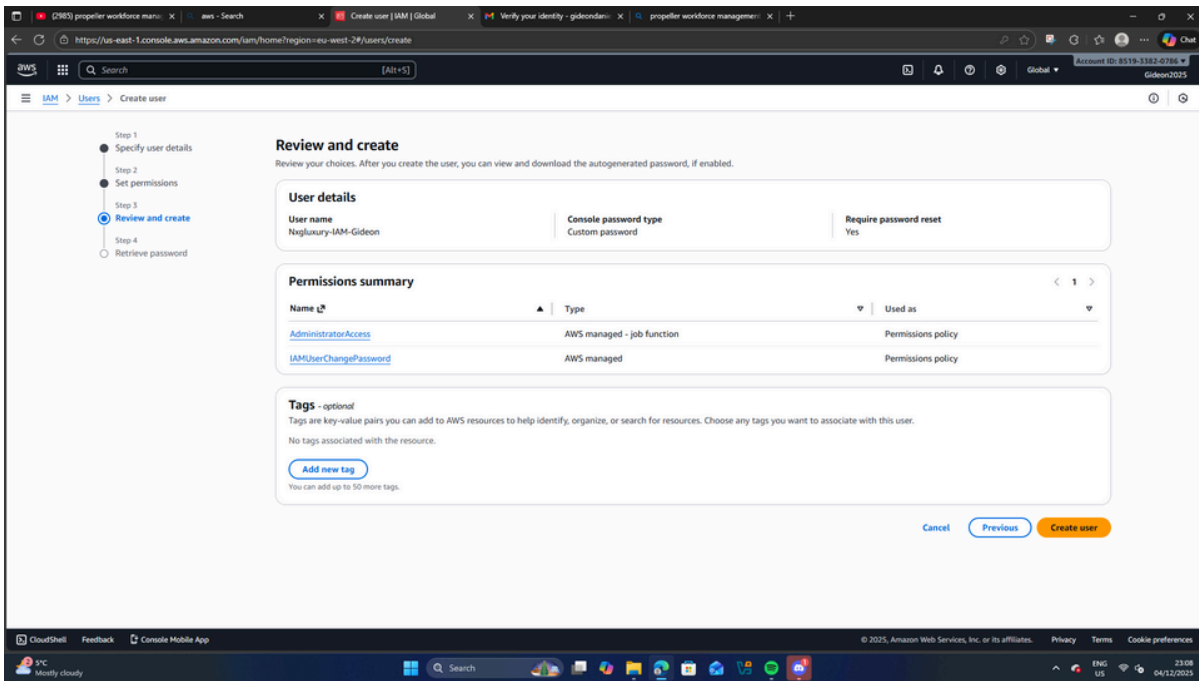
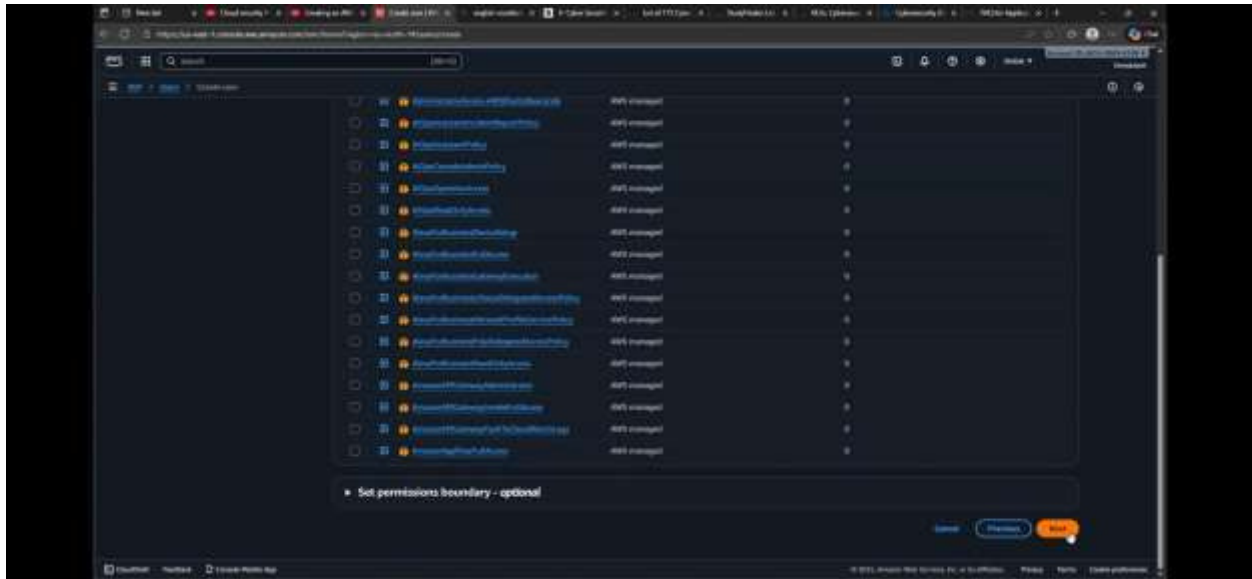
- Set Permission: Select Attach policies directly.



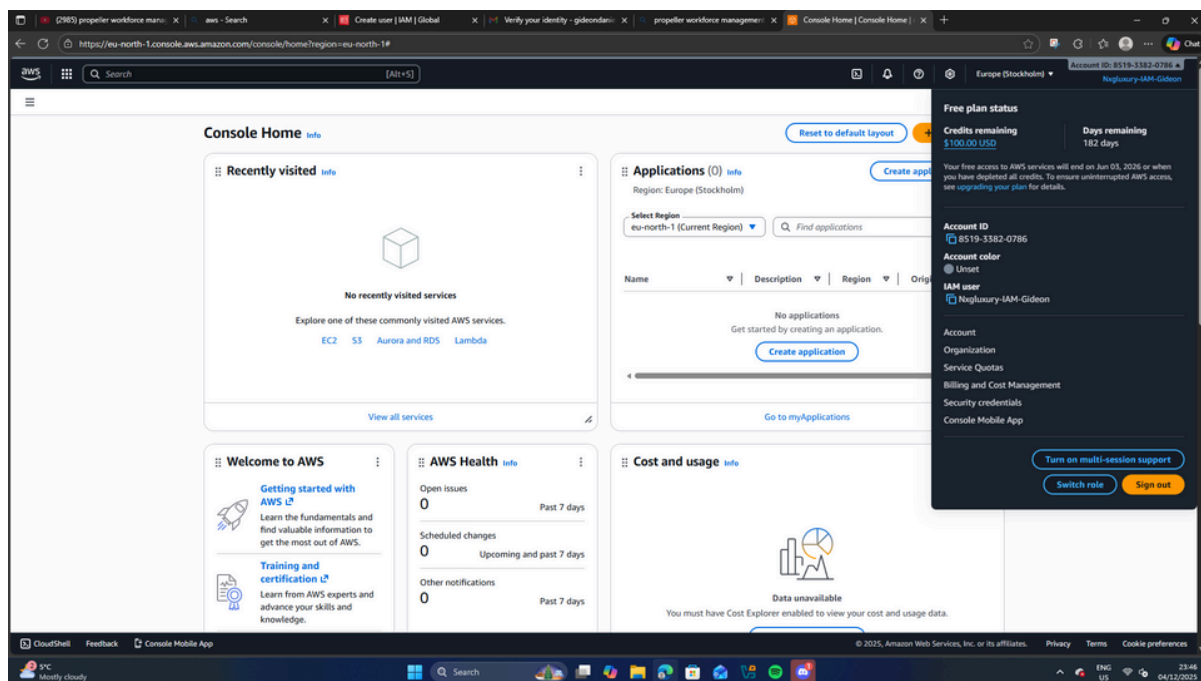
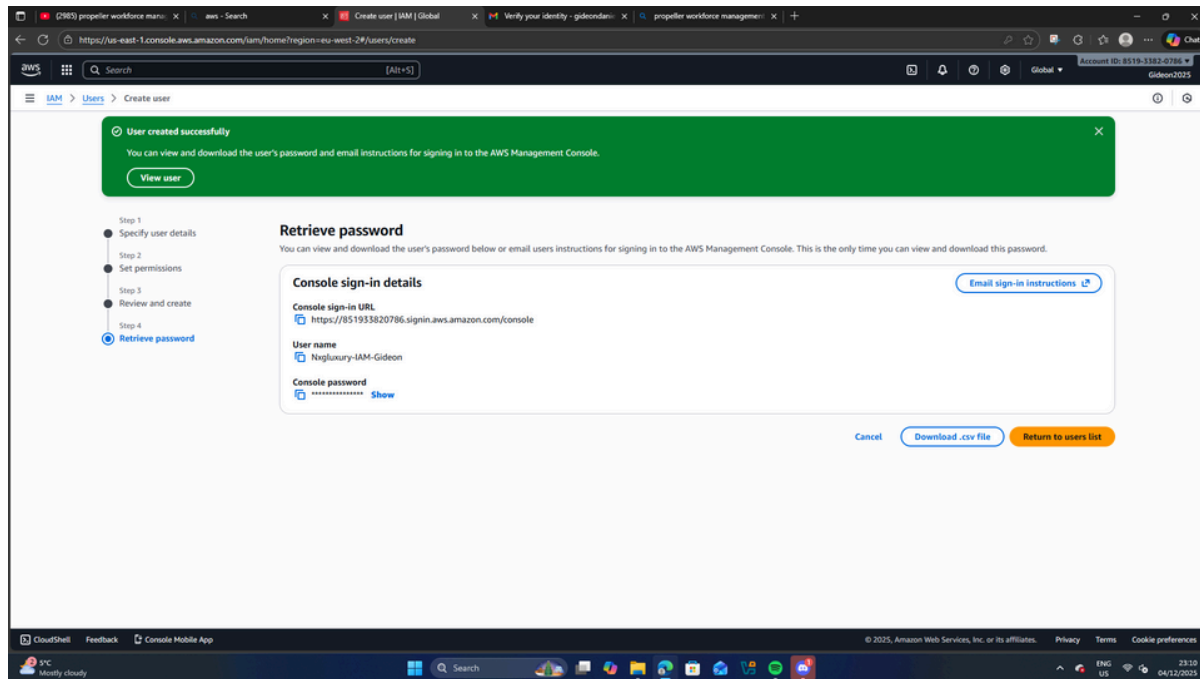
- Choose the Administrative Access policy to grant this user full control.



- Complete the creation steps. The user "GIDEON" has been created with administrative privilege.

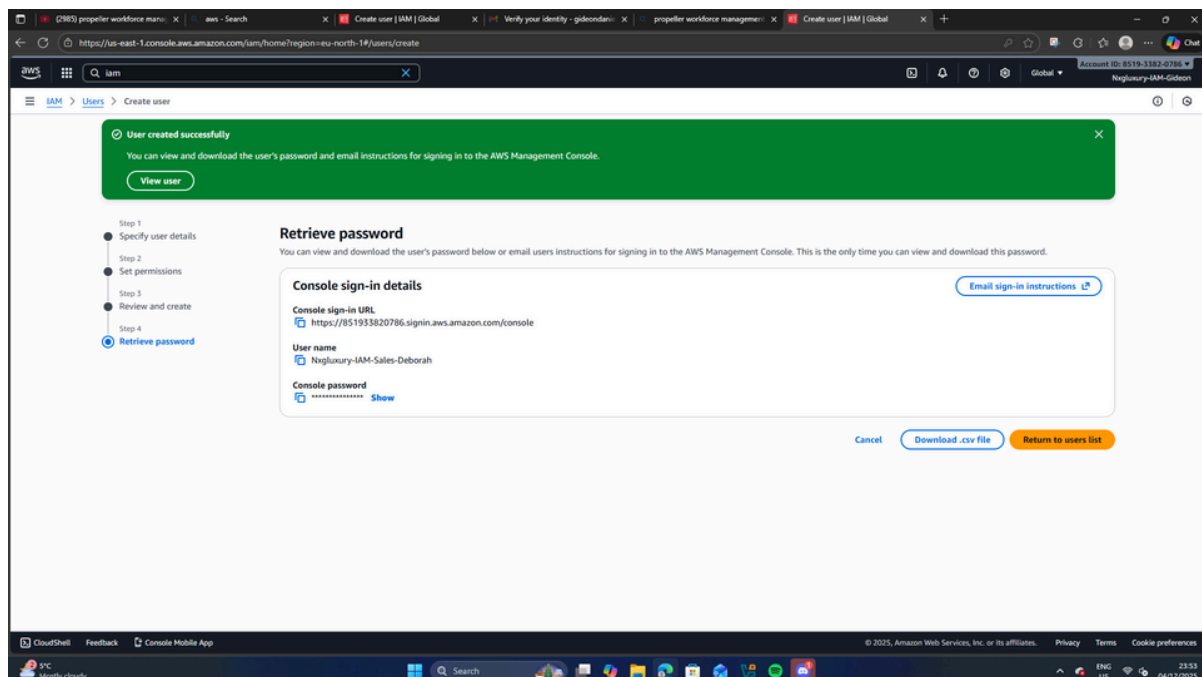
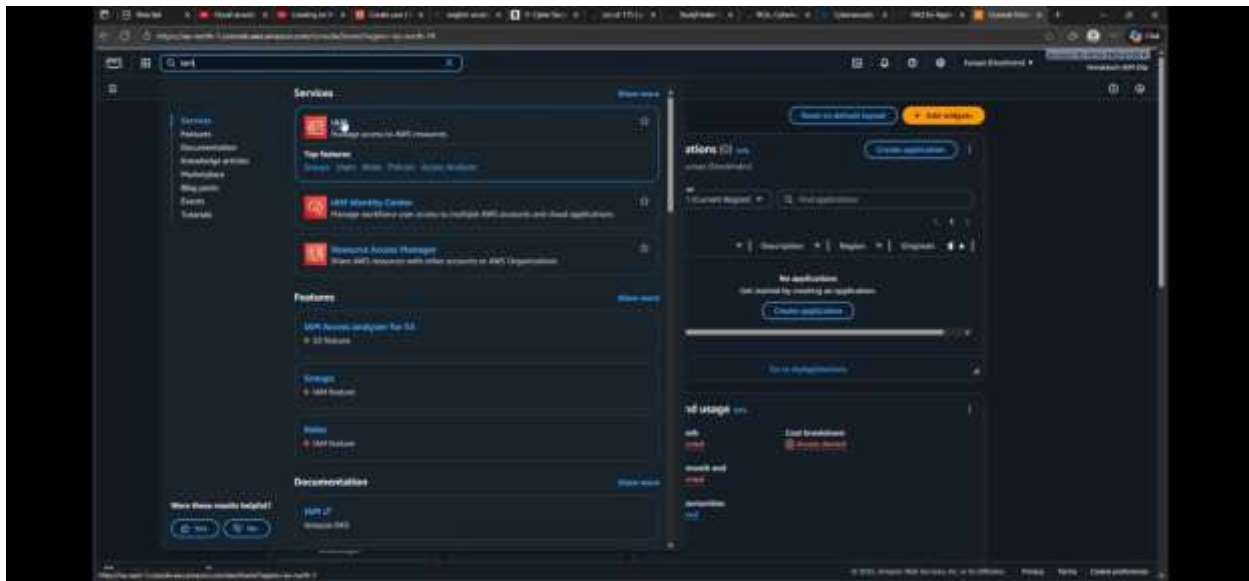


- Sign-in Verification: Copy the "**console sign-in URL**", paste it into a new browser, and sign in using the username and custom password. The dashboard will confirm the sign-in with the identity **Nxgluxury-IAM-Gideon** in the top-right corner.

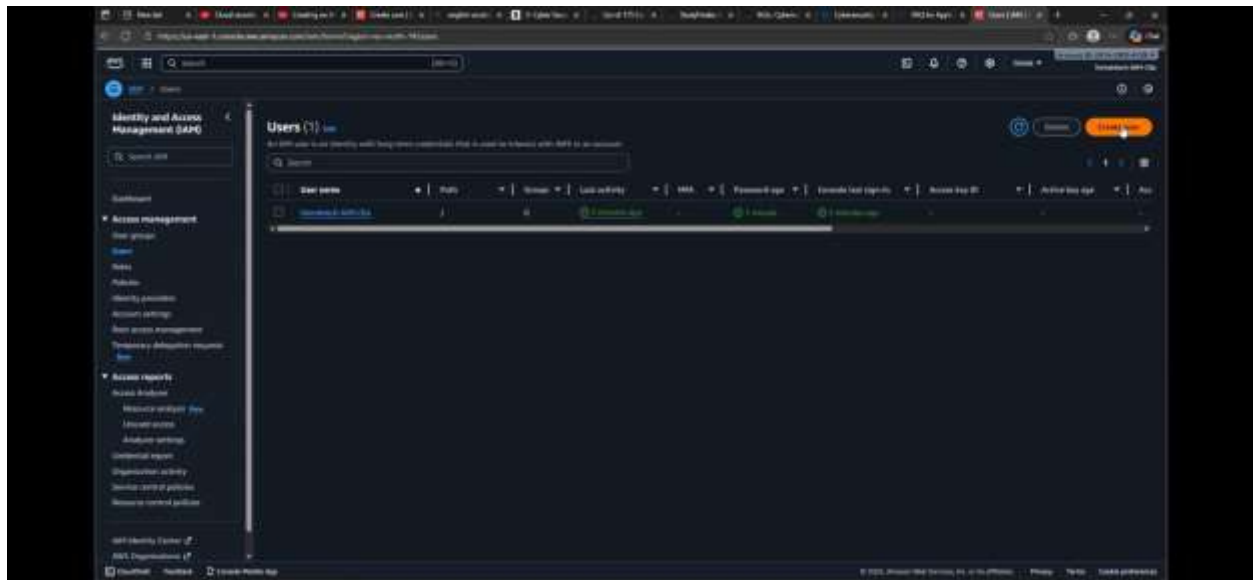


3. Create and Configure User 2 (Sales Group):

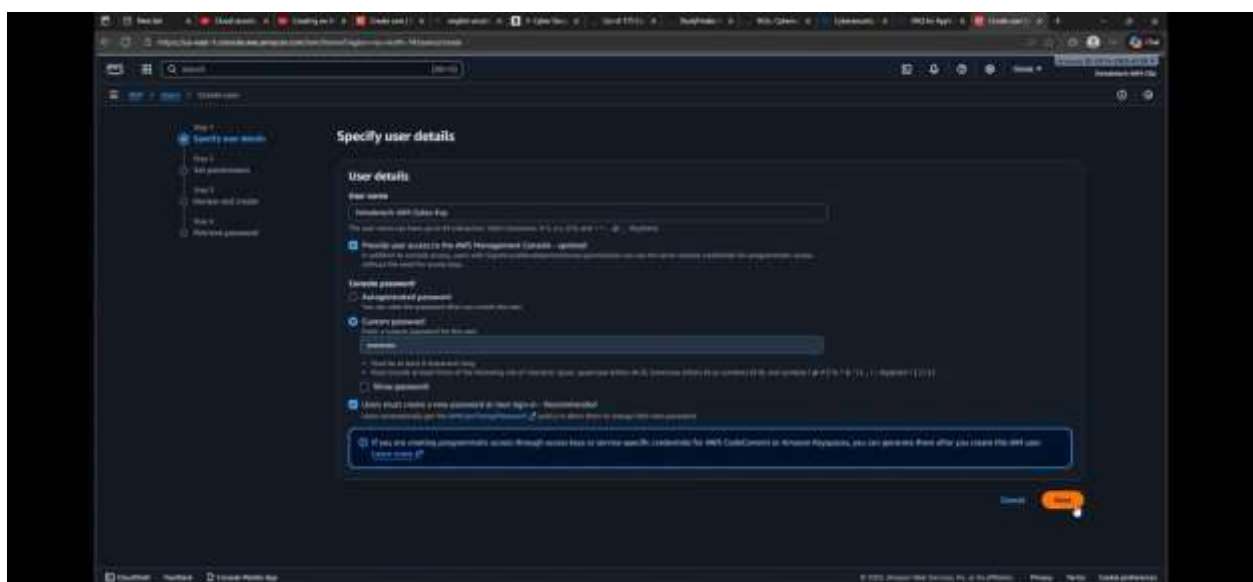
To create the first user. I navigate to the search button and search for IAM



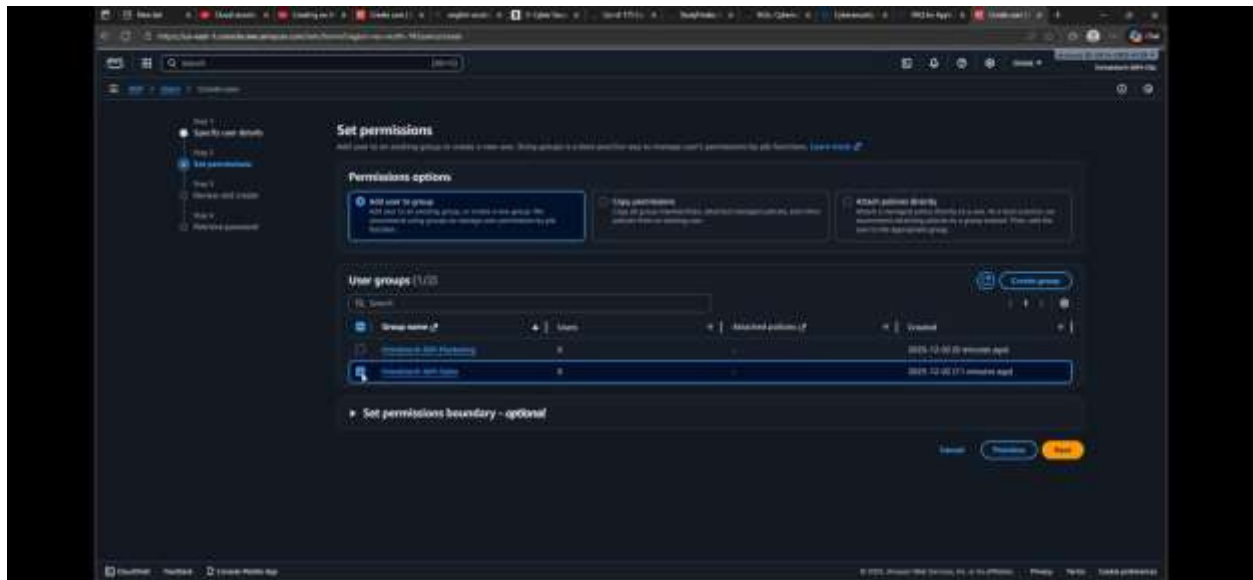
Then I clicked the user to open the user interface and when the user interface opened, I clicked Create User.



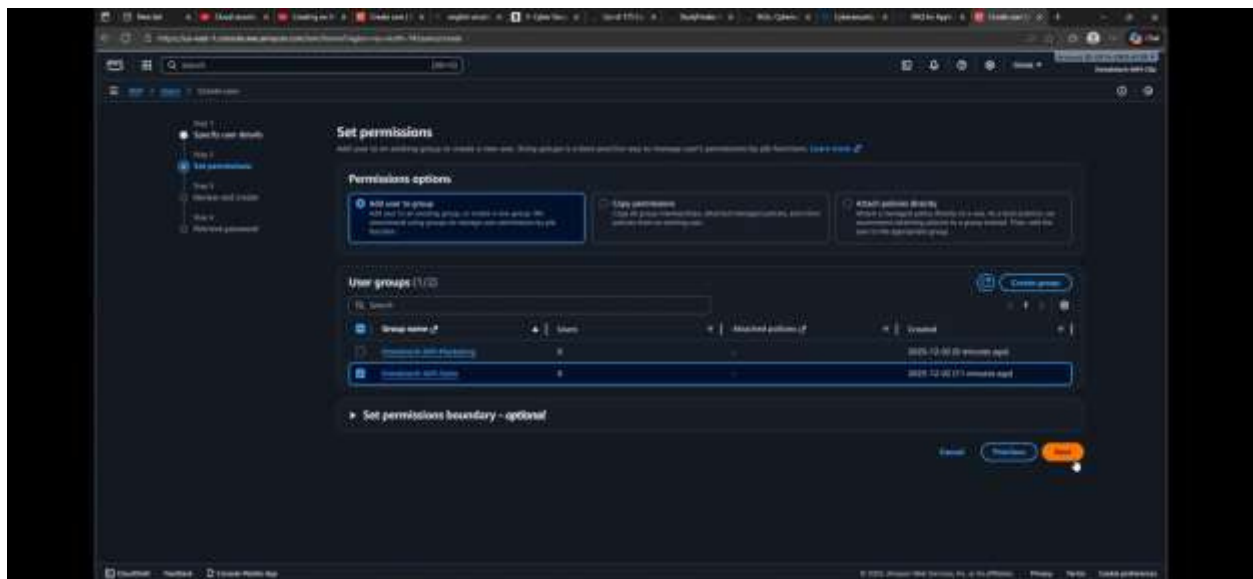
- Username: Nxgluxury-IAM-Sales-Deborah •Select the option to provide user access to the AWS Management Console.
- Choose Custom Password and input the required password.

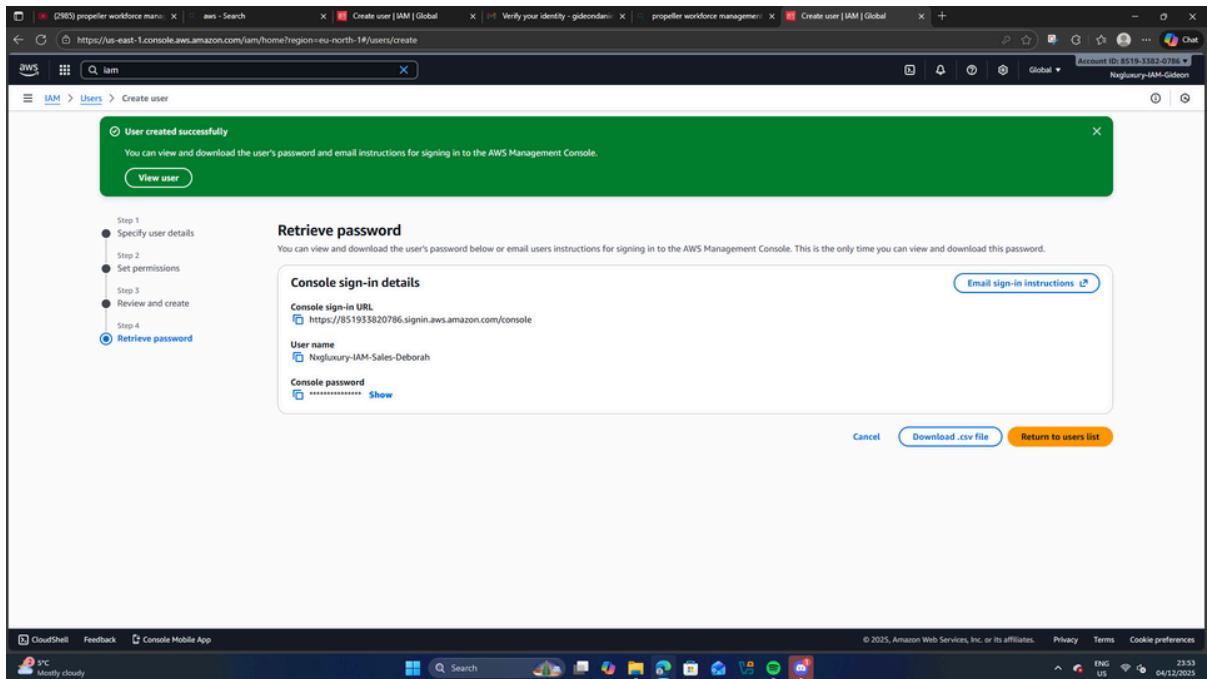
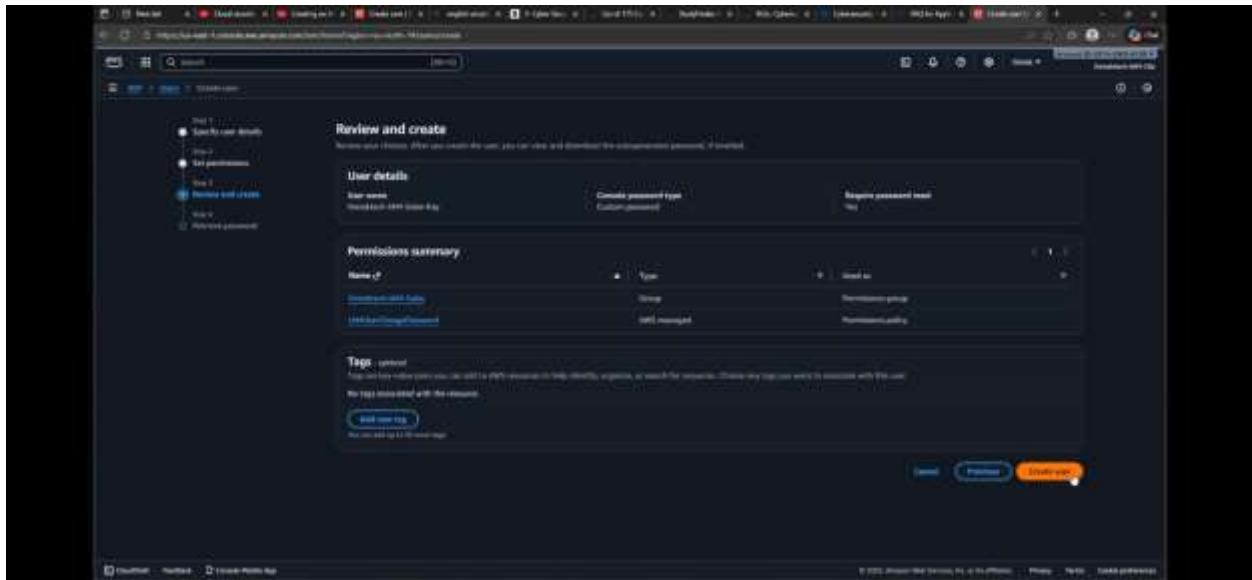


- Assign to Group: Select the Sales tab to assign the user to the Nxgluxury-iam-sales group.



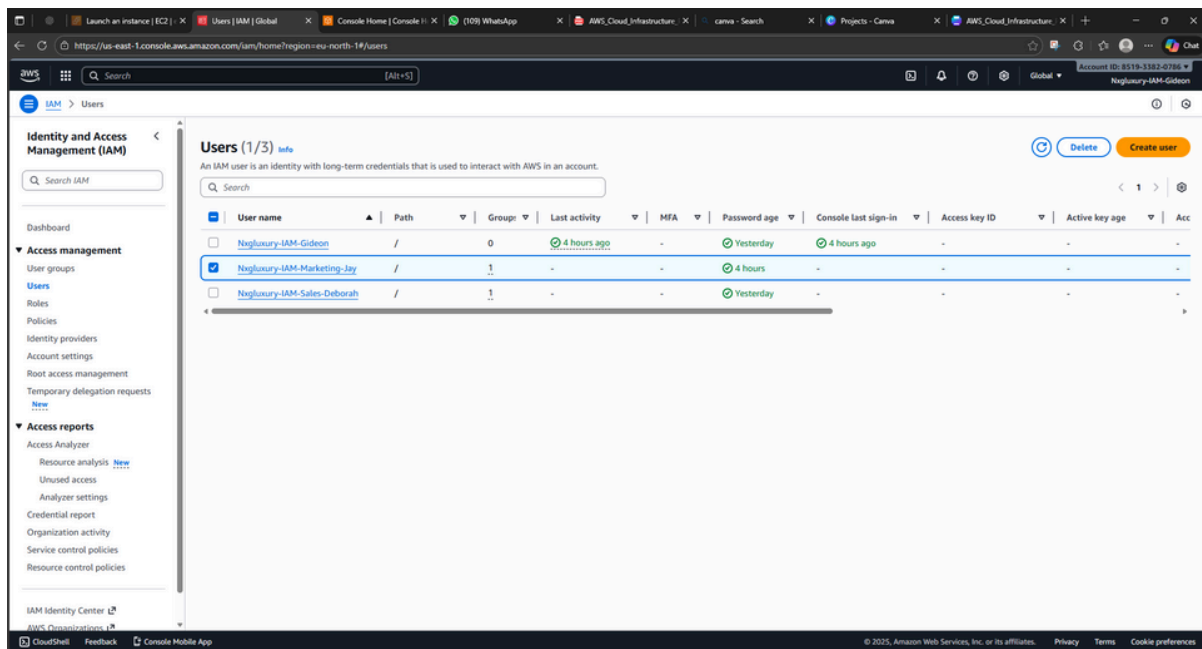
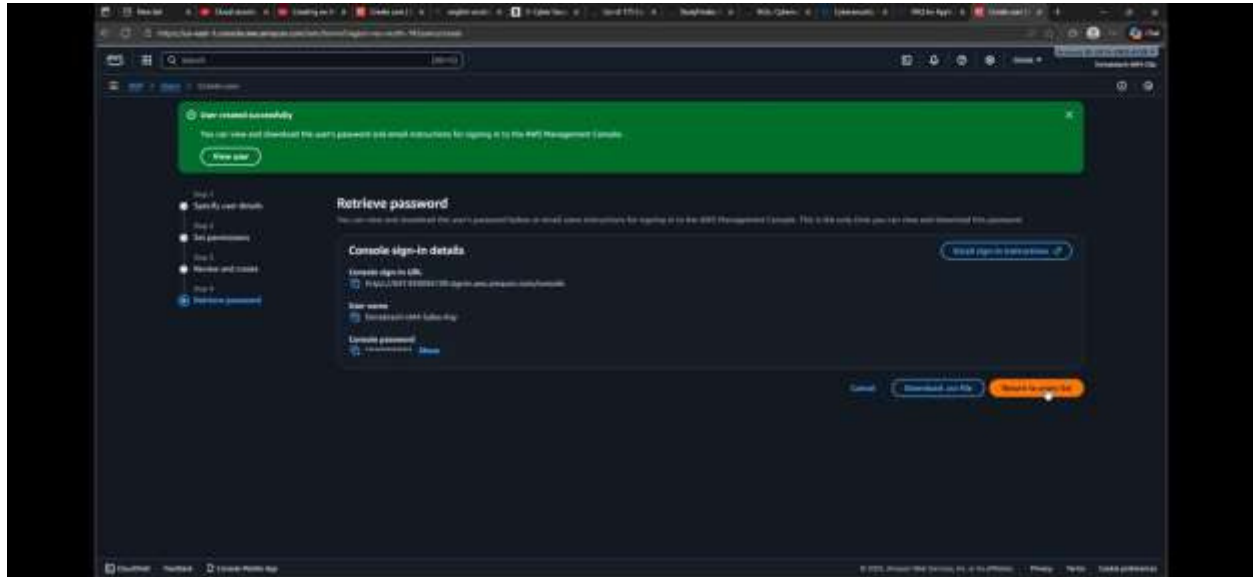
- Complete the creation steps



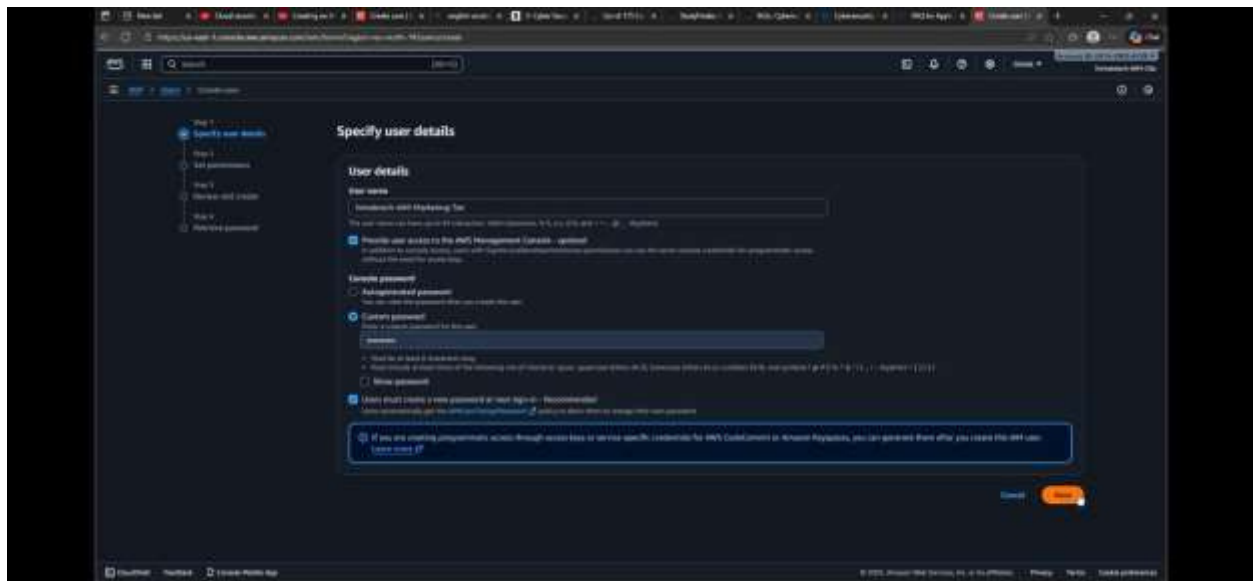


4. Create and Configure User 3 (Marketing Group):

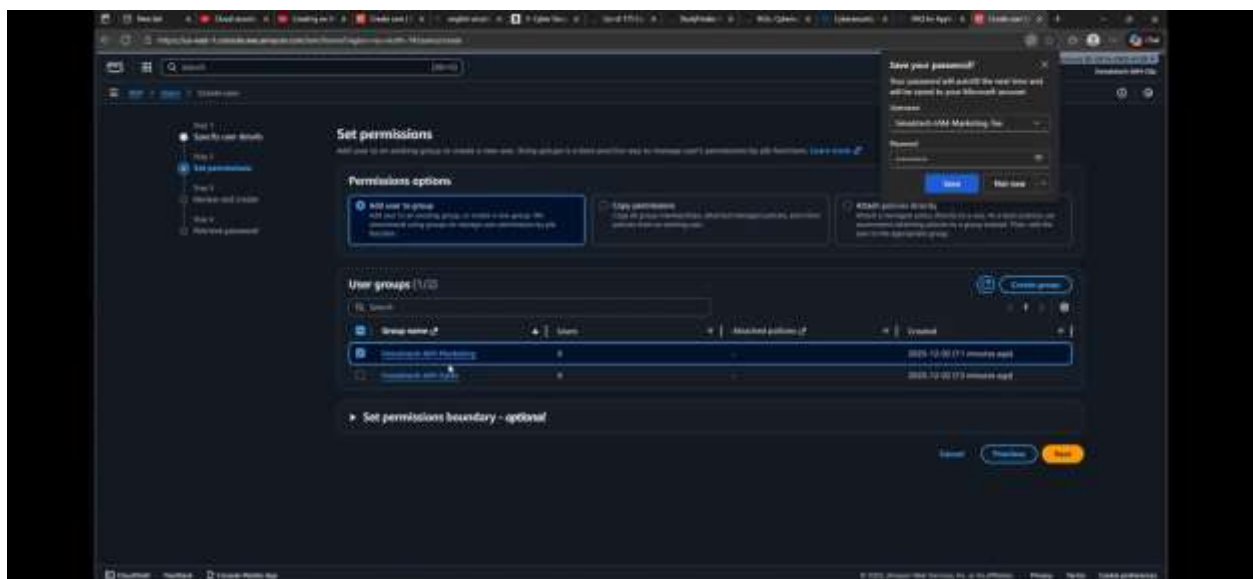
I return to the IAM user dashboard to create the 3 users



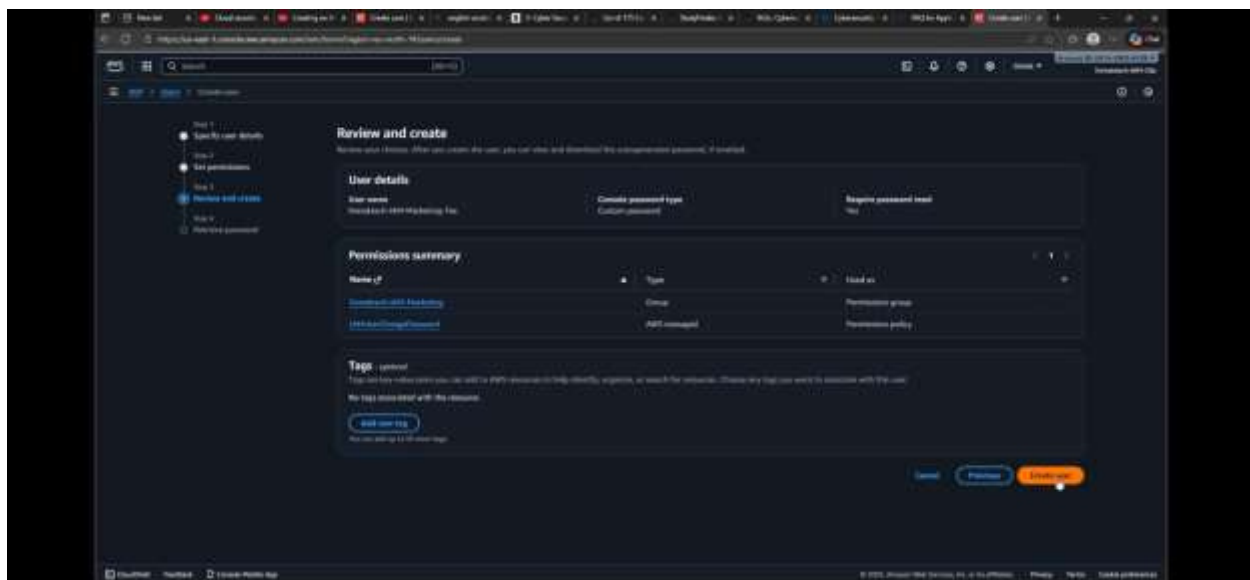
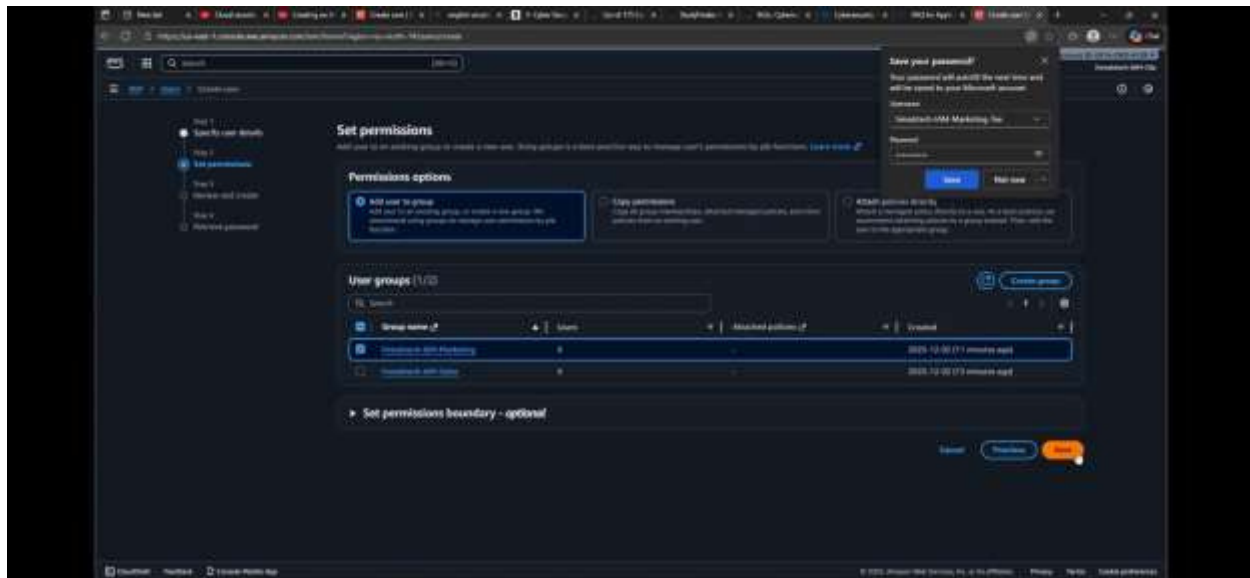
- Username: Nxg-IAM-Marketing-Jay • Select the option to provide user access to the AWS Management Console.
- Choose Custom Password and input the required password.

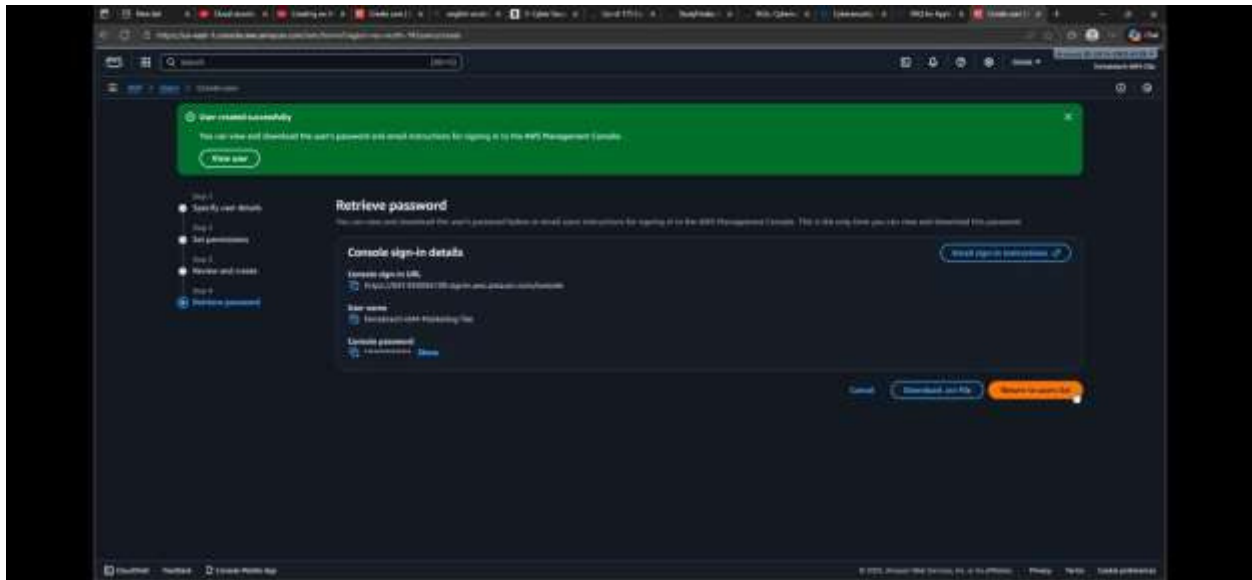


- Assign to Group: Select the Marketing tab to assign the user to the himoktech-iam-marketing group.

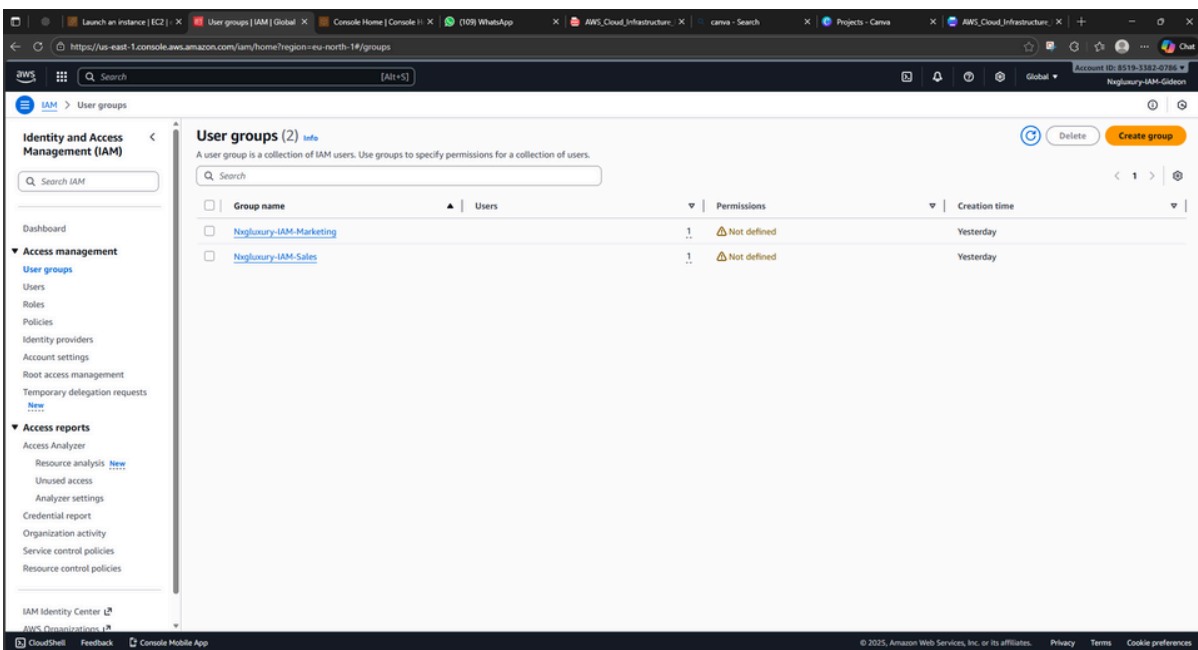


- Complete the creation steps.





After returning to the user interface, it was confirmed that each group user has been assigned to their respective groups



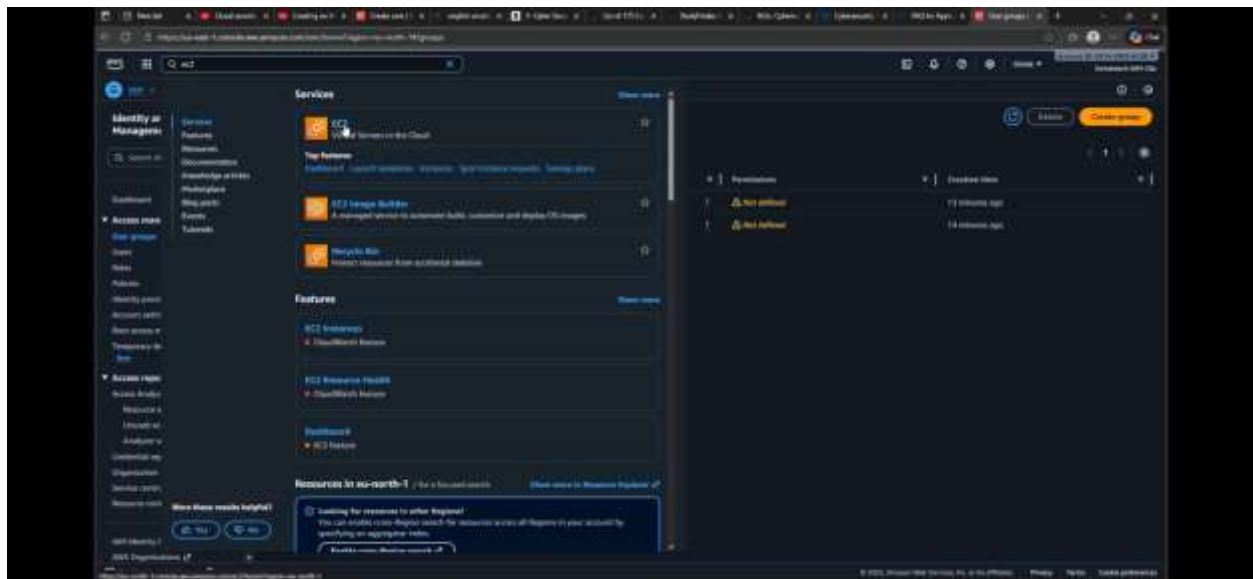
Phase B: Launching 2 EC2 Instances (Servers)

This phase involves creating the virtual servers (Instances) using the EC2 service. This is where teams can deploy projects and code.

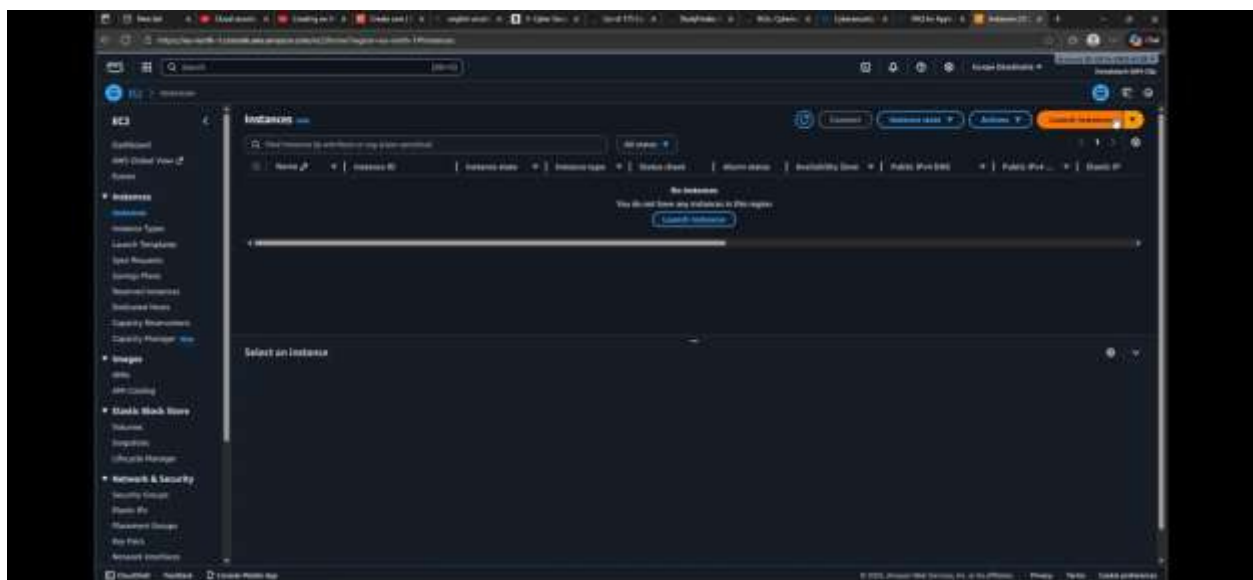
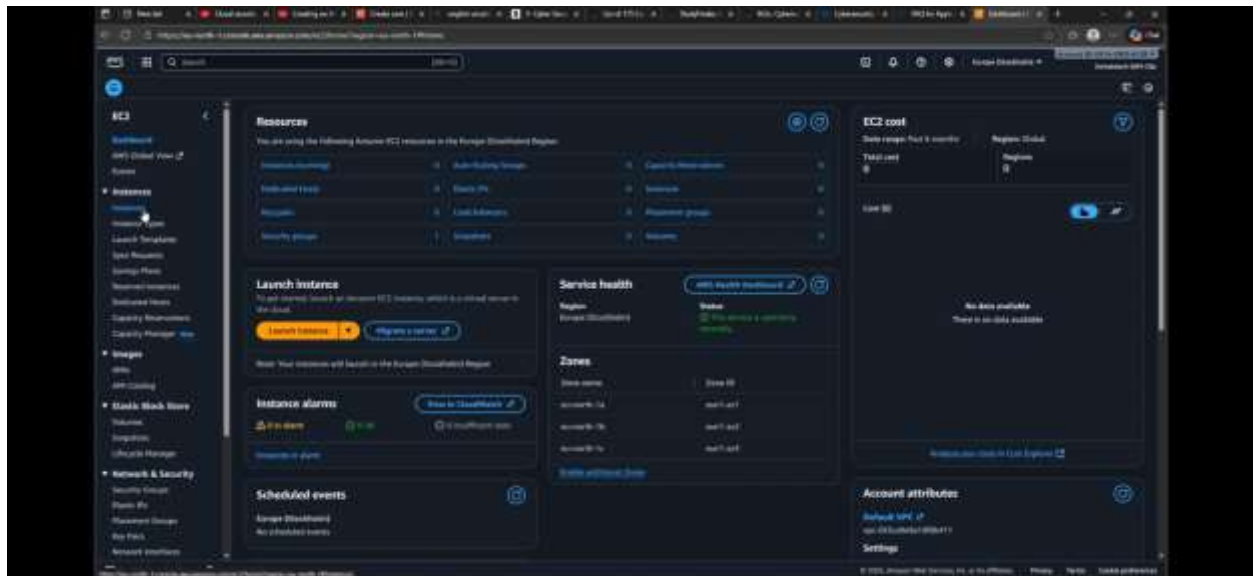
- EC2 Service (Elastic Compute Cloud): Provides resizable computing capacity (Virtual Servers) in the cloud.
- Instance: A virtual server running in the cloud.
- AMI (Amazon Machine Image): A template containing the required operating system and software for the server.
- Key Pair: A set of security credentials used to ensure data shared with the server is encrypted in transit.

Step 1: Create Sales Server Instance

1. Search for "**EC2**" and select the option for "**virtual servers in the cloud.**"

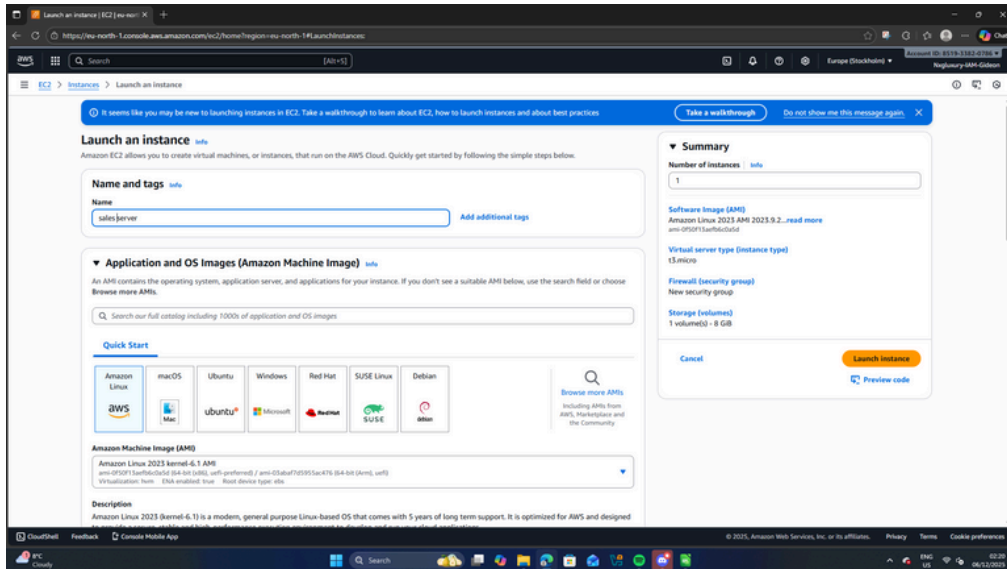


2. On the left-hand side, click "**instances**" and then "**launched instances**".

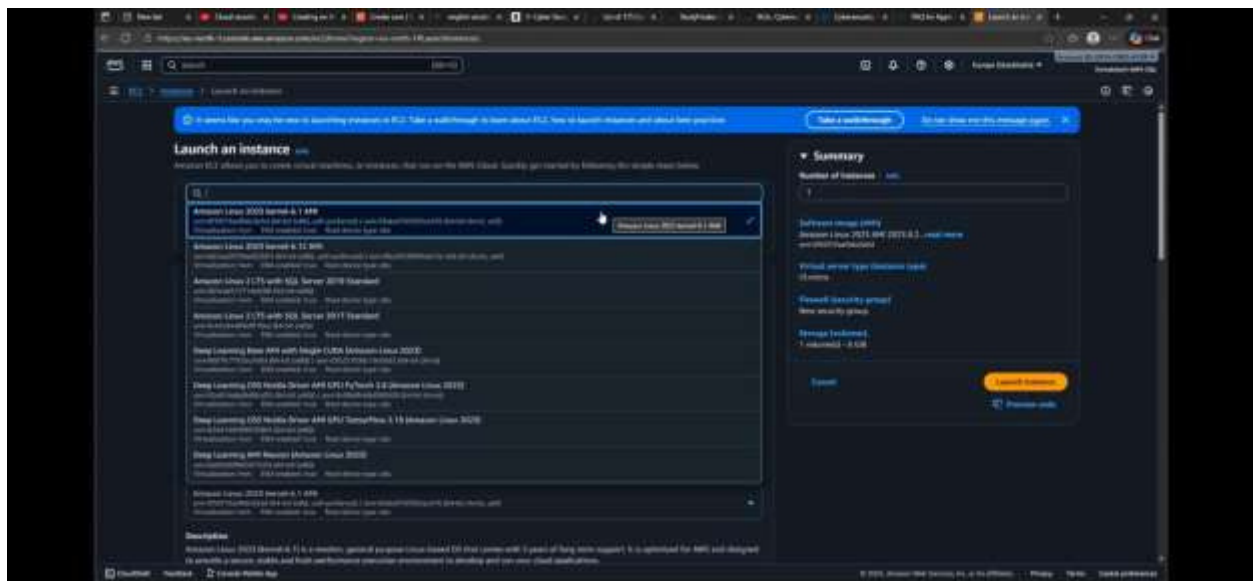


3. Configuration Details:

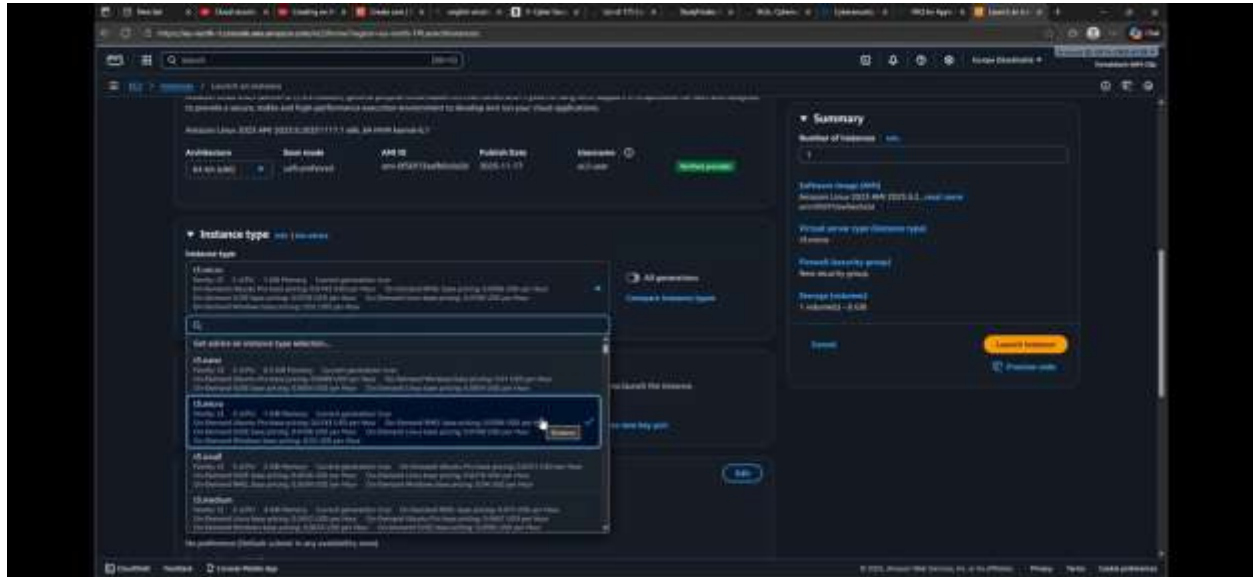
(a) Name: sales server



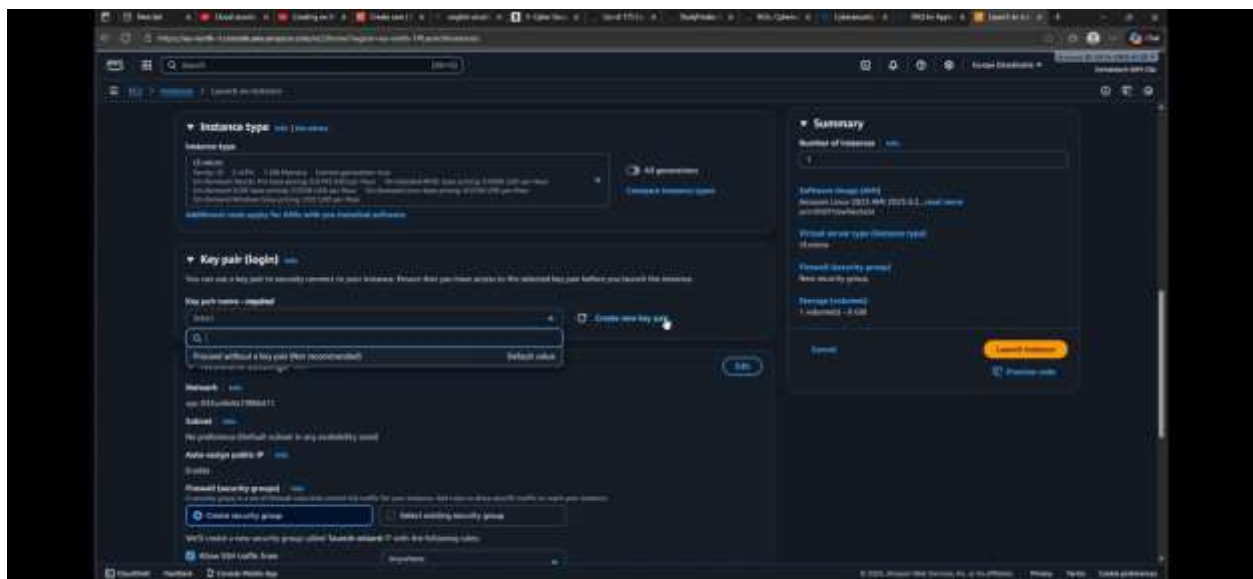
(b) Amazon Machine Image (AMI): Select a suitable Operating System, such as Amazon Linux 2023 kernel-6.1 AMI.



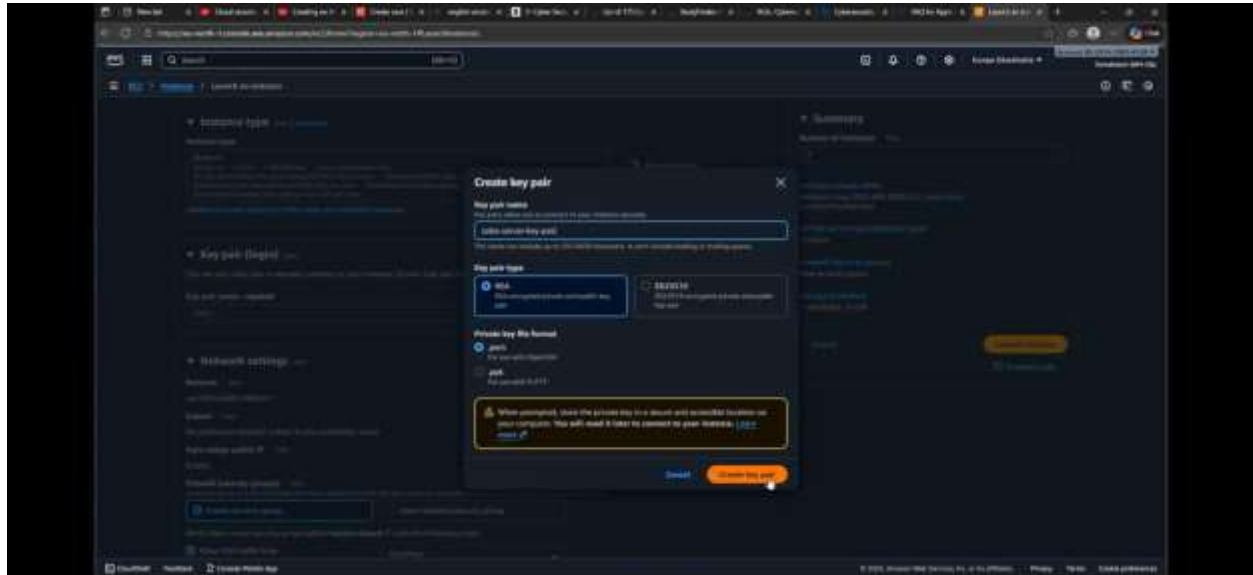
(c) Instance Type: Choose the required computing capacity; t3.Micro is selected for practice.



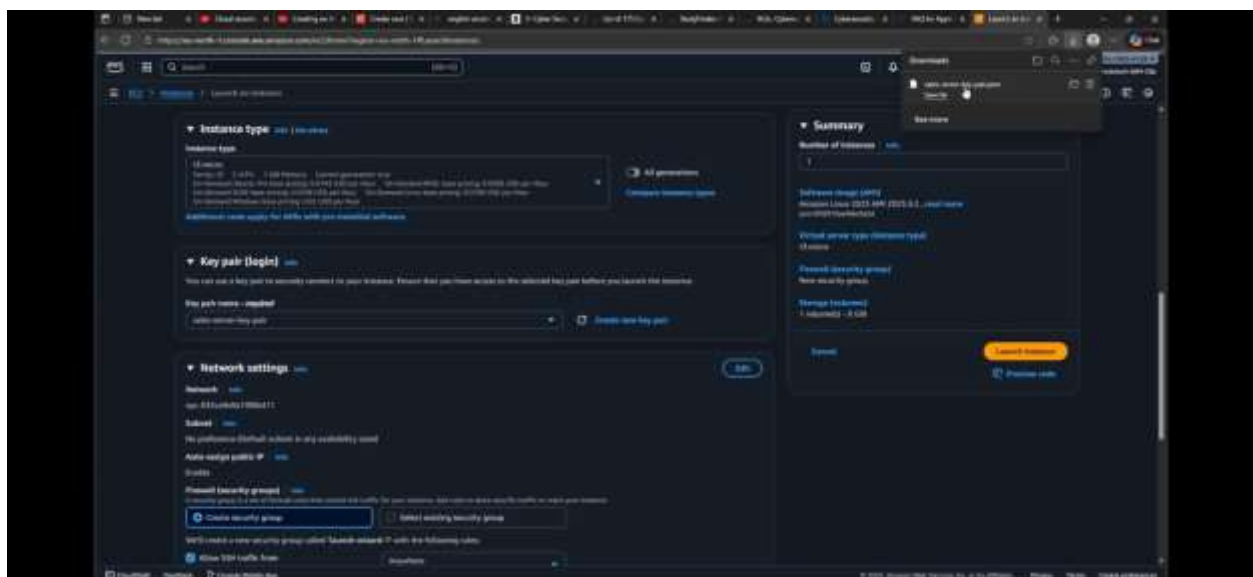
(d) Key Pair: Click create key pair for secure access:



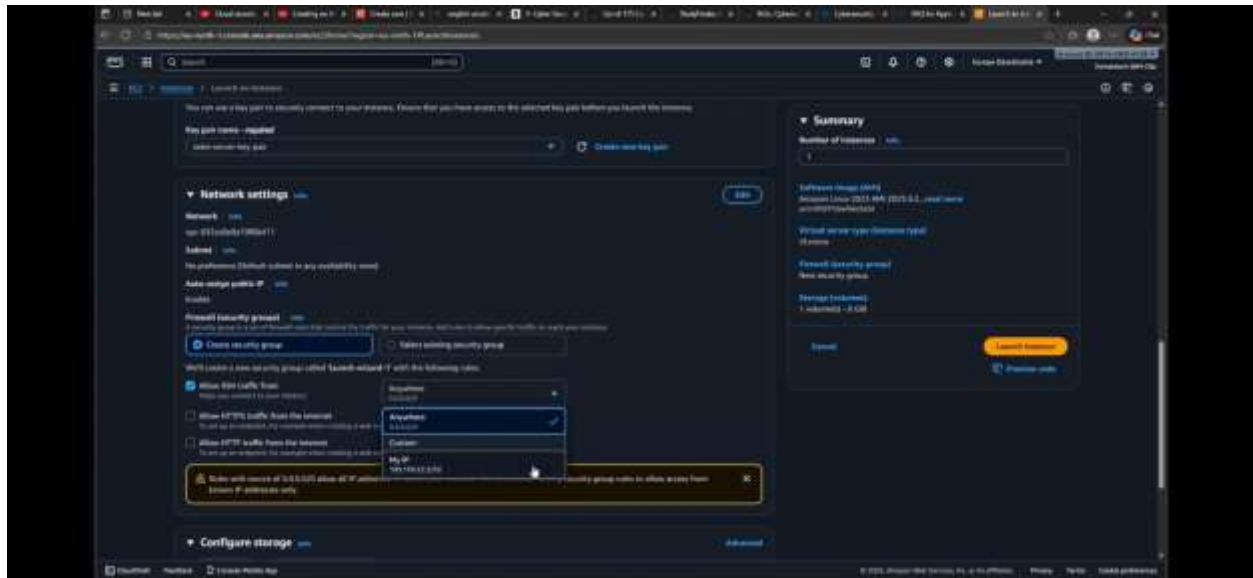
- Key Pair Name: sales server-server-key-pair
- Key Pair Type: Select RSA.



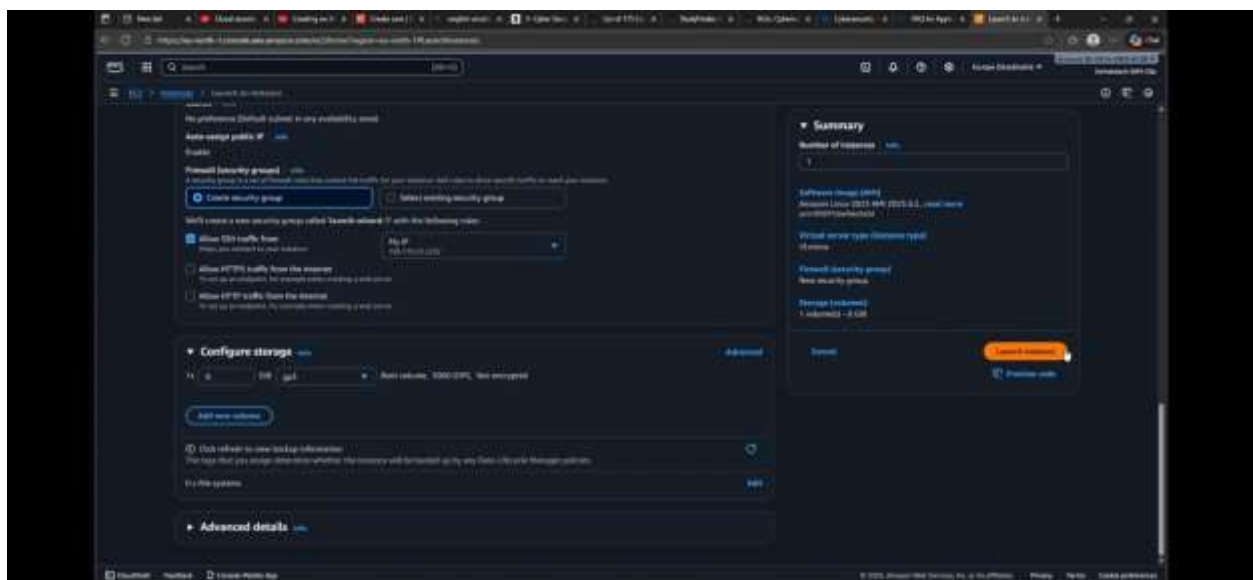
- The key pair is downloaded. (The key sharing medium should be different from the data sharing medium, and keys must not be stored in the same location as the data).

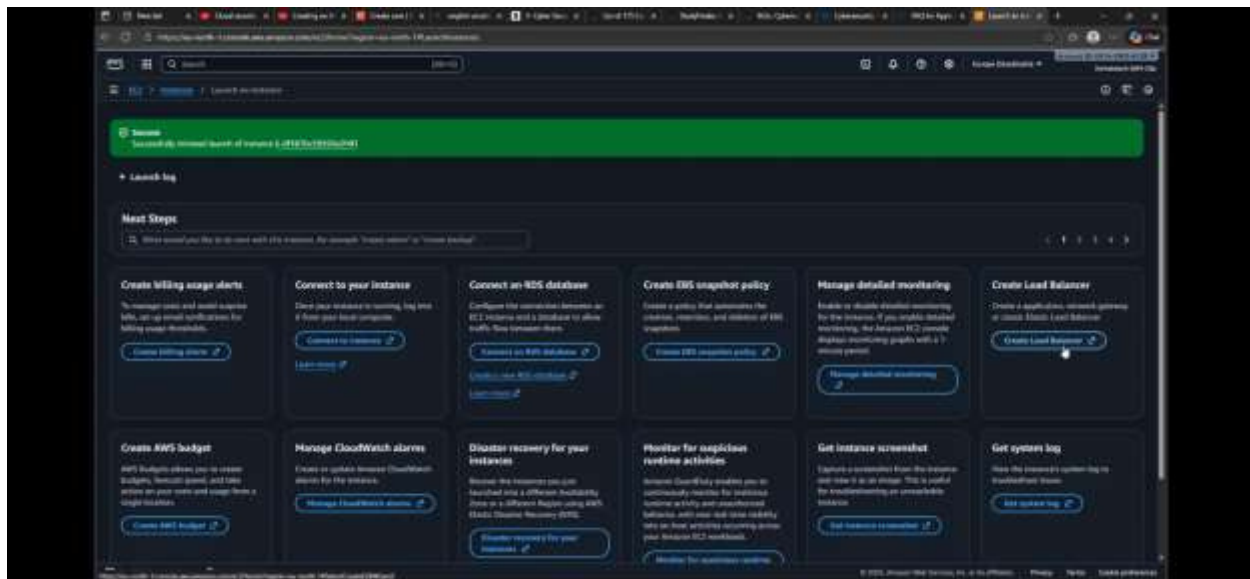


(e) Network Settings: Configure the SSH traffic source to "my ip" to enable secure console access via the Command Line Interface (CLI).



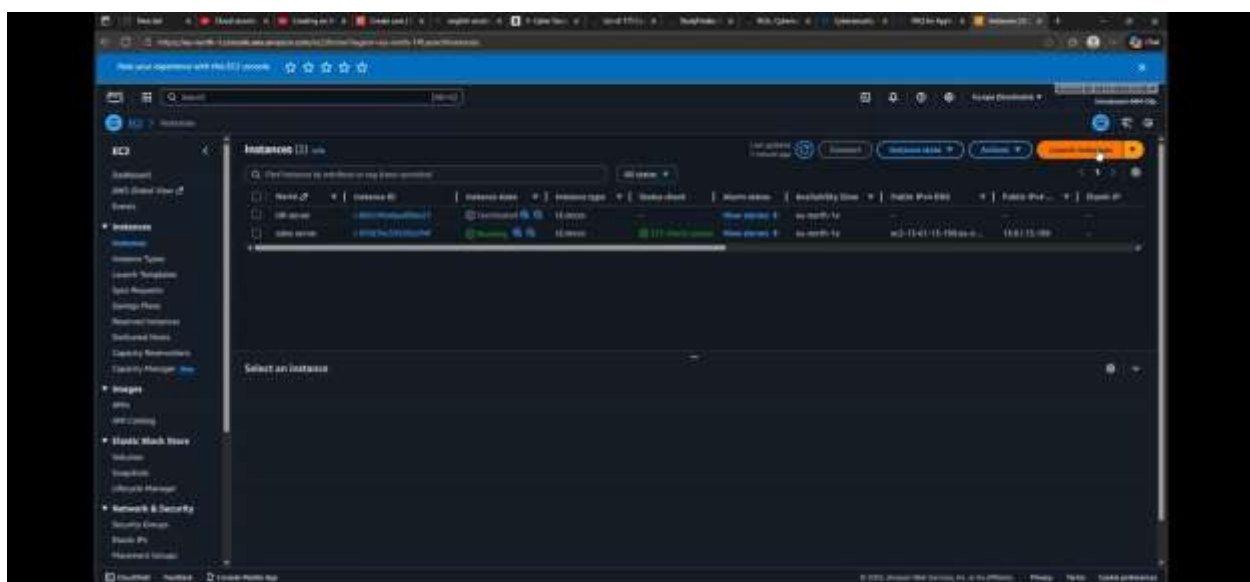
4. Click Launch Instance and verify the launch via the instance's log.

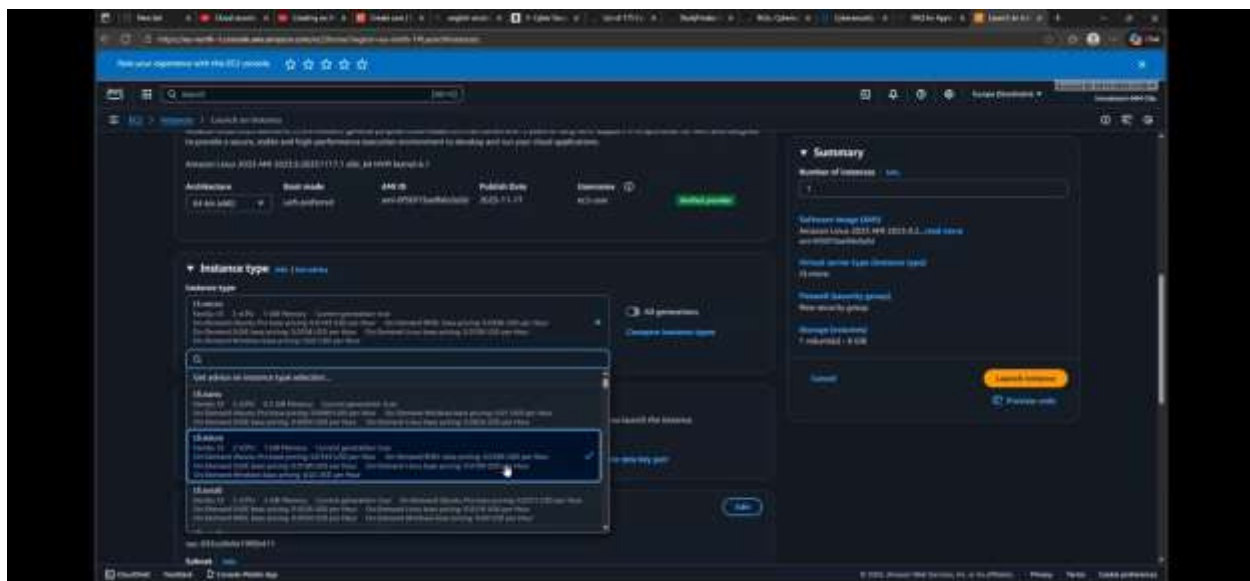
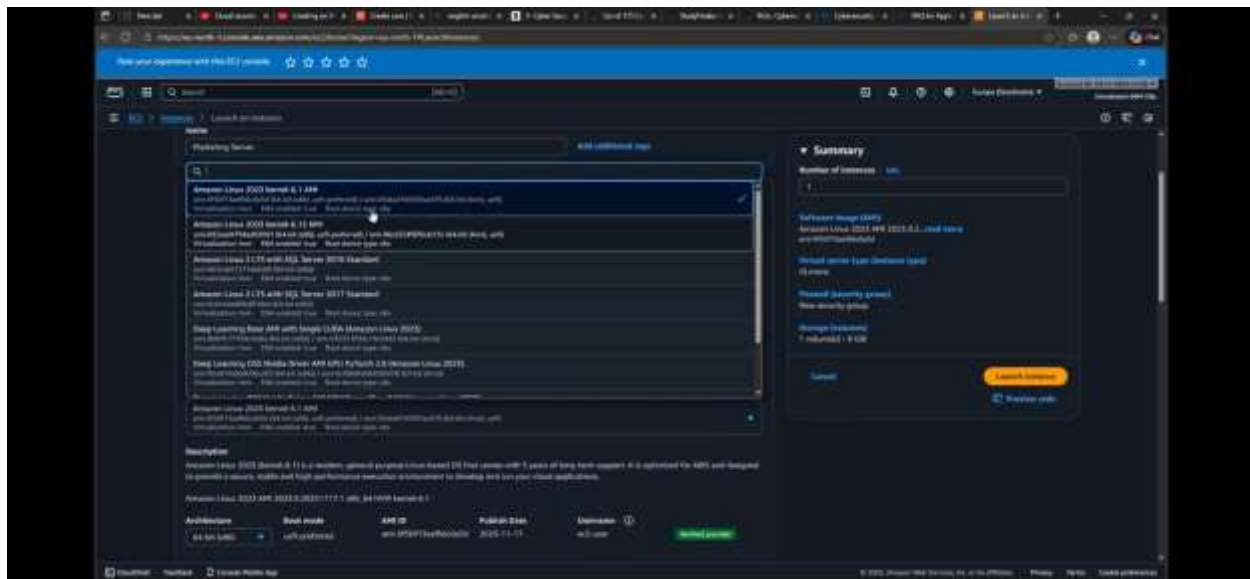


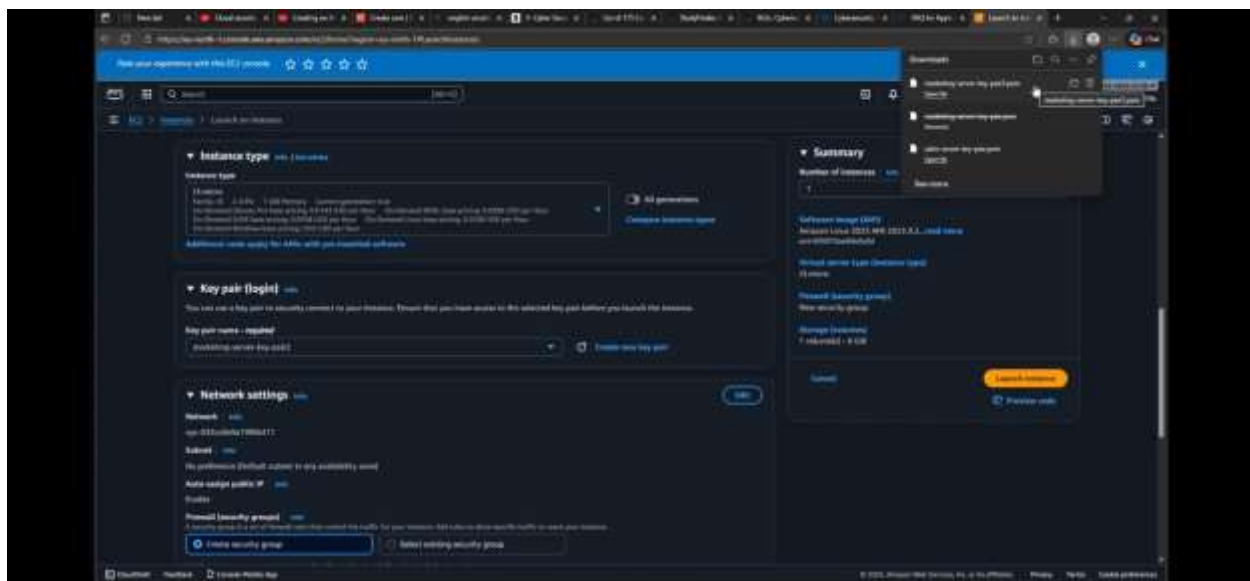
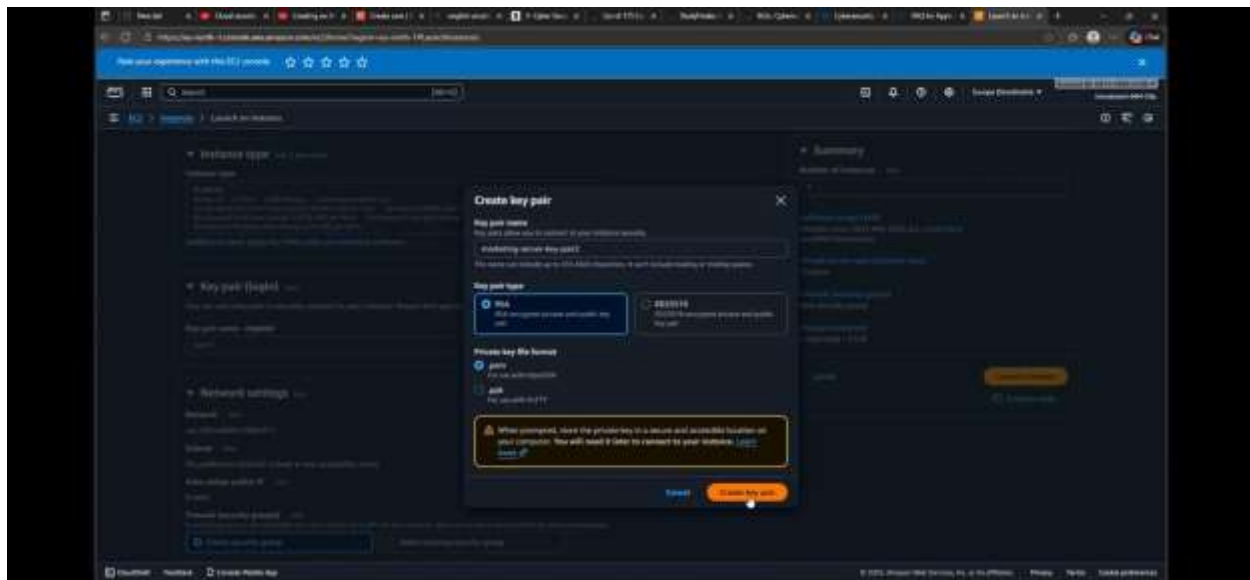


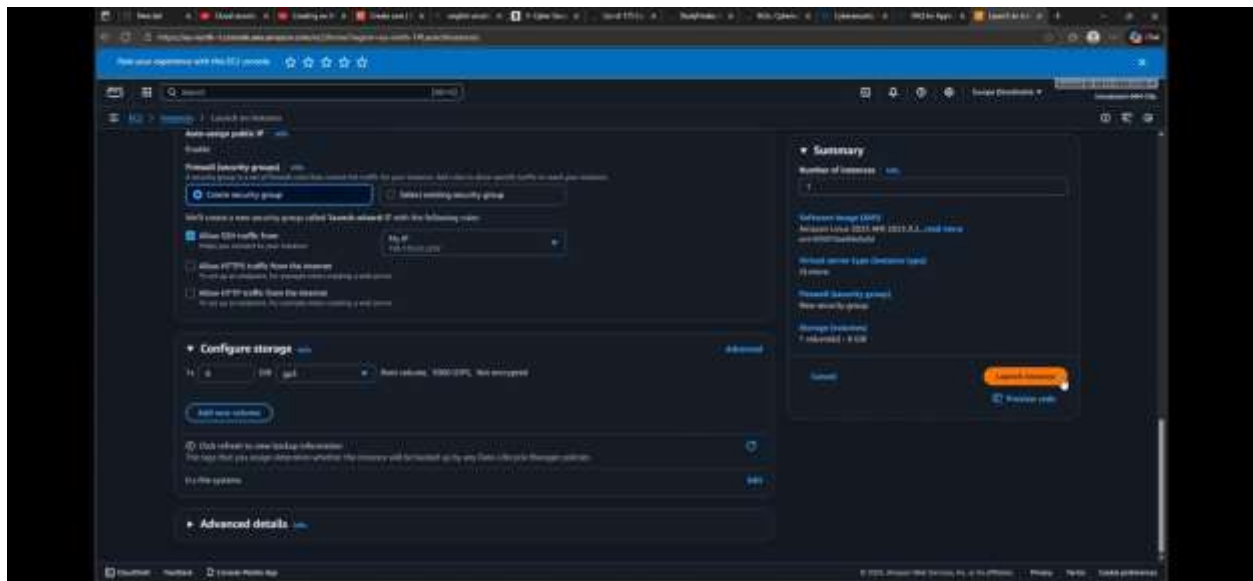
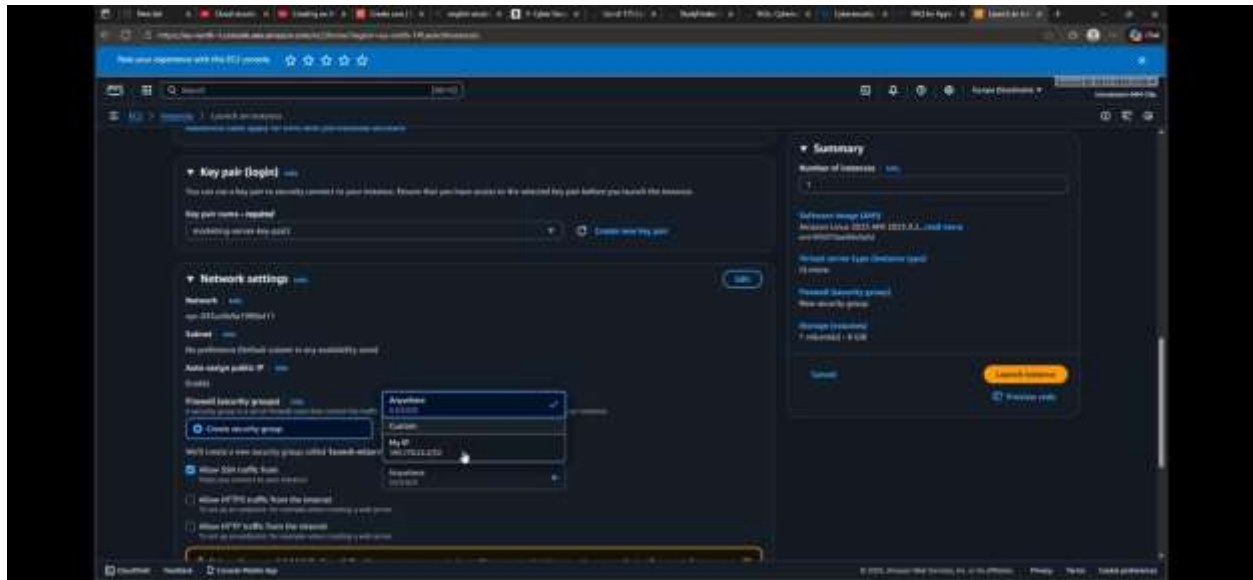
Step 2: Create Marketing Server Instance

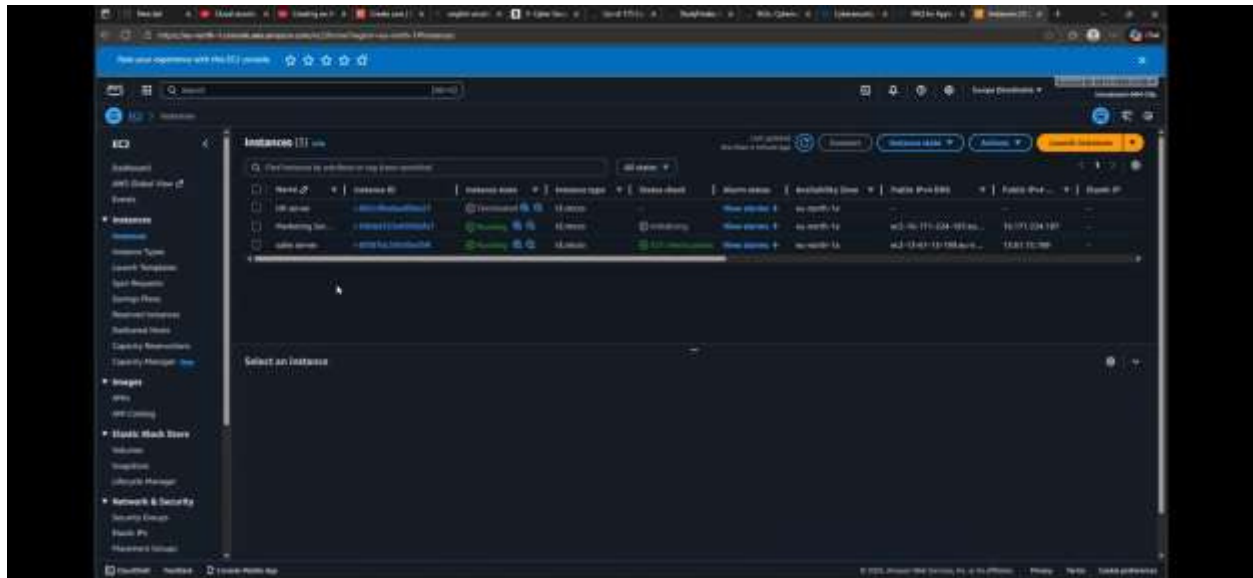
1. Repeat the entire process in Step 1 to create the marketing server instance.









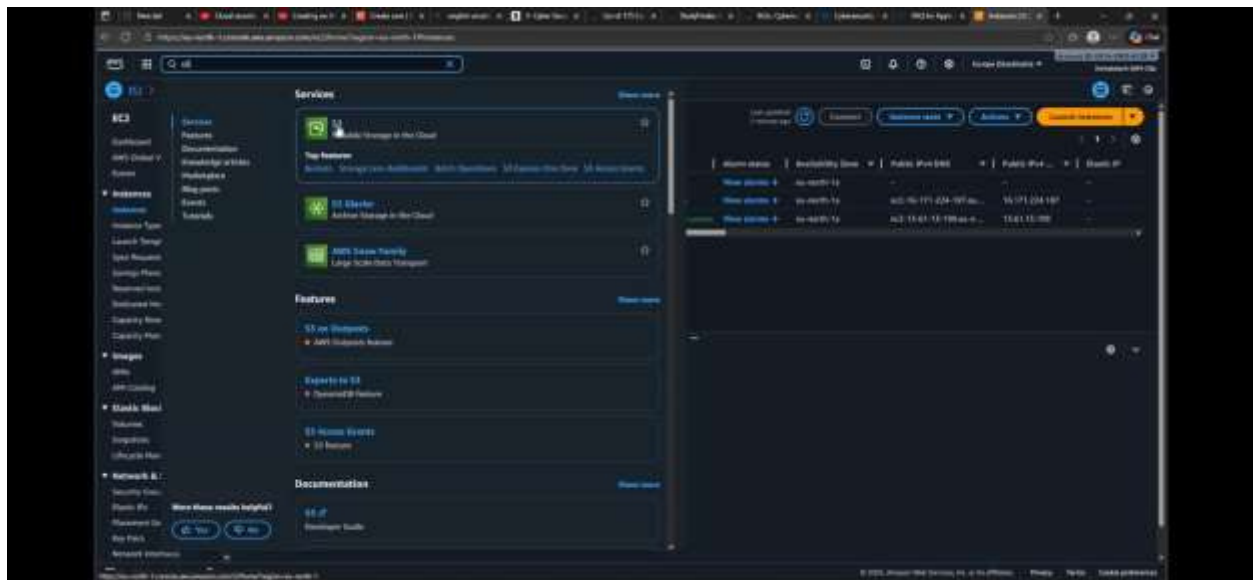


Phase C: Setting up a Storage Bucket (S3)

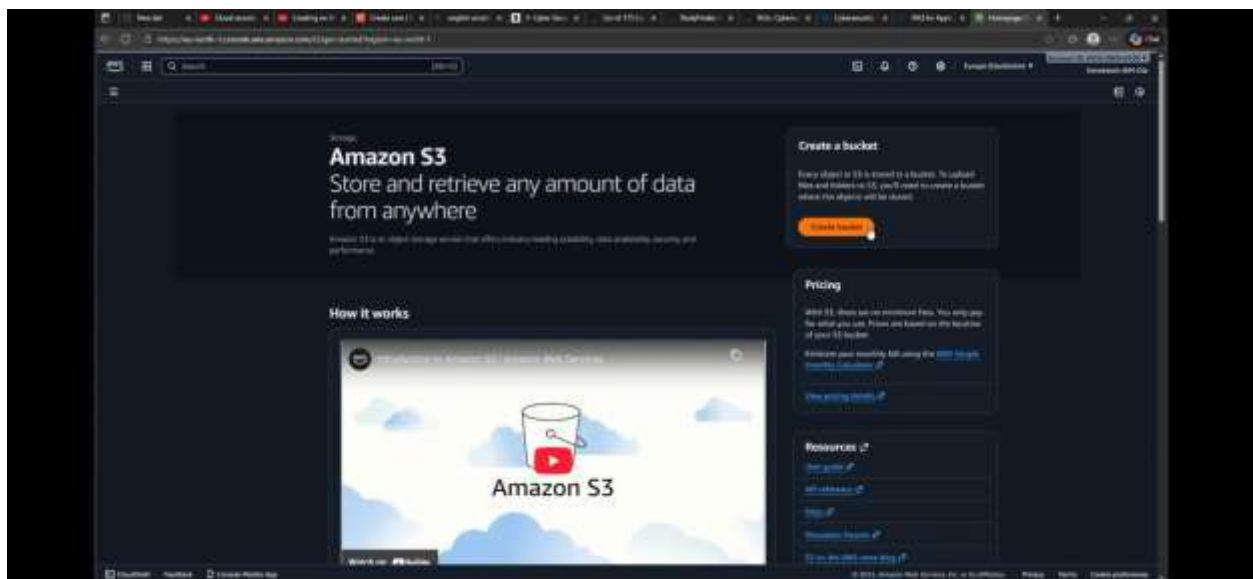
A Bucket provides scalable storage in the cloud, necessary for deployments, trials, and models.

- S3 (Simple Storage Service): An object storage service that provides storage capacity in the cloud.
- Bucket: A container for data stored in S3. Steps create “Sales Bucket”

1. Search for "**S3 storage in the cloud**" and choose the option for "**scalable**".

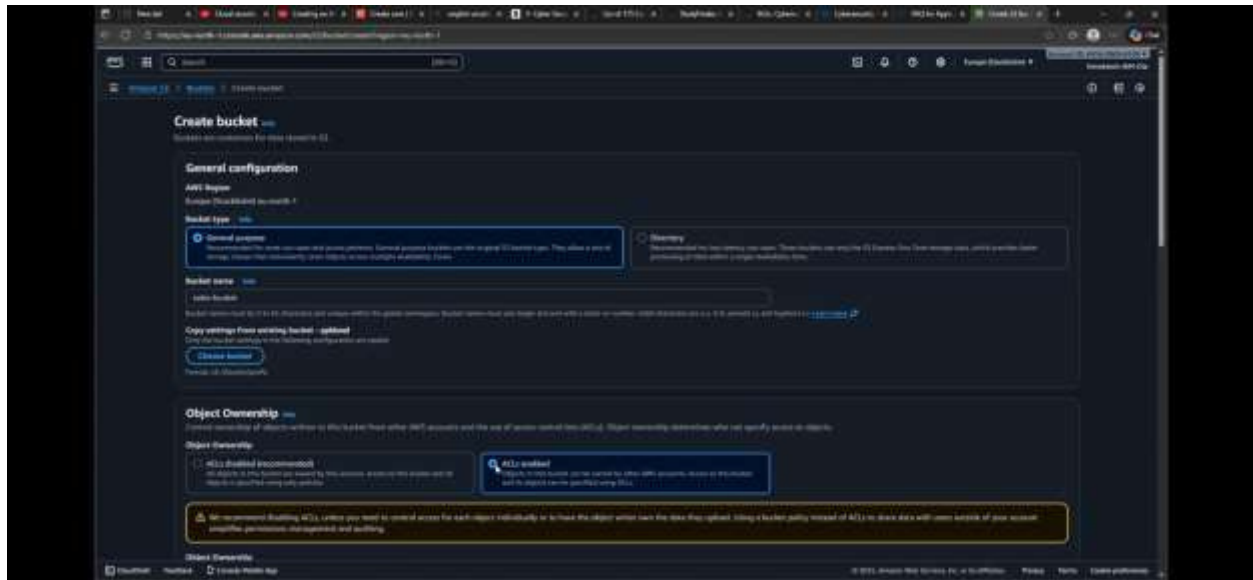


2. Click Create Bucket.

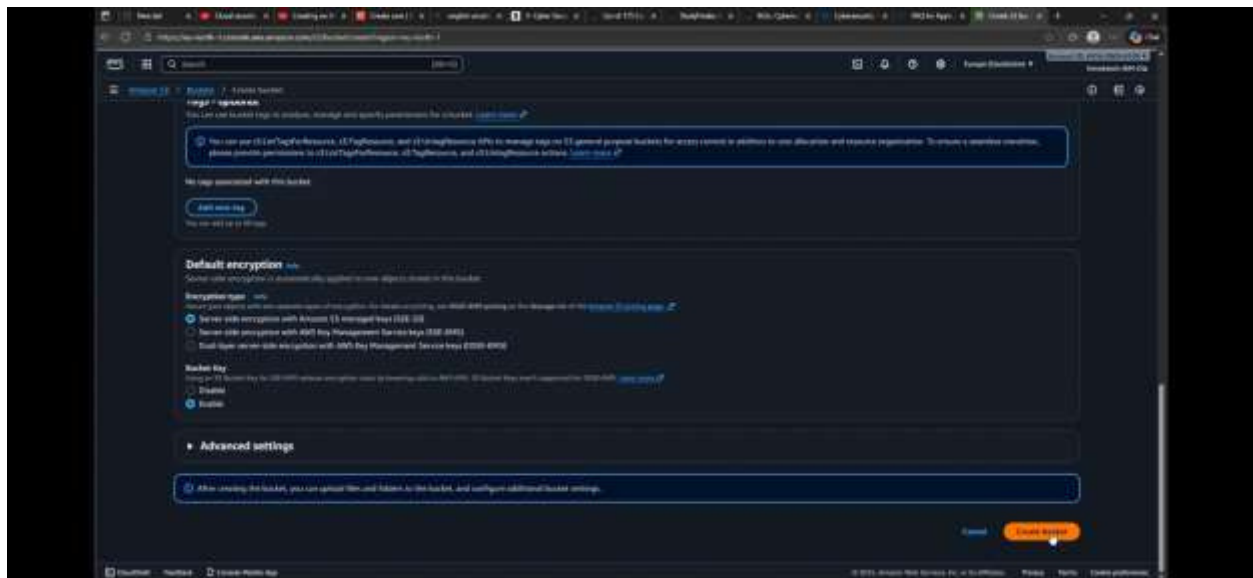


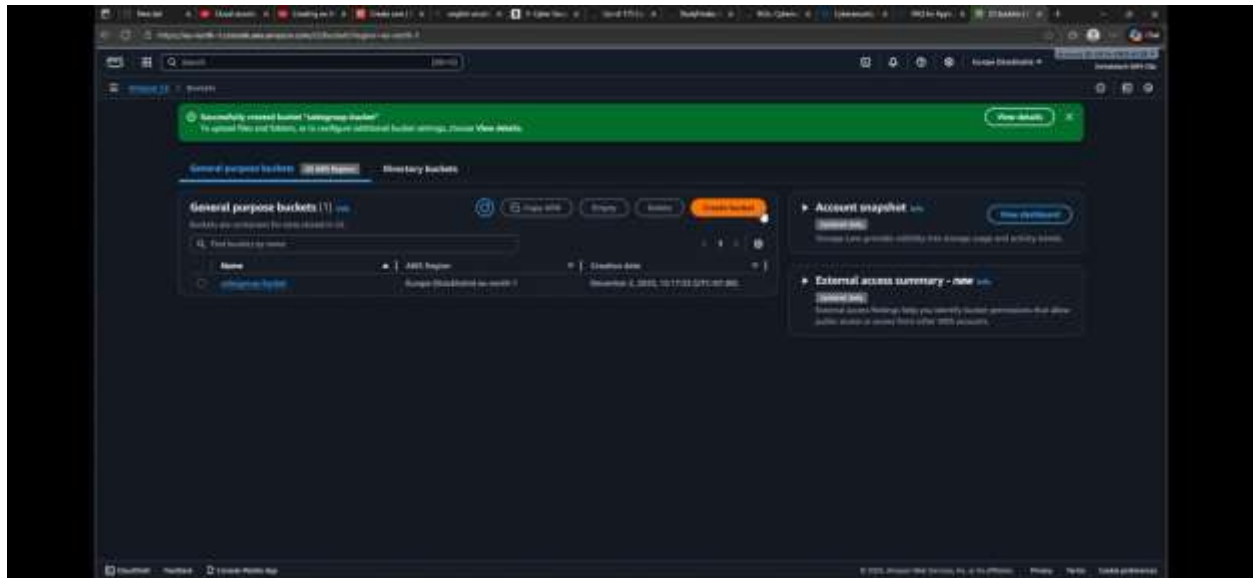
3. Configuration Details:

- (a) Bucket Name: sales bucket
- (b) Object Ownership: Select ACLs enable.



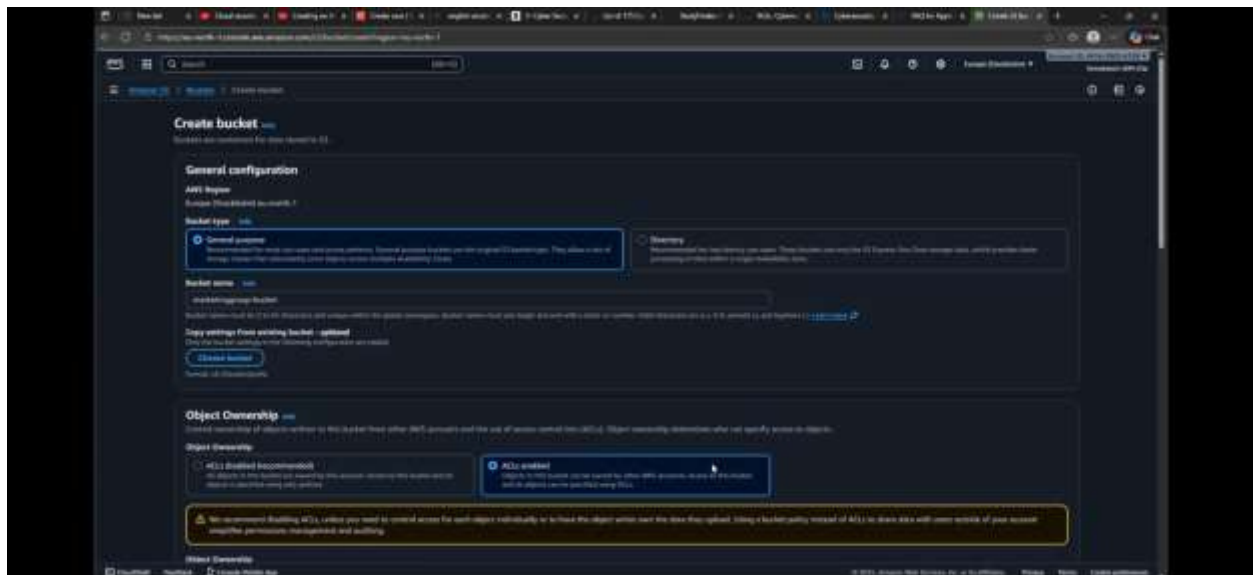
4. Click "create bucket".

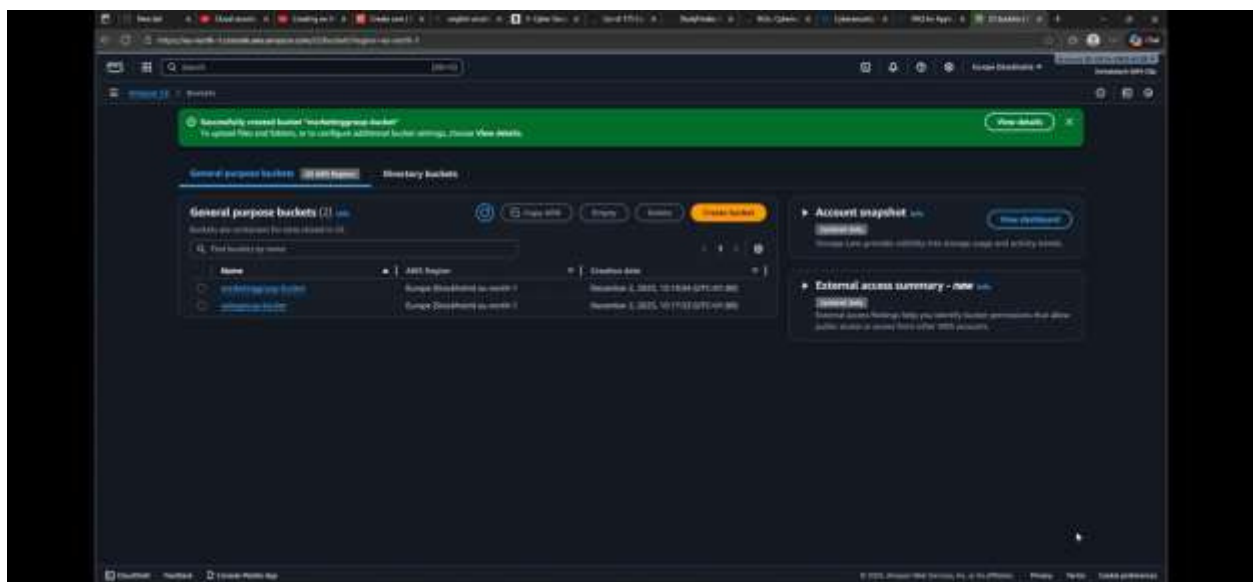
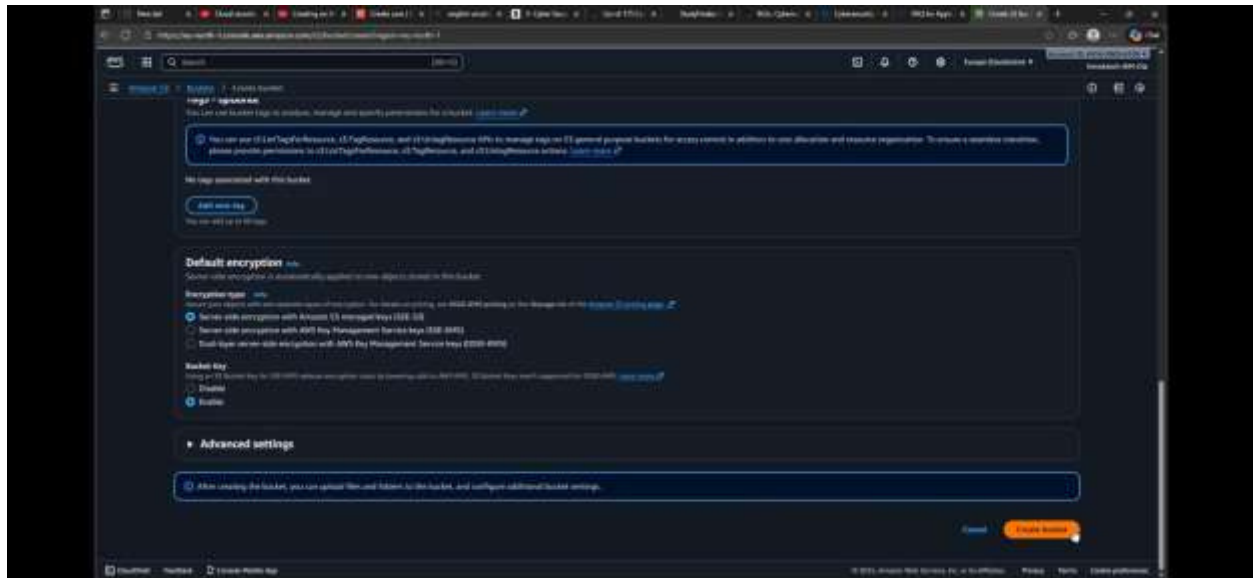




- Note: Documents that need to be passed to the bucket will be uploaded through the server created in Phase B.

I repeated the step to set up “Marketing group bucket”





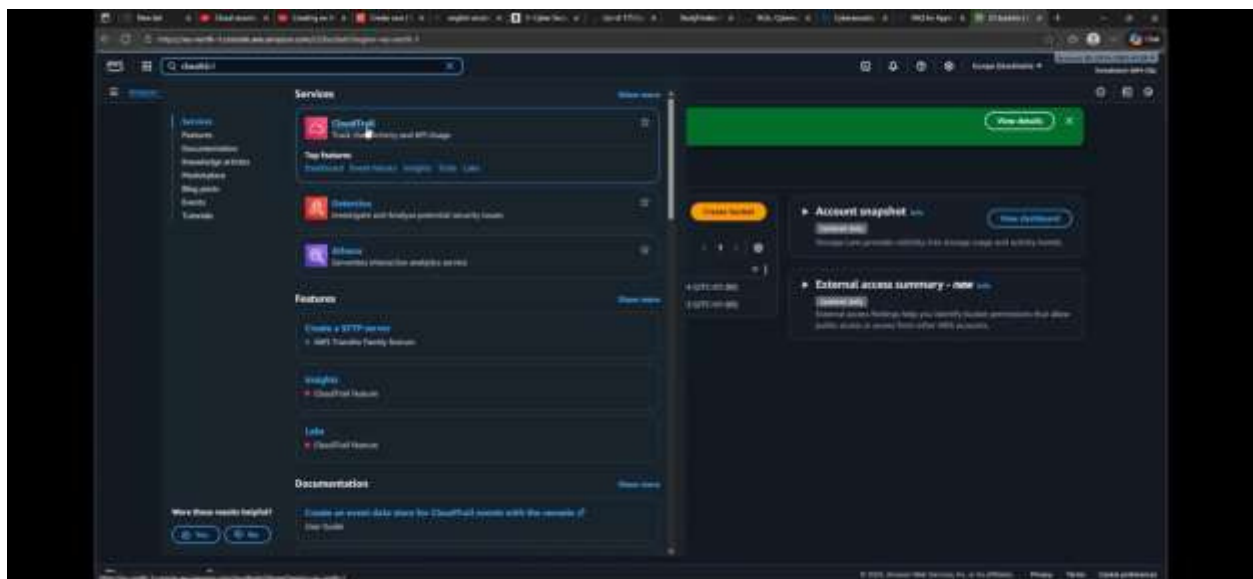
Sales group Bucket and Marketing group Bucket have been set up.

Phase D: Setting up the CloudTrail (Trail)

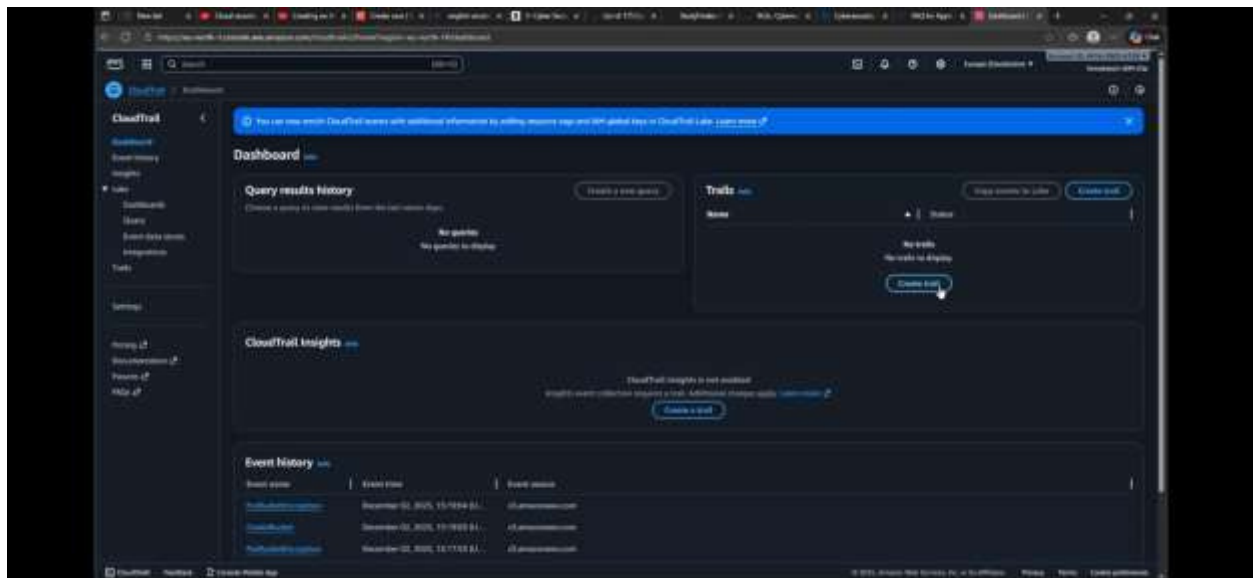
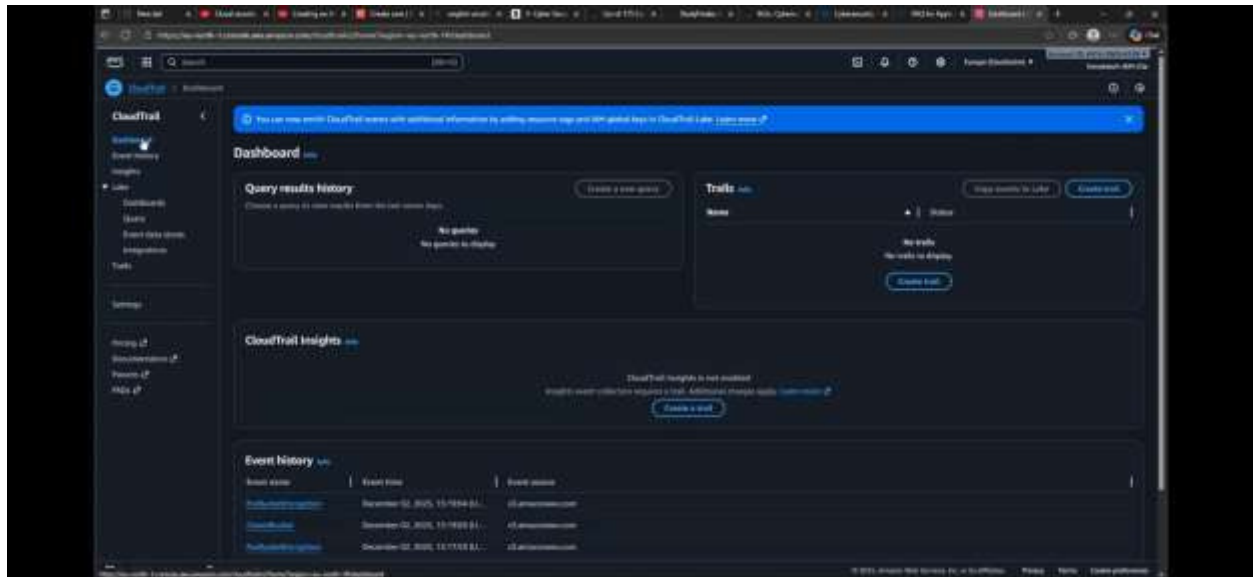
A Trail is like an event manager that records user activity and API usage in the AWS account.

- CloudTrail: A service that records information about user logins/logouts, services used, duration on the dashboard, etc.
- InsightEvents: Logs of unusual activity or A Plusage, which can flag security issues.
- API Error: An error received when a request is made to a service; often used as a flag for detection when unauthorized users try to access the infrastructure.

1. Search for "CloudTrail" and choose the option of **"CloudTrail (track user activity and API Usage) ."**



2. Click **“Dashboard”** and then click **"create trails"**.



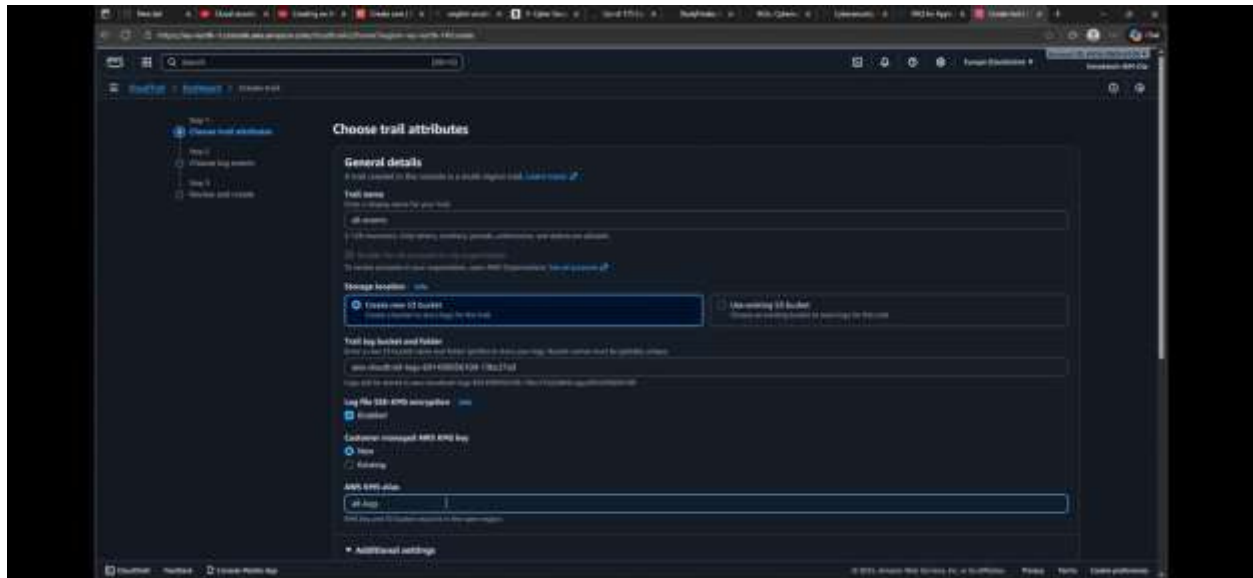
3. General Details:

(a) Trail Name: all events

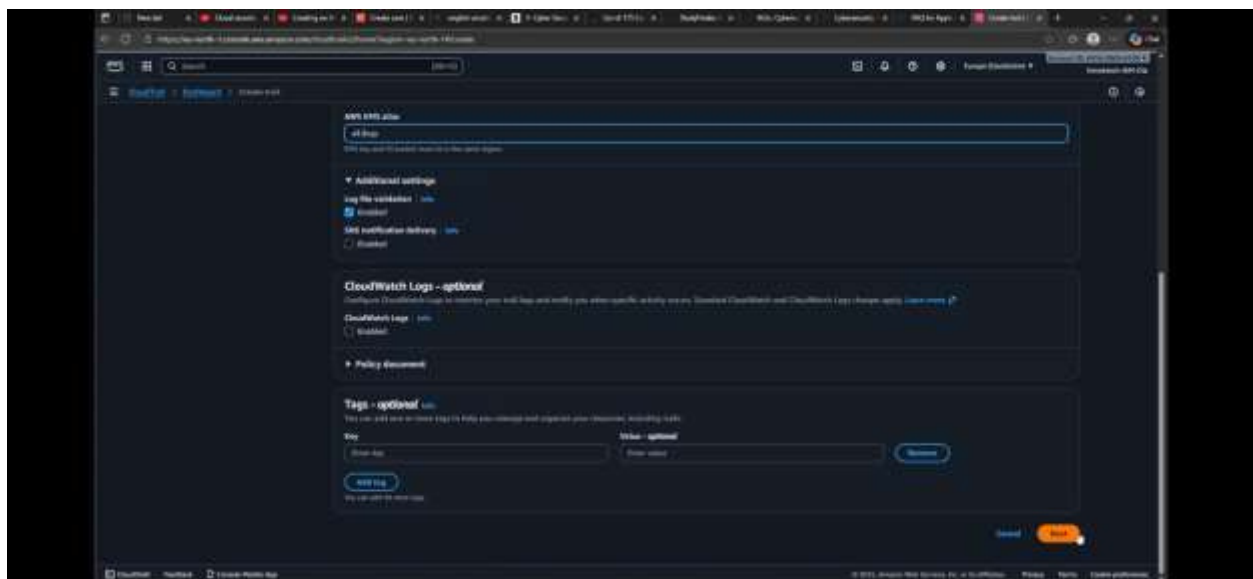
(b) Storage Location: Click Create new S3 bucket for log storage.

(c) Ensure encryption is enabled.

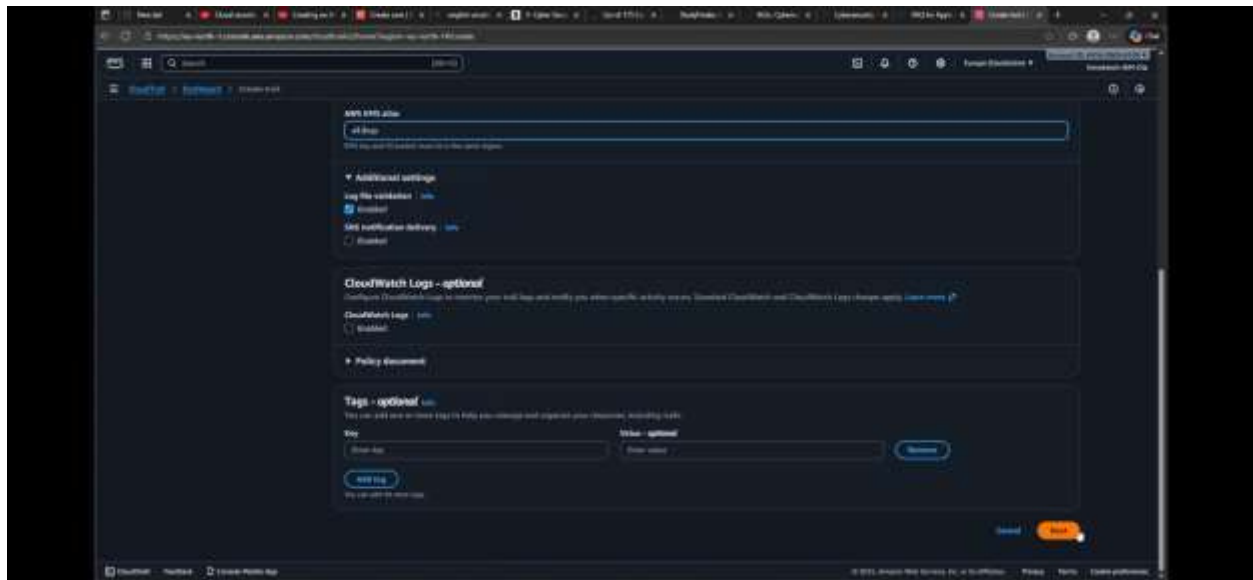
(d) AWS KMS alias: Name the key alias all log.



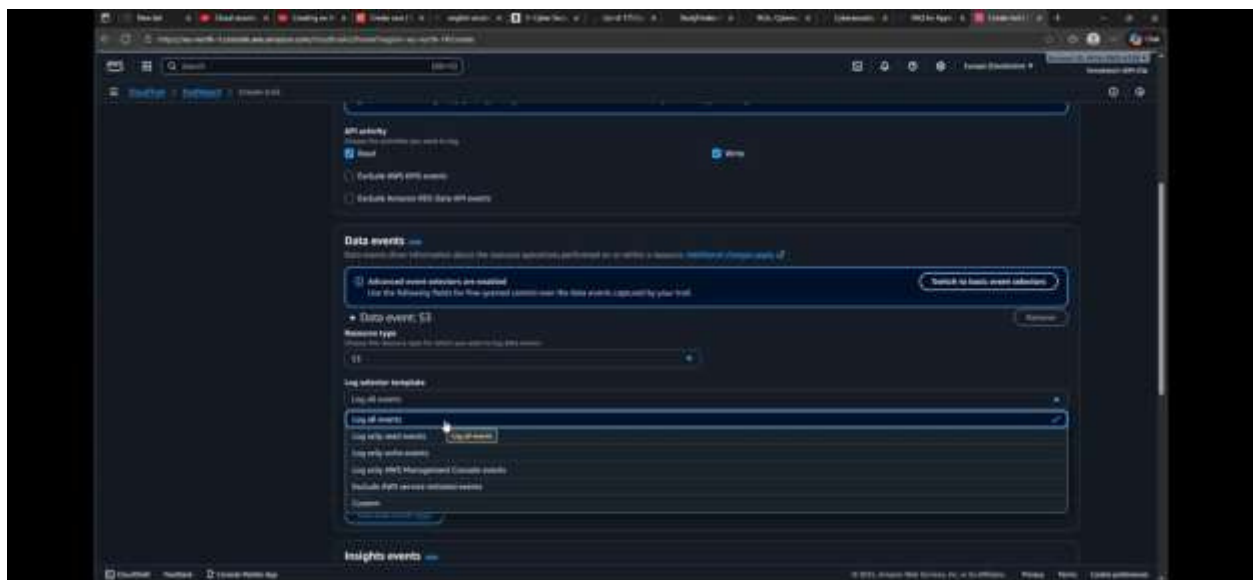
4. Event Selectors Configuration (Next Page):



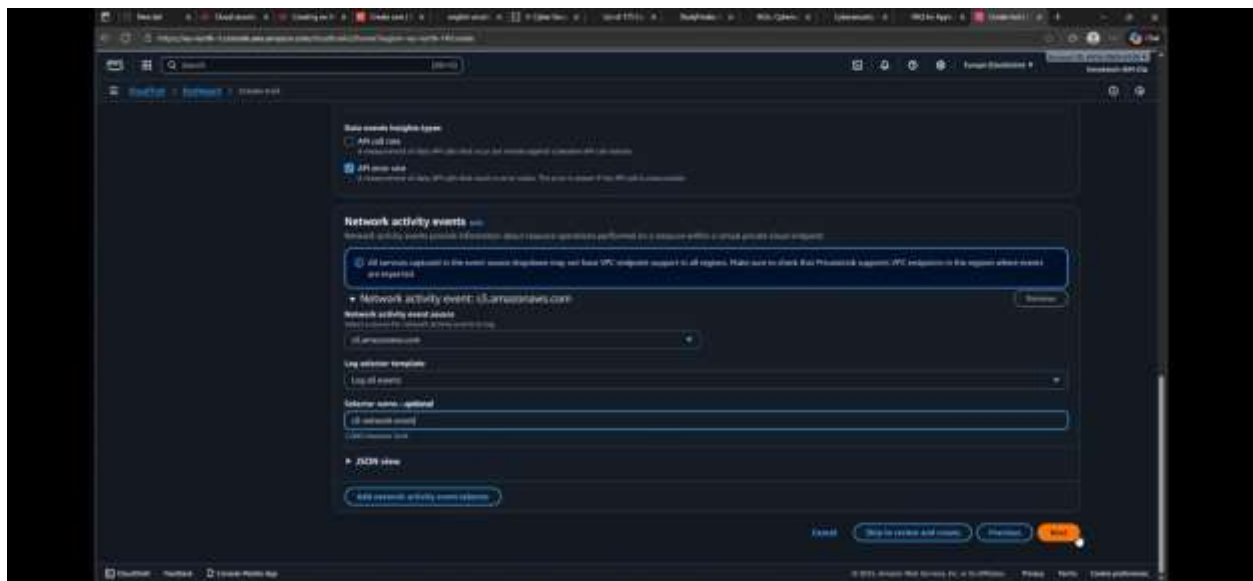
- Select the tabs for management events, data events, insight events, and network activity events.



- Under Data Events: Choose S3 to gather logs about S3 usage.
- Log Selector: log all events
- Selector Name: s3 event

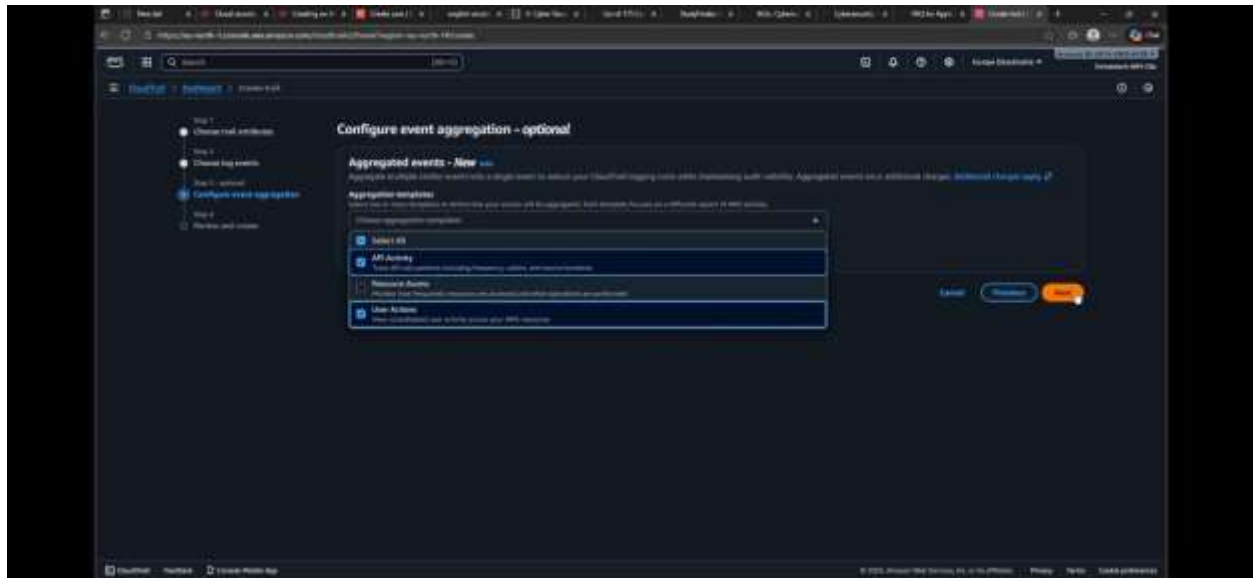


- Under Insight Events: Select "API error". (This helps flag potential unauthorized attempts).
- Under Network Event Source: Choose s3.amazonaws.com since S3 logs are being gathered.
- Log Selector: log all events
- Selector Name: s3 networkevents

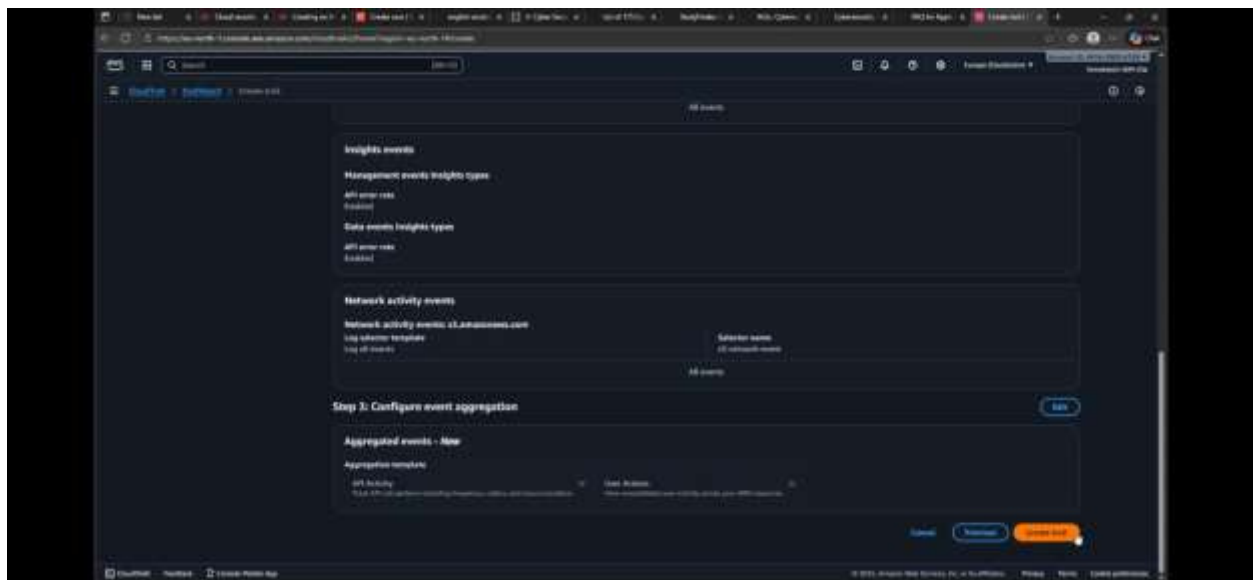


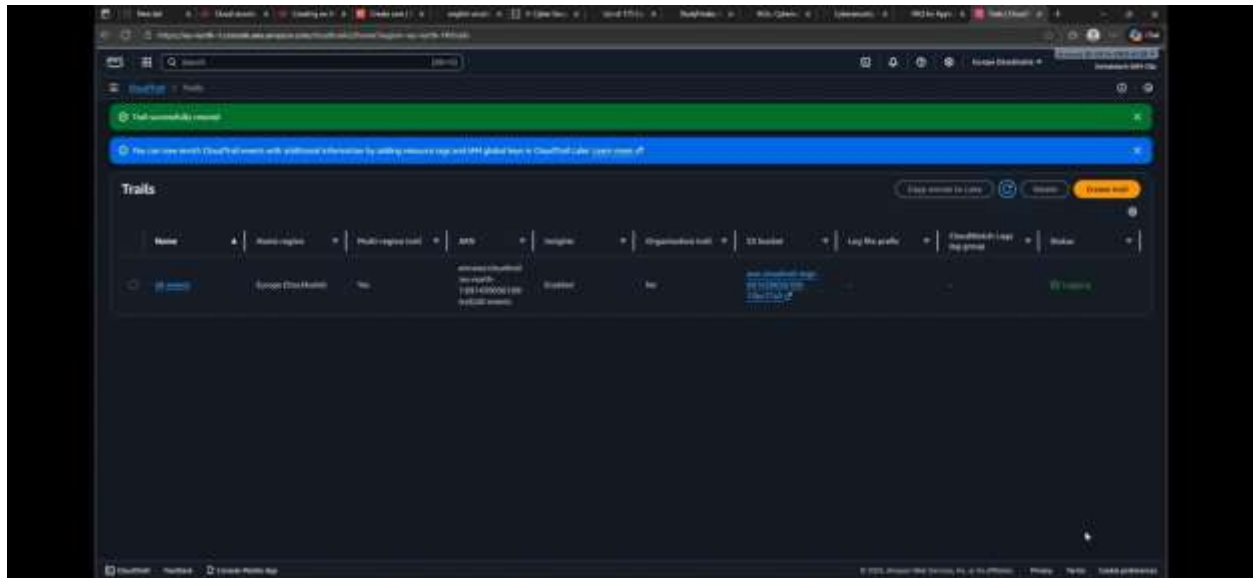
5. Aggregation Templates (Next Page):

- Select the drop-down for "aggregation templates" and choose API activity and user actions.



6. Click Create Trail.



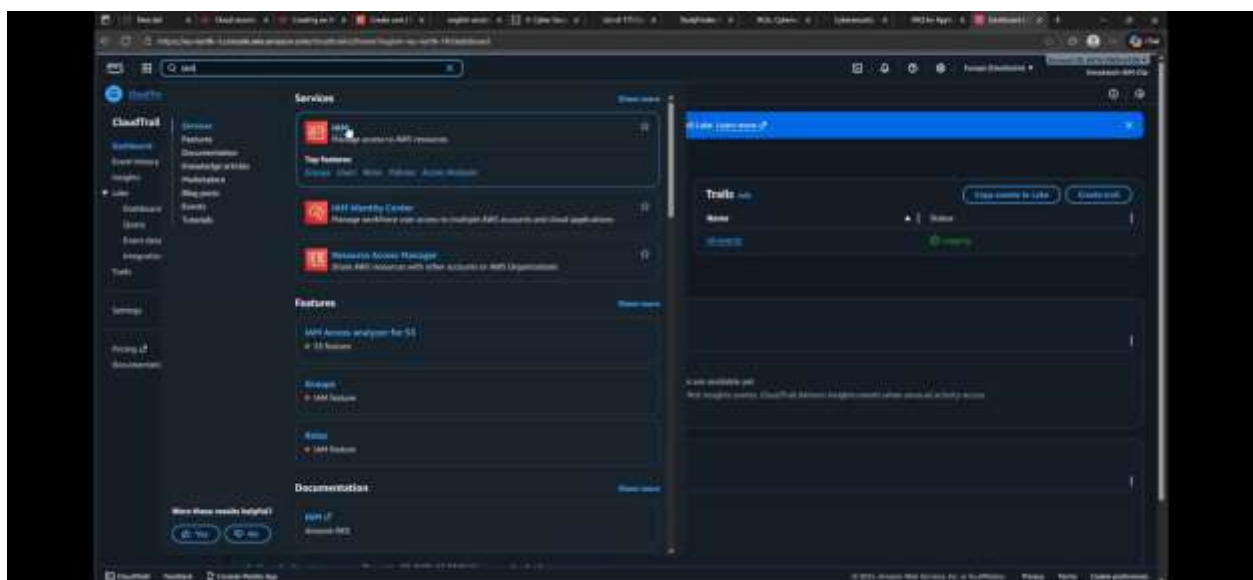


Phase E: Creating the IAM Policy

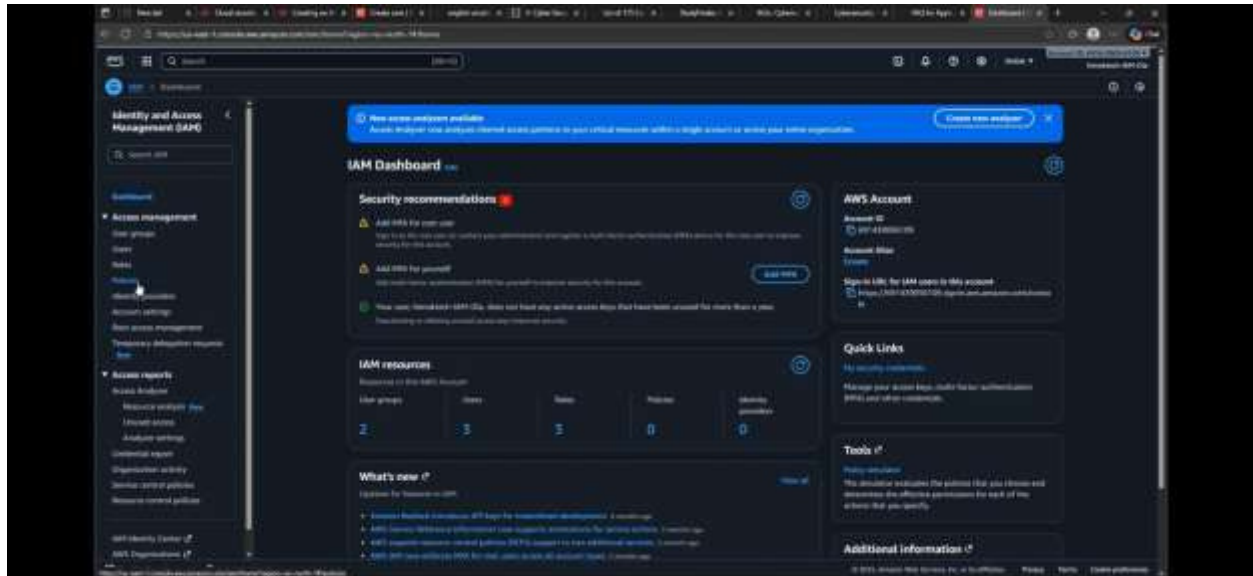
Policies are rules that govern the access users have to the resources in the cloud.

- Policy: A document that specifies the rules, ensuring that users working in marketing only have access to the marketing server, and users in sales only access the sales server and resources.
- JSON Editor: A standard format used to define these policies using code.

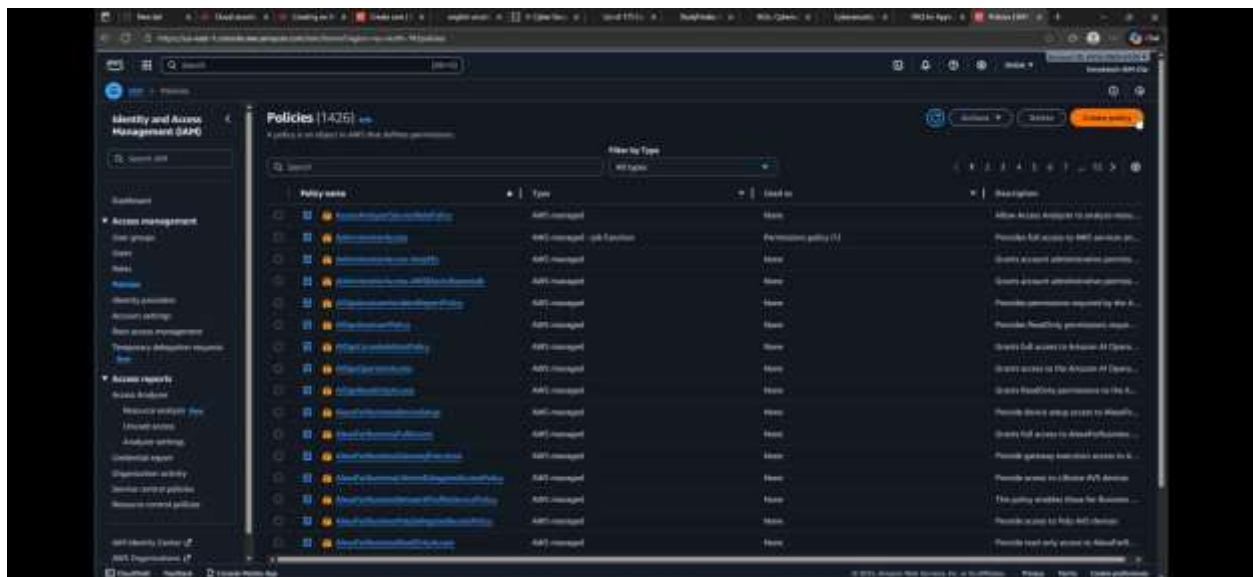
1. Navigate to the IAM service.



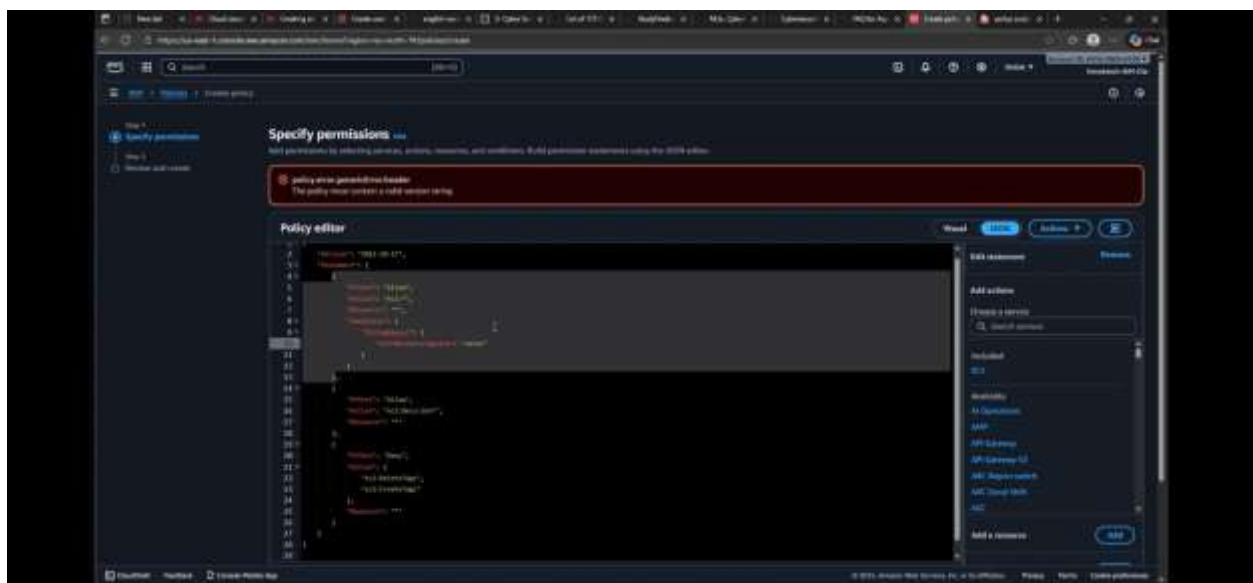
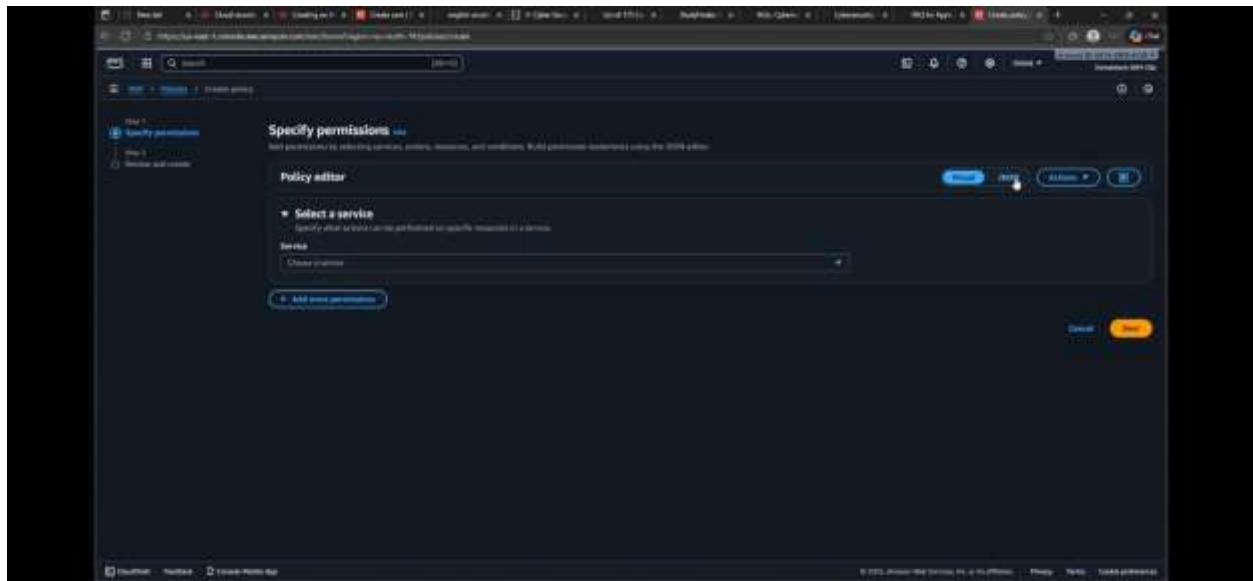
2. Under Access management, click "**policies**".



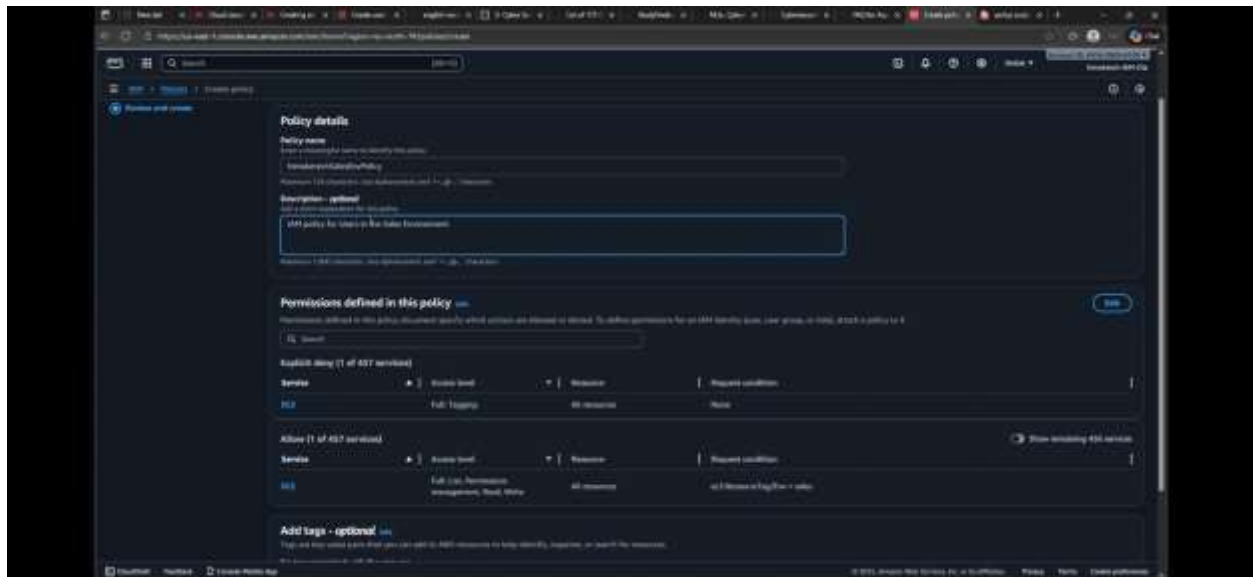
3. Click "create policy".



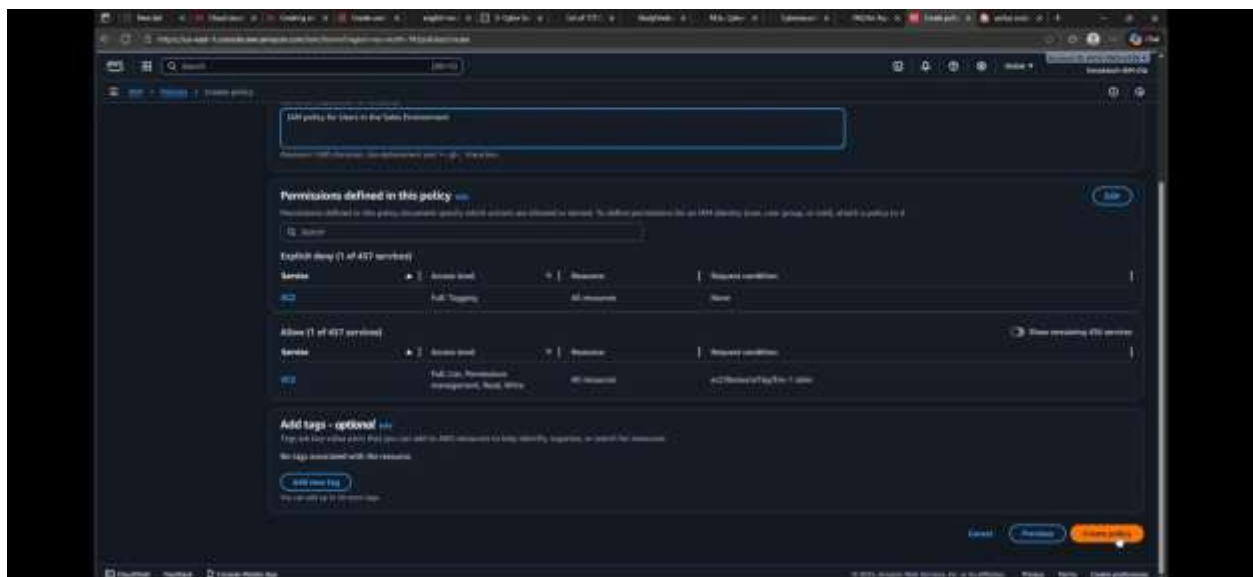
4. Use the JSON editor to upload the policy script that defines the necessary rules.



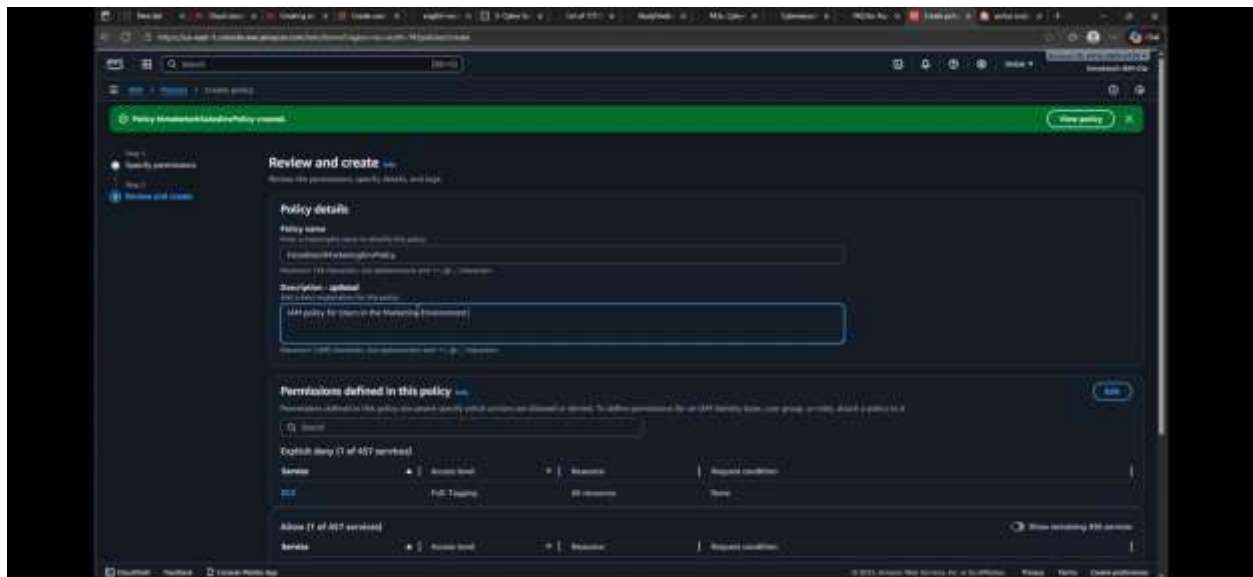
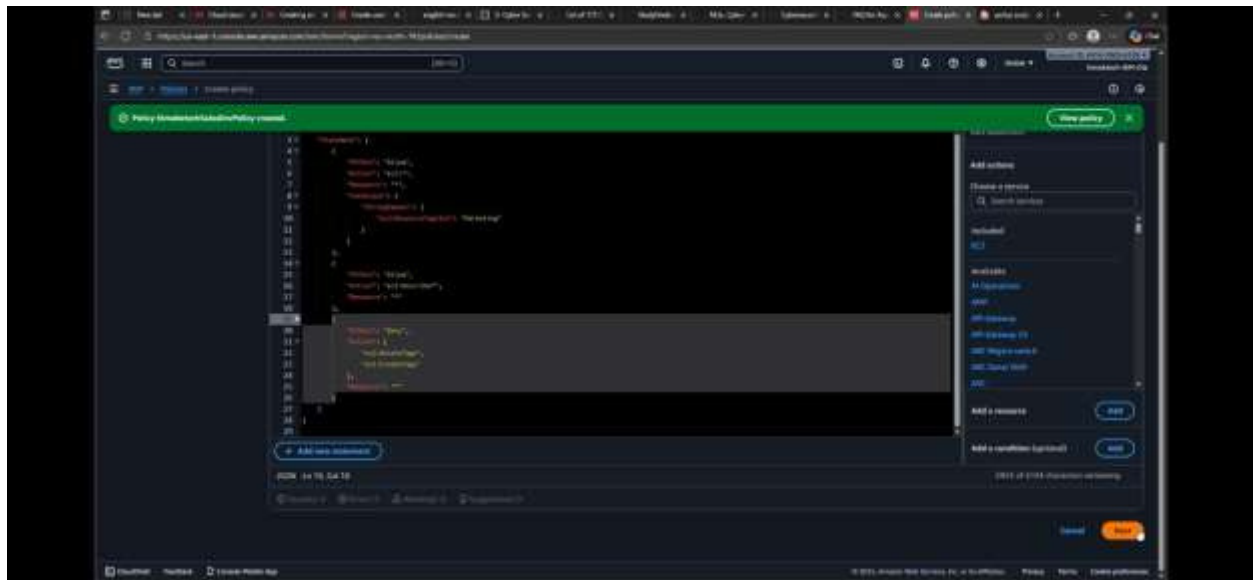
5. Policy Review: • Policy Name: cybertechsalesenvpolicy •
Description: IAM policy for users in the audit environment

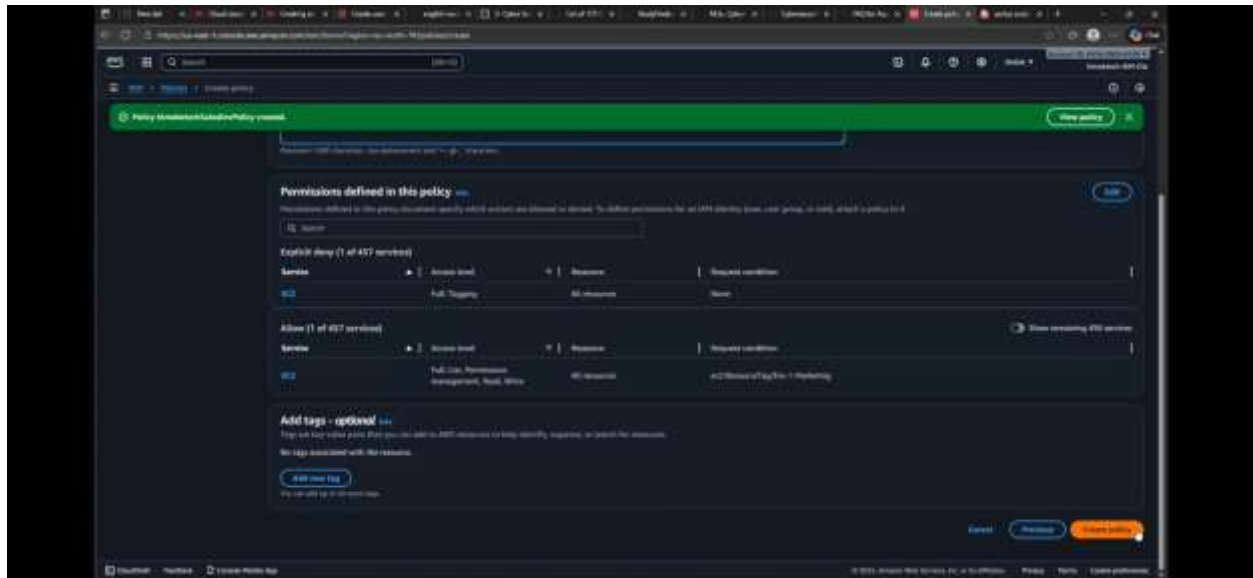


6. Finalize by clicking Create policy.



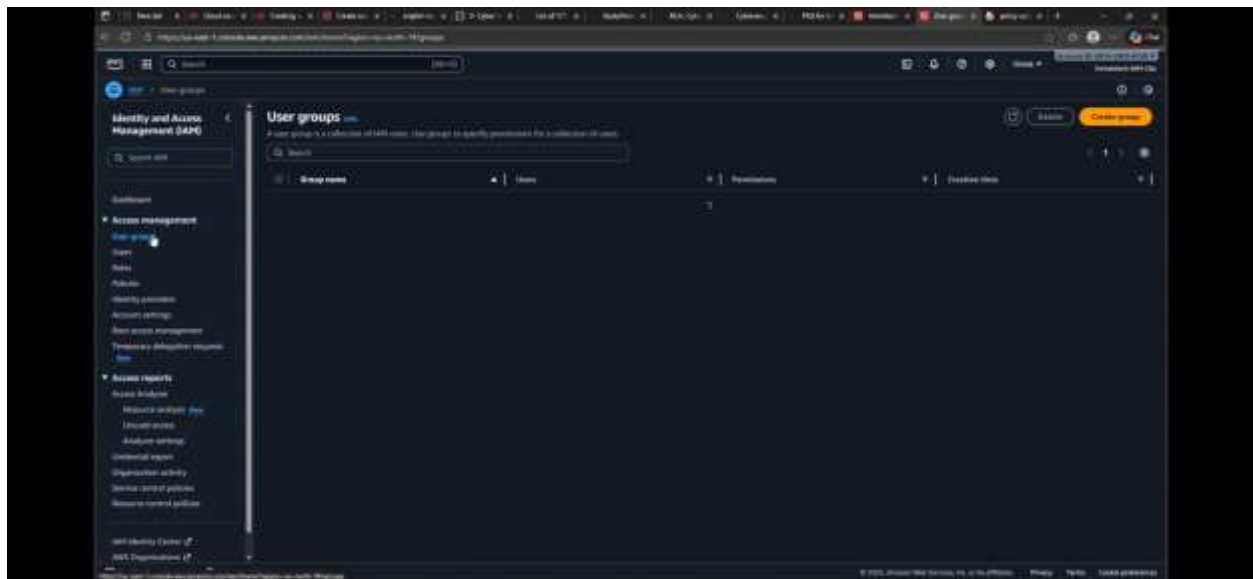
I repeated the process to create a policy for the marketing group also;

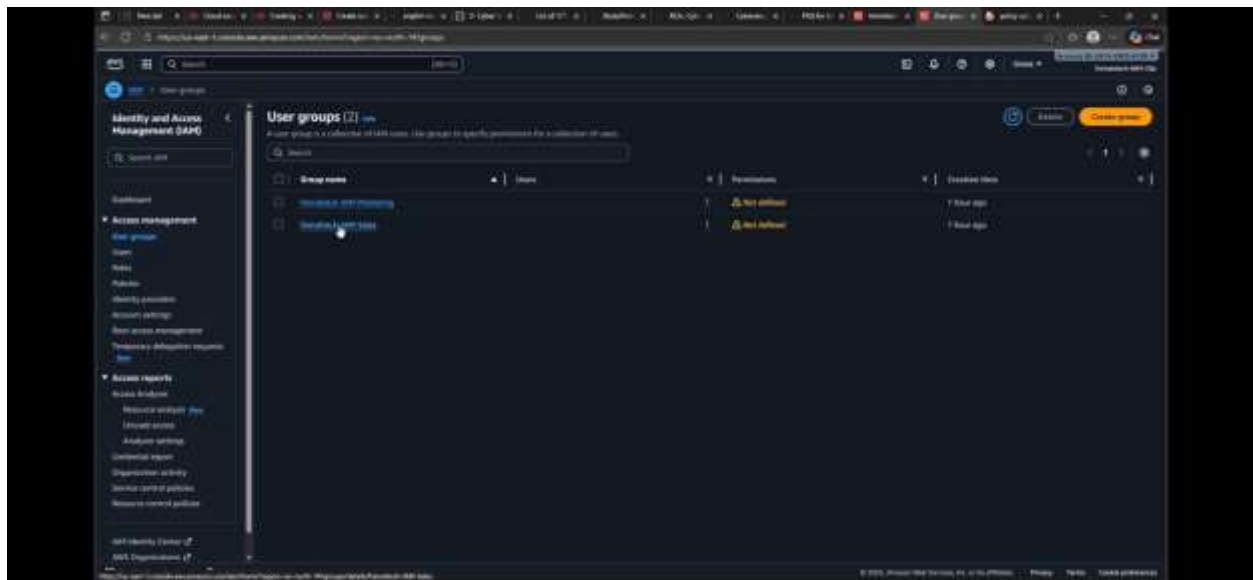




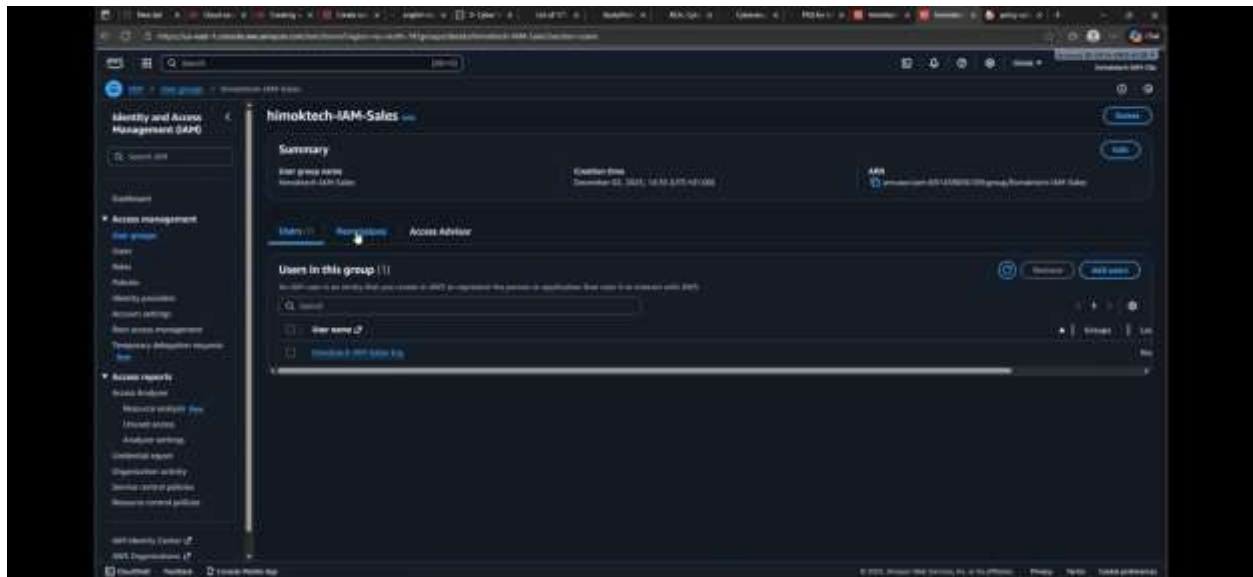
Assign Policy to Sales Group:

- Select the Nxgluxury-iam-sales user group.

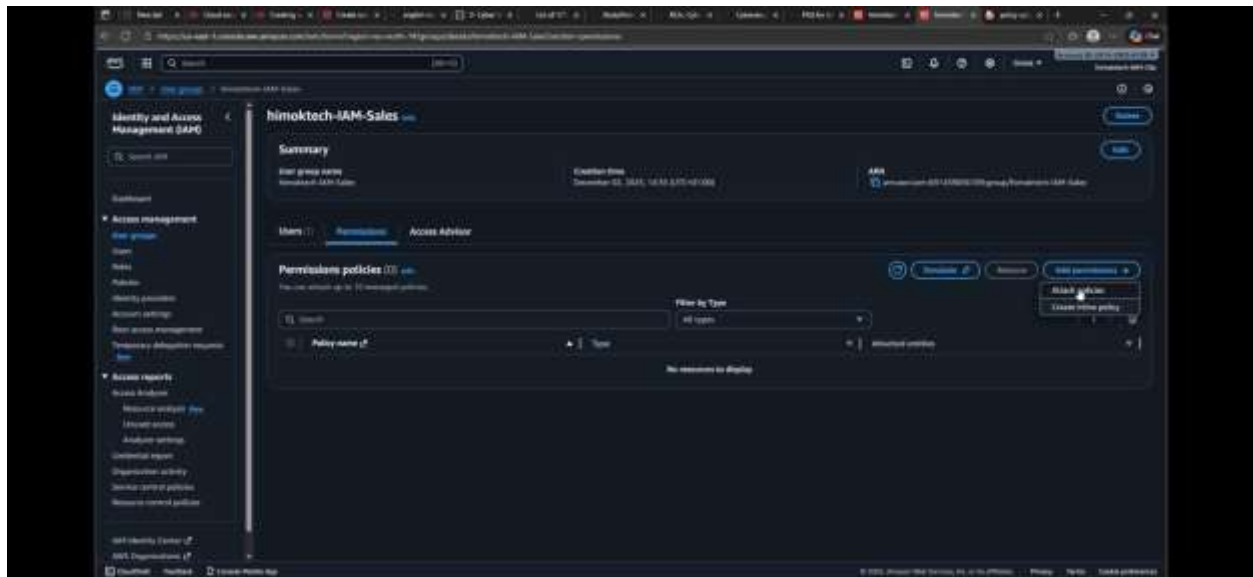




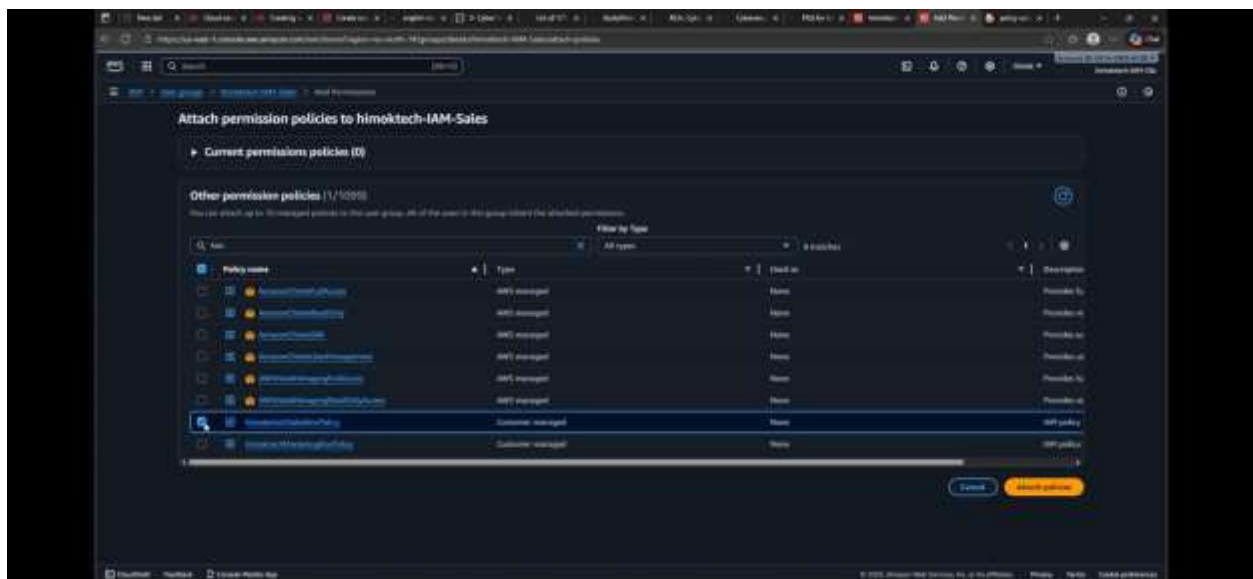
Then I clicked on permission;



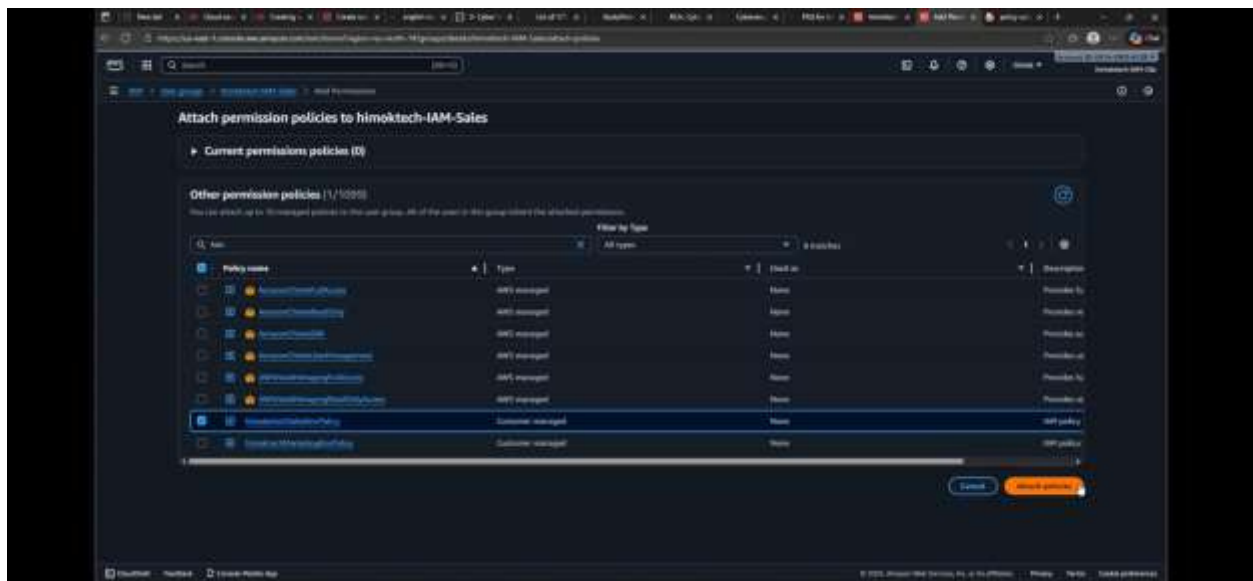
- Choose the option to attach a policy to the group.



- Search for and select the newly created policy: cybertechsalsesenvpolicy.

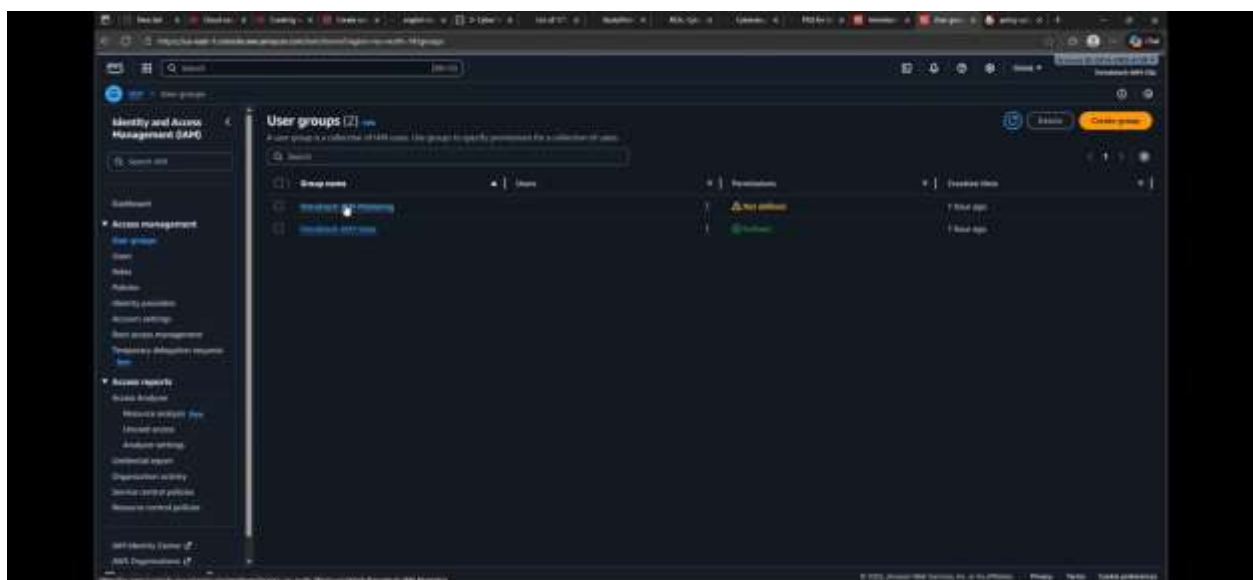


- This action immediately grants all users within the Sales group (e.g., Nxgluxury-iam-sales-Deborah) the defined permissions.

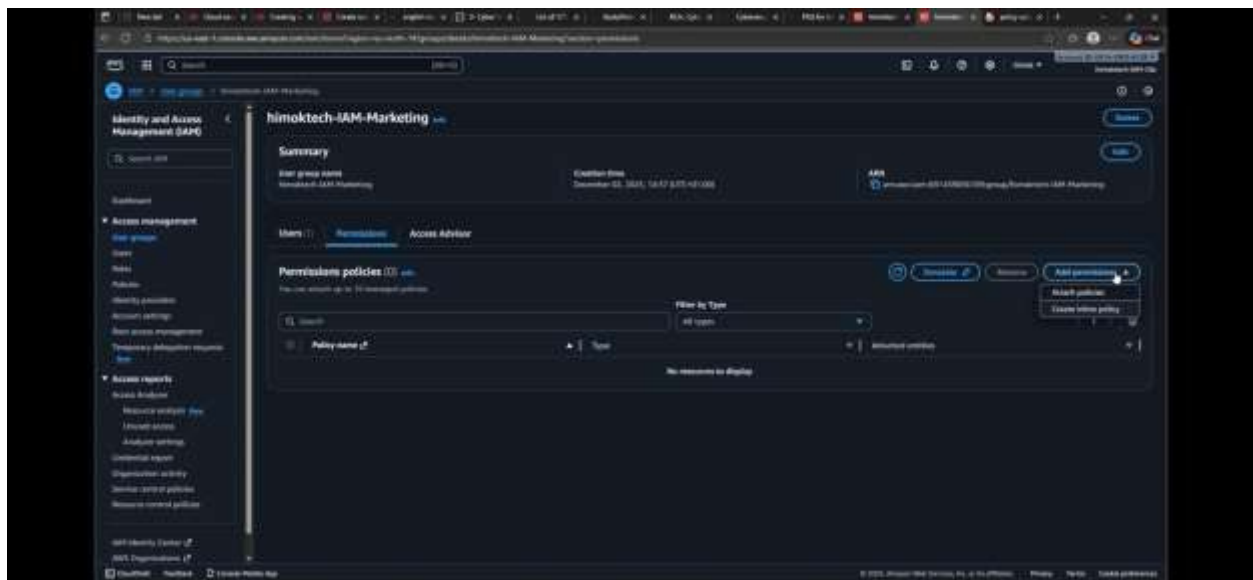
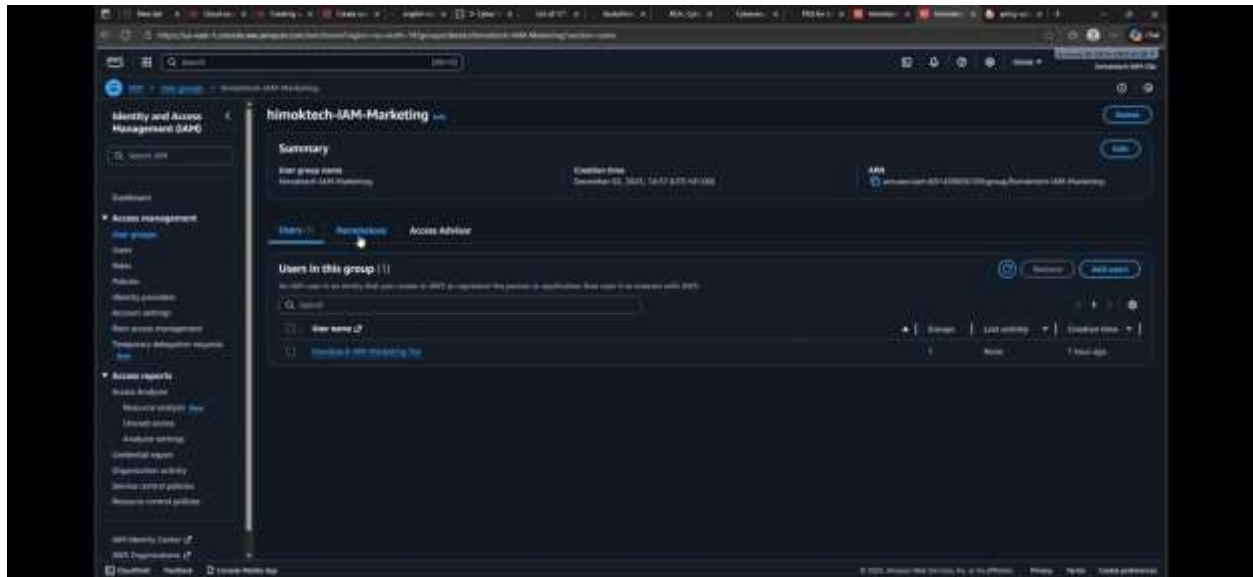


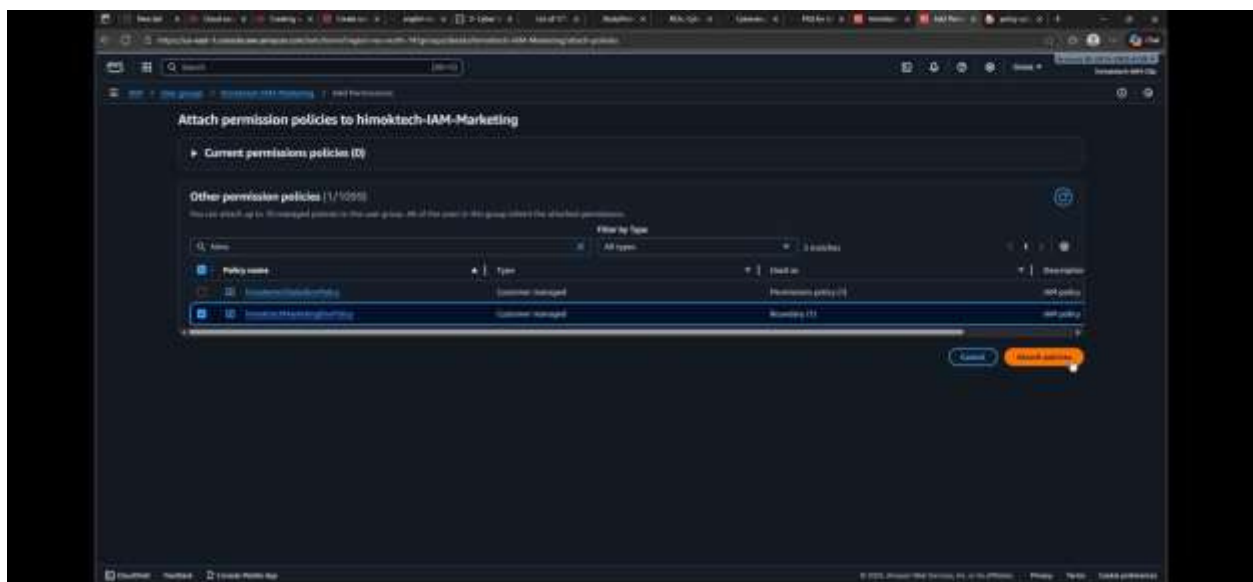
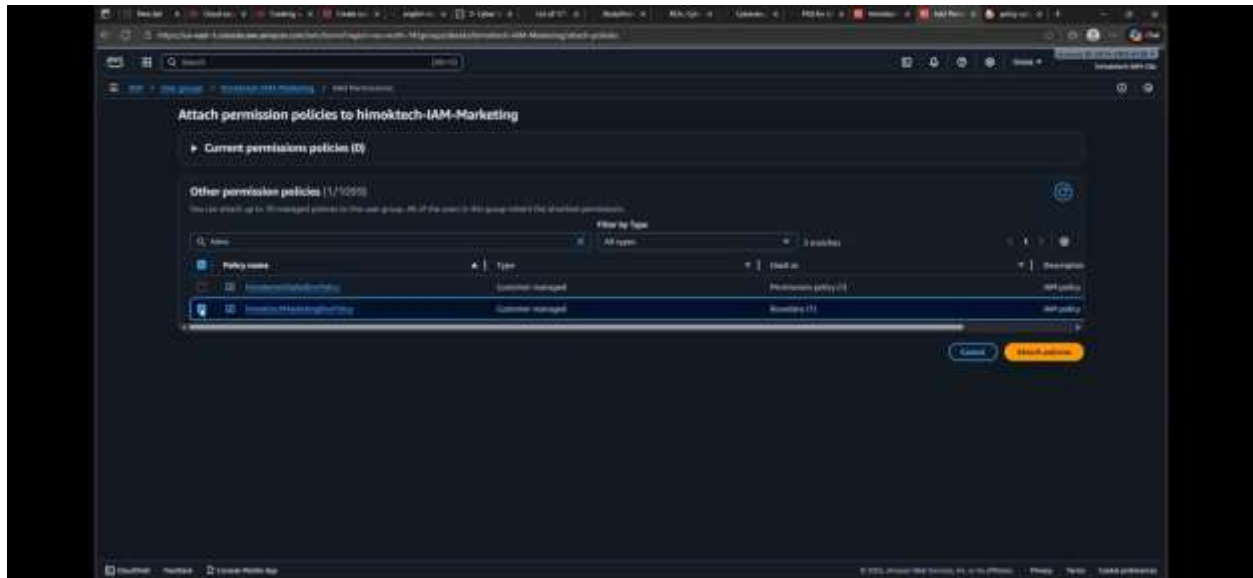
4. Assign Policy to Marketing Group:

- Select the Nxgluxury-iam-marketing user group.



Then click on permission;





- *policy has been assigned to the sales and marketing groups.*

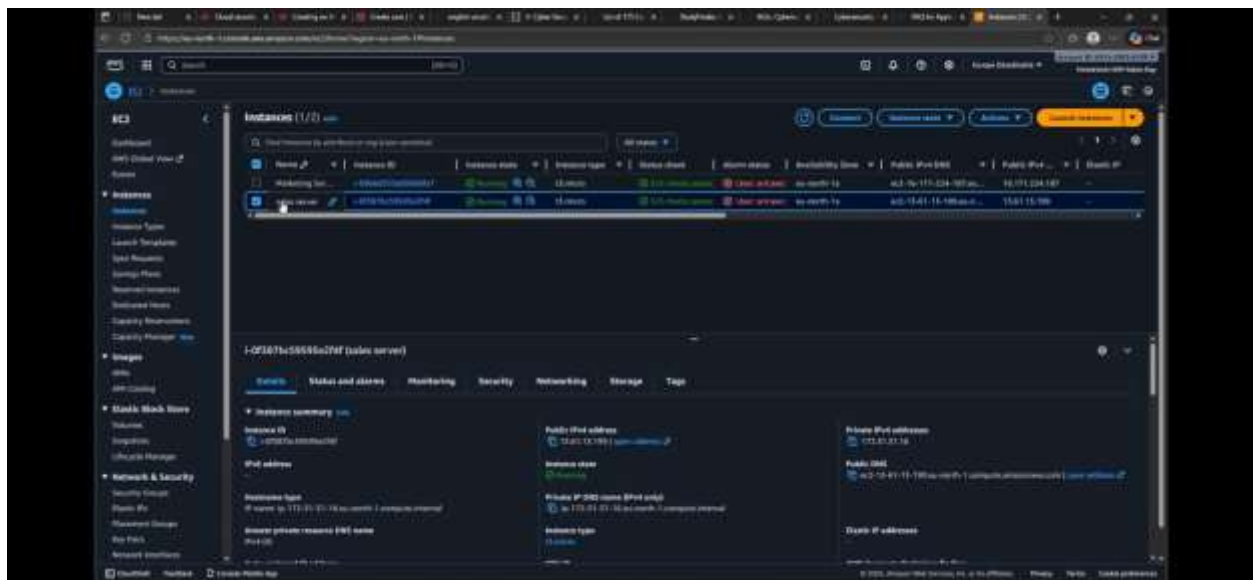
Phase F: Policy Testing and Validation

The final phase involves rigorously testing the custom-created IAM Policy (Phase E) to ensure that the permissions assigned to the respective user groups are correctly enforced. This step validates the security controls for the project, confirming that users cannot take unauthorized actions on critical resources.

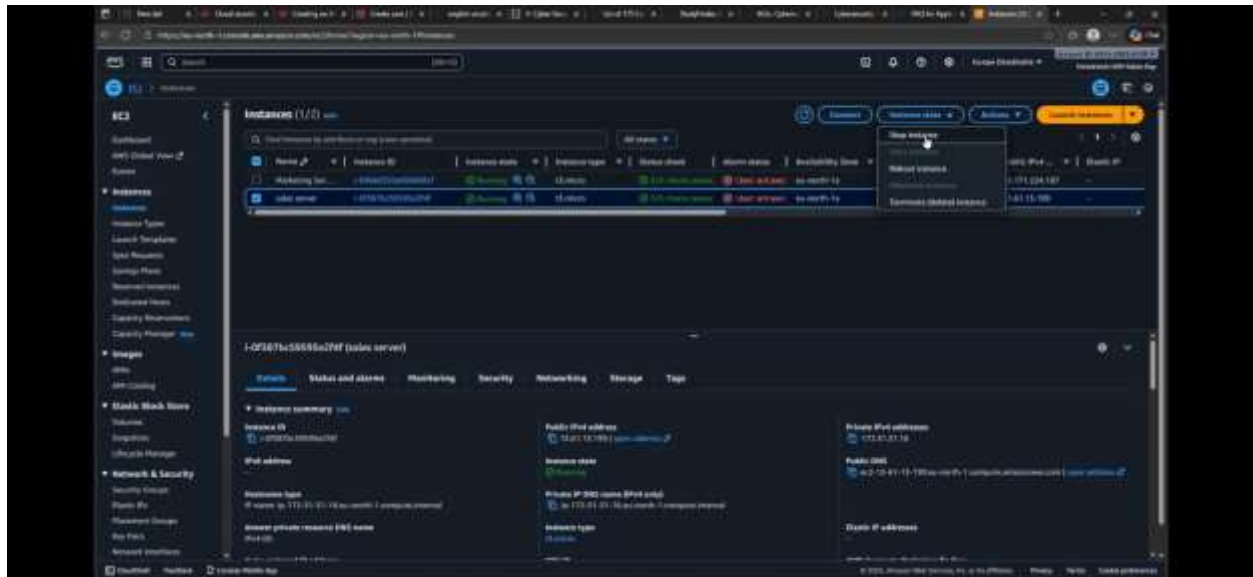
Objective: Enforce the Principle of Least Privilege

The goal of this test is to verify that the Sales user, who is only authorized for sales-related resources, is explicitly denied the permission to modify (stop or terminate) the dedicated compute instances that fall outside their scope.

- Validation Steps**
1. Access Attempt: Sign in to the AWS Management Console using the restricted user credentials: Nxgluxury-iam-sales-Gideon.
 2. Unauthorized Action: Navigate to the EC2 Service and attempt to perform a critical infrastructure management action on the target server.

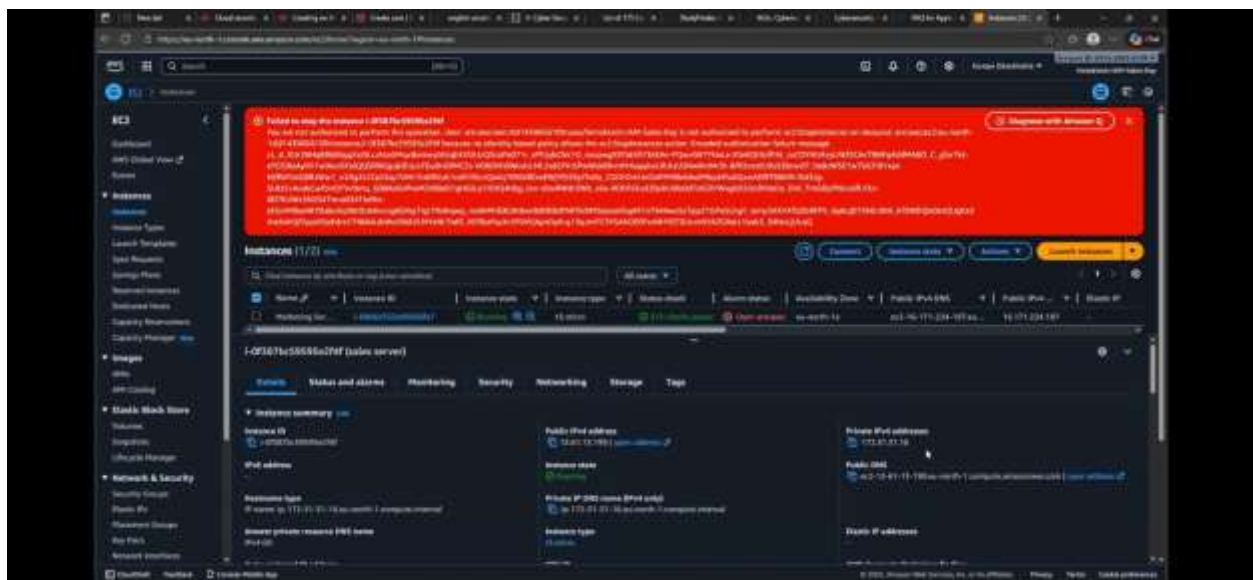


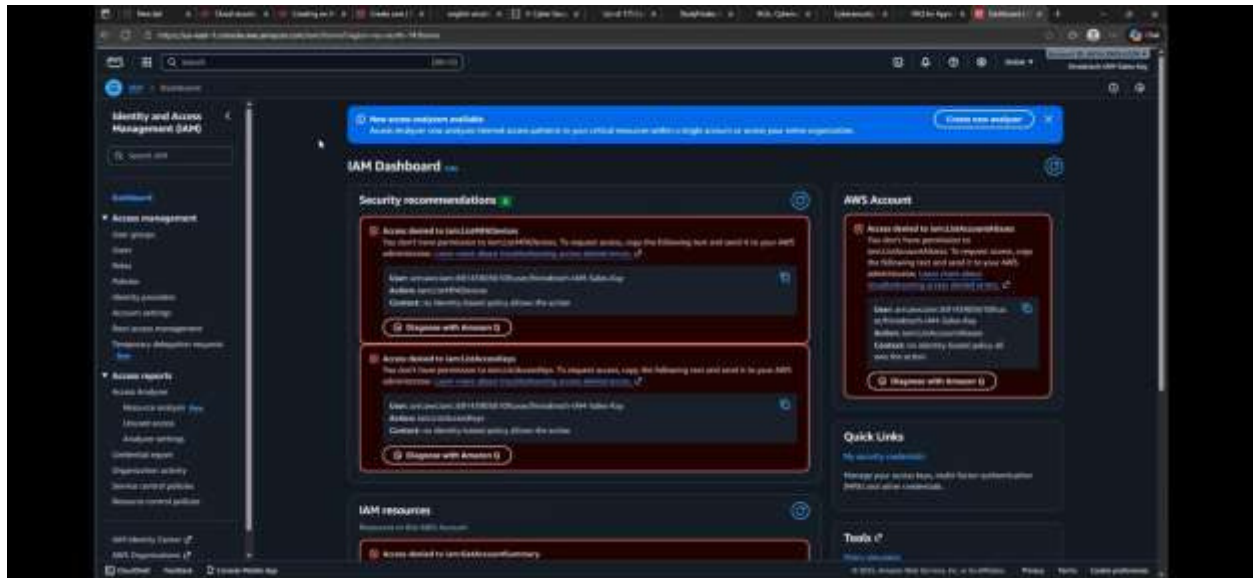
- **Action:** Attempt to stop the running Sales Server instance.



Result and Confirmation:

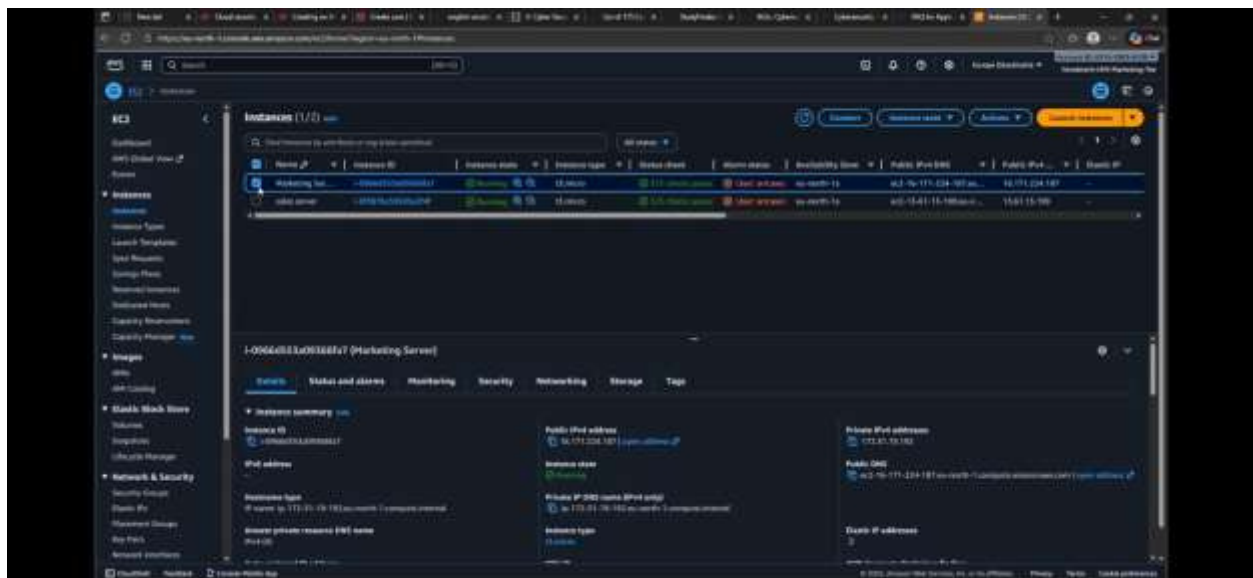
- The attempted action resulted in an Access Denied error message.





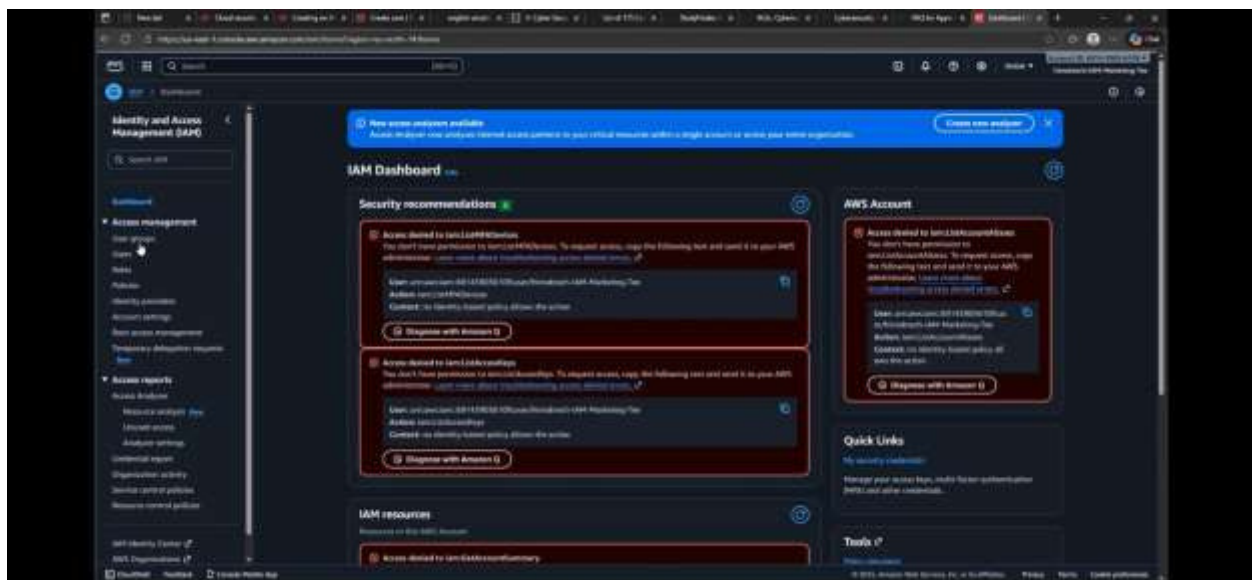
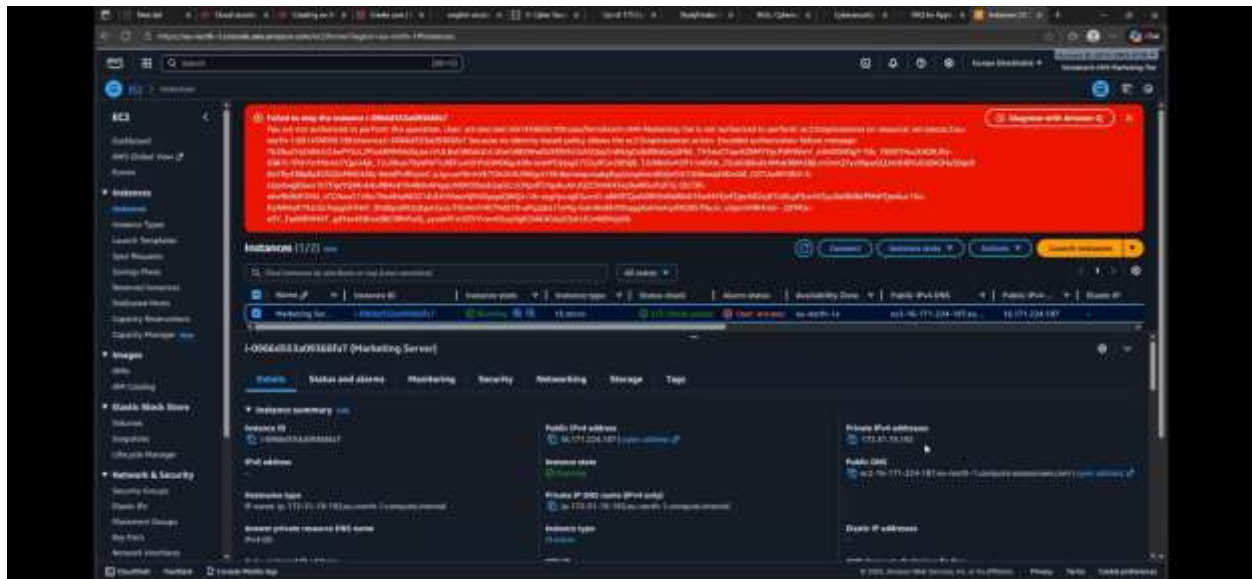
- This successful denial confirms that the custom IAM Policy is properly associated with the user's group (Nxgluxury-iam-sales) and is functioning as designed.

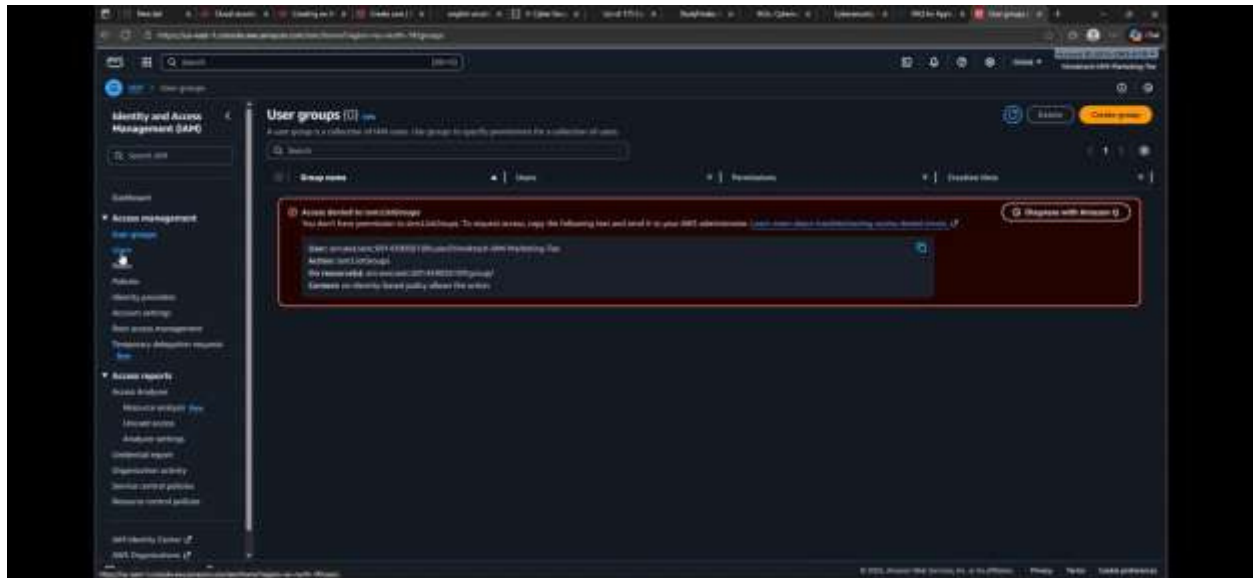
Action: Attempt to stop the running Marketing Server instance.



Result and Confirmation:

- The attempted action resulted in an Access Denied error message

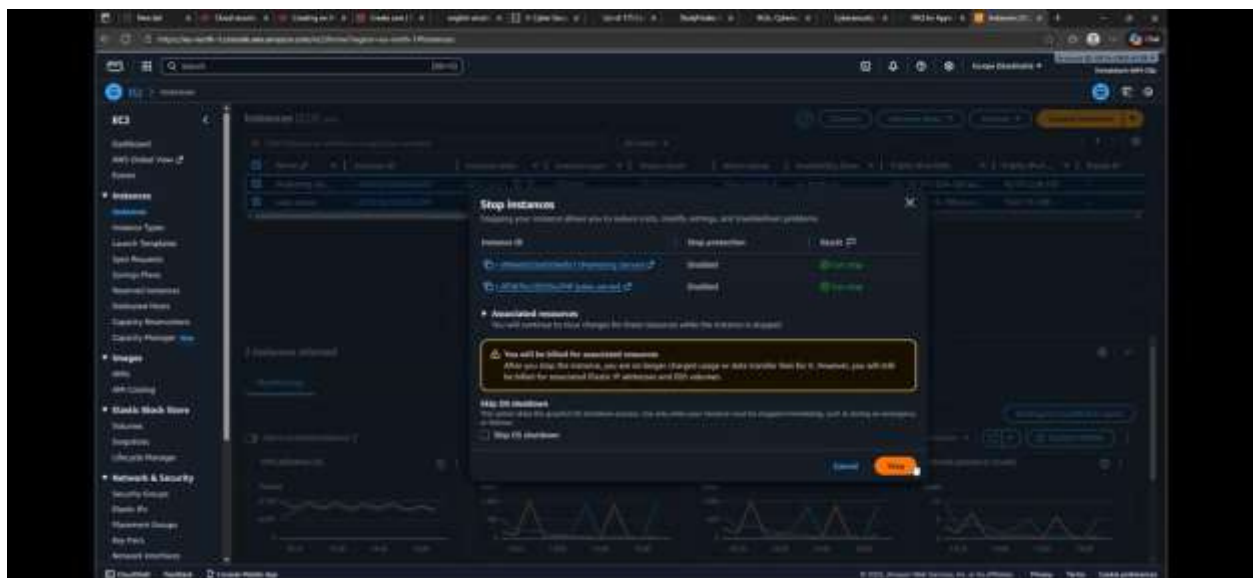
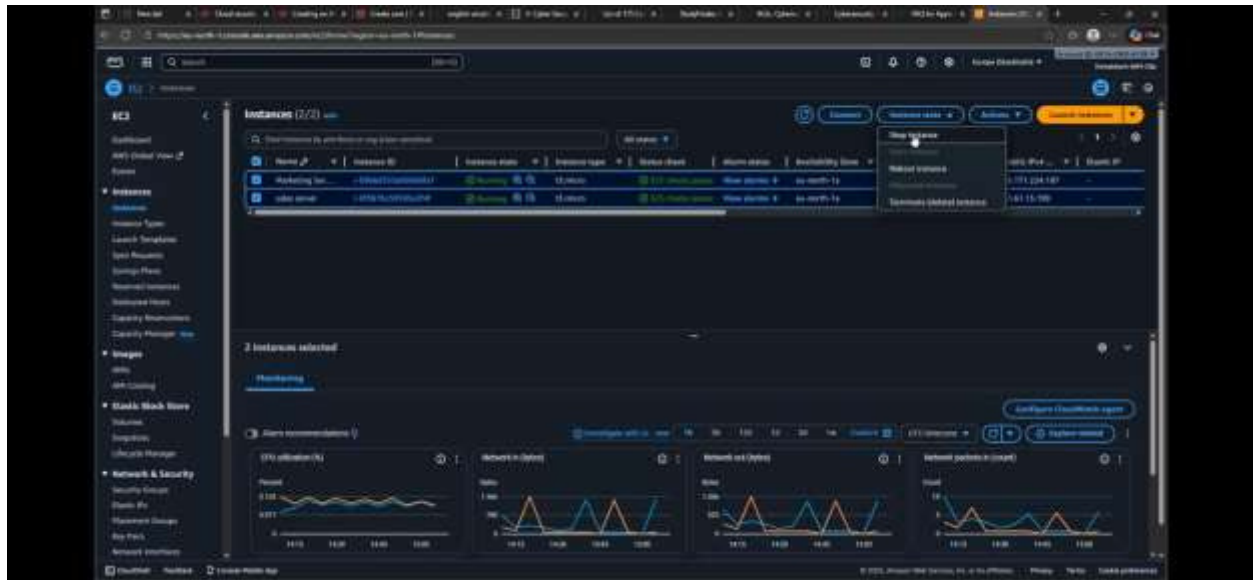




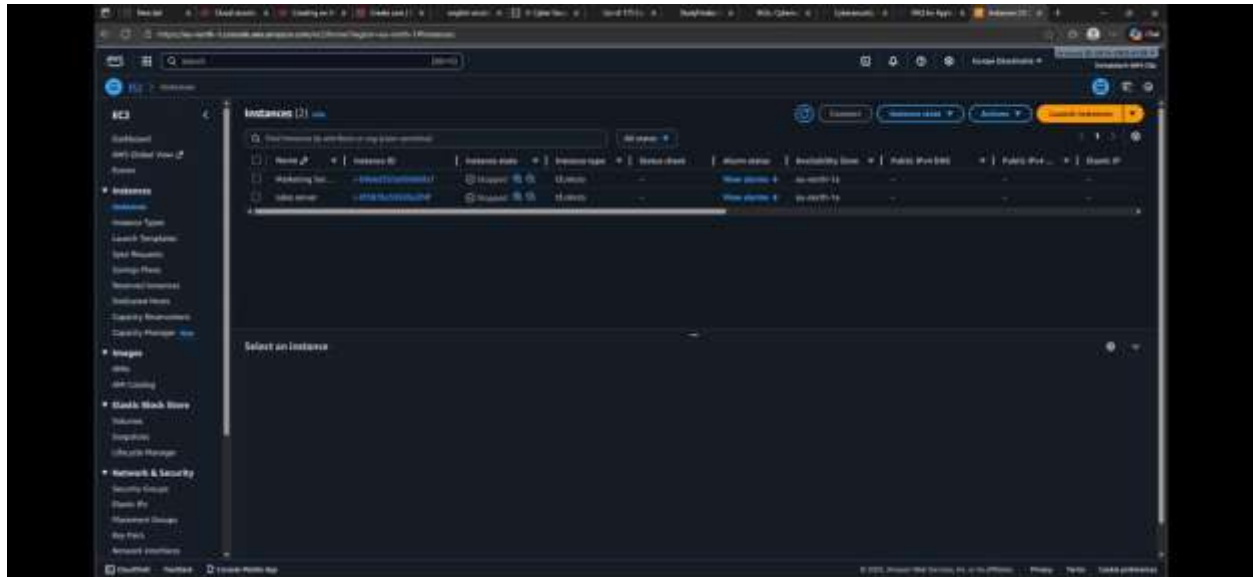
This successful denial confirms that the custom IAM Policy is properly associated with the user's group (nxgluxury-iam-Marketing) and is functioning as designed.

Access Attempt through the admin user: Sign in to the AWS Management Console using the user credentials for the admin: Nxgluxury-IAM-Gideon.

Action: Attempt to stop the running instances through the admin.



Result and Confirmation:



- The attempted action through the user designated as admin, resulted in an Access granted message.
- This successful denial confirms that the custom IAM Policy is properly associated with the user's group (himoktech-iam-sales and himoktech-iam-Marketing) and is functioning as designed.

Conclusion of Testing

The policy test successfully demonstrated that the established Governance framework controlled by IAM is robust. The user's inability to perform actions beyond their defined scope confirms that the project has successfully implemented Identity Segmentation and the Principle of Least Privilege.